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Title:	Shipwrecks can mirror predator assemblages of pelagic pinnacles but may lack the trophic balance of natural reef environments.
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1 **Shipwrecks can mirror predator assemblages of pelagic**
2 **pinnacles but may lack the trophic balance of natural reef**
3 **environments.**

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12 Editing.

13 **ABSTRACT**

14 Artificial reefs are widely deployed to enhance fish populations, yet their ecological equivalence
15 to natural reefs remains uncertain, partly because comparisons are typically limited to fringing
16 coral reefs. Such assessments may overlook how structural features and varying reef
17 morphologies influence ecological outcomes. Fish assemblages were compared across
18 shipwrecks, pelagic pinnacles, and fringing reefs around an island in the Gulf of Thailand to
19 evaluate how artificial structures of differing relief align with natural community and trophic
20 patterns. Bayesian multivariate models applied to 350 surveys from 14 sites showed that
21 shipwrecks resembled pelagic pinnacles in mesopredators and higher trophic level predators, but
22 had consistently lower representation of grazers and invertivores relative to both natural reef
23 types. This functional divergence likely reflects reduced benthic complexity and forage
24 availability; shipwrecks may support predator biomass but not fully function as ecological
25 analogues of natural reefs. Findings highlight the importance of aligning artificial reef design
26 with ecological objectives and considering structural context in management strategies across
27 tropical reef systems worldwide.

28 **Keywords:** artificial reefs, fish assemblages, trophic structure, Bayesian analysis, coral reefs

29 1. INTRODUCTION

30 Artificial reef performance is typically evaluated by comparison with nearby fringing coral reef
 31 systems, although such comparisons may overlook structural influences on ecological outcomes.
 32 Fringing coral reefs, which dominate nearshore tropical coastlines, are generally low in vertical
 33 relief and often embedded in sediment-rich, human-impacted environments (Bell & Galzin 1984,
 34 Letessier et al. 2019). In contrast, many artificial reefs, particularly shipwrecks, share greater
 35 physical similarity with pinnacles (steep-sided bathymetric features similar to seamounts),
 36 including high vertical relief, current exposure, and spatial isolation (Letessier et al. 2019,
 37 Paxton et al. 2023, Galbraith et al. 2022). These features influence foraging opportunities,
 38 predator access, and planktonic input, shaping fish community composition and trophic balance
 39 (White et al. 2007, Galbraith et al. 2023). Despite these parallels, few studies have examined
 40 whether shipwrecks more closely resemble the ecological characteristics of pinnacles, align
 41 instead with fringing reef communities, or represent a functionally novel assemblage.

42 Artificial reefs are deployed worldwide to mitigate habitat loss and enhance fish populations
 43 (Rilov & Benayahu 2000, Paxton et al. 2020a, Harvey et al. 2021, Pondella et al. 2022, Elrick-
 44 Barr et al. 2022, Baillie et al. 2024). Among the various designs, shipwrecks are frequently used
 45 due to their complex structure, tourism appeal, and availability through both intentional
 46 deployment and maritime accidents (Mattos & Yeemin 2020, Medeiros et al. 2022, Hickman et
 47 al. 2024). Intentional shipwreck deployment in Thailand is managed under the national Artificial
 48 Reef Program (since 1978) by the Thai Department of Marine and Coastal Resources (DMCR) to
 49 reduce diver pressure on natural reefs and enhance fisheries, reflecting the high social and
 50 economic value placed on artificial reefs by recreational users and fishers (Yeemin et al. 2006,
 51 Mattos & Yeemin 2020, Harvey et al. 2021, Elrick-Barr et al. 2022, Baillie et al. 2024).

The Gulf of Thailand contains three primary reef habitat types relevant to this study: nearshore fringing reefs that slope gradually toward sandy bottoms (Yeemin et al. 2009), deeper offshore pinnacles with substantial vertical relief, and a growing number of intentionally deployed artificial structures such as shipwrecks (Mattos et al. 2023). Despite the prevalence of these artificial reefs, few studies have assessed their ecological functioning, particularly their capacity to support trophic structures comparable to those of natural reefs and replicate the ecological roles of natural reef systems (Fowler & Booth 2012, Simon et al. 2013, Streich et al. 2017, Medeiros et al. 2022, Paxton et al. 2023, Schram et al. 2024).

Where evaluations exist, artificial reefs often support higher fish abundance and species richness than nearby natural reefs (Fowler & Booth 2012, Folpp et al. 2013, Schulze et al. 2020); however, these metrics (total abundance, richness) can obscure differences in community and trophic structure (Ross et al. 2016, Paxton et al. 2020a, Paxton et al. 2020b, Paxton et al. 2023). Predator biomass is frequently elevated, while lower trophic guilds (grazers, invertivores) tend to be underrepresented on artificial reefs (Fowler & Booth 2012, Schram et al. 2024). This imbalance may limit processes such as algal control, nutrient cycling, and trophic transfer, showing that high abundance alone does not imply ecological equivalence.

Such divergences are influenced by multiple ecological and structural factors, including vertical relief, habitat complexity, reef age, and proximity to natural systems (Rilov & Benayahu 2000, Simon et al. 2013, Streich et al. 2017, Paxton et al. 2023). High relief, isolated shipwrecks favour aggregation of upper trophic level species while offering limited support for lower trophic guilds (Harvey et al. 2021, Medeiros et al. 2022, Schram et al. 2024). Over time, these dynamics may produce assemblages that diverge functionally and compositionally from adjacent natural reefs (Paxton et al. 2020b, Sibley et al. 2023). Distinguishing species-level similarity from broader

75 ecological performance is essential when evaluating artificial reef outcomes (Simon et al. 2013,
76 Medeiros et al. 2022, Schram et al. 2024).

77 Variation among natural reef types provides important context for interpreting artificial reef
78 performance. Isolated pelagic pinnacles, often exposed to strong hydrodynamic flow (Leitner et
79 al. 2021), tend to support high densities of mesopredators and larger predators such as *Caranx*,
80 *Lethrinus*, *Lutjanus*, and *Epinephelus* spp. (Letessier et al. 2019, Galbraith et al. 2022, Cresswell
81 et al. 2023, Galbraith et al. 2023, Mattos et al. 2023). These physical features enhance planktonic
82 input and nutrient delivery, increasing productivity at lower trophic levels and facilitating
83 biomass accumulation at higher trophic levels (White et al. 2007, Leitner et al. 2021, Galbraith et
84 al. 2022, 2023). High relief and structural complexity also provide refuge and foraging habitat
85 across trophic groups (Letessier et al. 2019, Galbraith et al. 2021, 2022, 2023, Pondella et al.
86 2022).

87 In contrast, fringing reefs and other nearshore systems are more strongly shaped by local benthic
88 variation, hydrodynamic exposure, and human disturbance (Done 1983, Bell & Galzin 1984).
89 These environments typically support higher relative abundance of grazers and invertivores that
90 maintain reef resilience and benthic regulation (Koeck et al. 2014, Boaden & Kingsford 2015,
91 Streit et al. 2015, Schram et al. 2024). Comparisons between artificial reefs and adjacent natural
92 reefs, including fringing systems, have consistently shown that shipwrecks support higher
93 densities of mid- to higher-trophic fish, but reduced densities of lower trophic guilds (Fowler &
94 Booth 2012, Paxton et al. 2020a, Medeiros et al. 2022, Baillie et al. 2024, Schram et al. 2024).
95 These patterns likely reflect limited food availability for grazers and invertivores on artificial
96 substrates, as well as differences in benthic microhabitats and structural complexity (Rilov &
97 Benayahu 2000, Schram et al. 2024, Gress et al. 2025).

This study builds on previous assessments of fish assemblages in the Gulf of Thailand (Mattos & Yeemin 2020, Harvey et al. 2021, Mattos et al. 2023, Baillie et al. 2024), to evaluate whether intentionally deployed shipwrecks replicate the community structure and trophic composition of natural reef analogues or represent a functionally novel assemblage. Unlike most artificial coral reef studies limited to fringing-reef comparisons, this study tests shipwreck similarity across two contrasting natural reef types: fringing reefs representing shallow coastal systems and pelagic pinnacles representing deep, high-relief offshore reefs, to determine which they more closely resemble. Given their structural similarity to pinnacles, shipwrecks were hypothesised to support predator-dominated assemblages more closely aligned with pinnacles than with fringing reefs, though limited food availability and reduced benthic complexity may constrain lower trophic guilds such as grazers and invertivores compared to both natural reef types (Fowler & Booth 2012, Simon et al. 2013, Schram et al. 2024). Characterising such patterns is critical to evaluating the ecological role of artificial reefs in tropical marine systems and understanding how reef geomorphology shapes biodiversity, trophic structure, and ecological function. As shallow and coastal reefs continue to degrade under increasing human pressure, isolated offshore features such as pinnacles and shipwrecks may become increasingly important refuges for reef-associated communities and potential sources of ecological resilience. By comparing functional group and species-level patterns across reef types, and relating these to structural features such as vertical relief and complexity, the study aims to clarify the ecological significance of shipwrecks and inform context-specific deployment strategies in Thailand and other tropical regions pursuing artificial reef development (Paxton et al. 2020a, Baillie et al. 2024).

2. MATERIALS AND METHODS

2.1. Study Sites

121 Surveys were conducted across three reef categories: fringing reefs ($n = 5$ sites), pelagic
 122 pinnacles ($n = 6$ sites), and artificial reefs (shipwrecks; $n = 3$ sites), totalling 14 individual sites
 123 (Figure 1, Table S1). All sites were located around Koh Tao (10.10980° N, 99.81180° E) in the
 124 Gulf of Thailand, the largest semi-enclosed area in the central Indo-Pacific (Mattos et al. 2023).
 125 Koh Tao lies approximately 65 km offshore and provides access to a range of reef structures for
 126 comparative study.

127 The three shipwrecks studied were decommissioned Royal Thai Navy vessels: the landing craft
 128 *HTMS Sattakut*, which was deployed in June 2011, and the two fast-attack vessels *HTMS*
 129 *Supharin* and *HTMS Hanhak Sattru*, both deployed in September 2023. All three were
 130 intentionally sunk by the DMCR to enhance social and economic benefits to key stakeholders, in
 131 this case divers and recreational fishers (Elrick-Barr et al. 2022). At 48m in length and 7m in
 132 width, *HTMS Sattakut* sits on sandy sea floor in the west of Koh Tao (10.10169° N, 99.81537° E)
 133 at a maximum depth of 30m and provides 12m of vertical relief from the base of the hull to the
 134 highest point. *HTMS Suphairin* and *HTMS Hanhak-Sattru* were deployed on sand in the north-
 135 west (10.1166° N, 99.80993° E), at a maximum depth of 30m, and north-east (10.09239° N,
 136 99.85292° E), at a maximum depth of 31m, of the island respectively. *HTMS Suphairin* and
 137 *HTMS Hanhak-Sattru* have identical measurements at 45m in length and 7m in width, and
 138 provide approximately 16 m of vertical relief from the hull base to the highest point.

139 Each site was surveyed multiple times from March 2019 to June 2025. Surveys were planned at
 140 least monthly outside the November–December monsoon pause, though frequency varied by site
 141 and year, with lower effort in 2020–22. In total, 350 fish surveys were completed, with 150 on
 142 fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at

predetermined locations within each reef type to ensure consistency across dives. Per-site counts, first and last survey dates, and median intervals are provided in Table S1.

2.2. Survey Protocol

All surveys followed a standardised underwater census (UVC) methodology. Divers conducted eight minute timed-swim surveys (used in Baillie et al. 2024; adapted from Hill and Wilkinson 2004) using a predetermined survey species list (Table S2), which encompasses multiple functional groups (grazers, invertivores, mesopredators and higher trophic level predators) and ensures consistency in species identification and data collection. Survey species were assigned to these four functional groups using precedent consistent with Baillie et al. (2024); the full mapping is provided in Table S2. The list was derived from pilot surveys and regional checklists and was held constant across sites and months to standardize counts, reduce identification error, and focus on conspicuous taxa that are reliably detected within a 10 m inclusion radius (Baillie et al. 2024). Pomacentrids were excluded as their small size and high abundance make them poorly suited to timed-swim methods; accurate assessment would require stationary point counts, which were beyond the scope of this study. Eight minutes was selected as the survey duration based on multiple pilot surveys conducted on the *HTMS Sattakut*, where one full circuit of the wreck could be completed without overlap, minimizing double counting.

Each survey commenced from the same designated starting point on every dive site. The survey team consisted of two divers working in a buddy team on SCUBA. From the fixed start point, the transect path followed the structural contour at the target depth; the end point varied because surveys were time standardised. On shipwrecks, one diver maintained a position ~1 m above the hull edge while the buddy remained 1–2 m into open water at the same depth. On pinnacles, one

diver remained 1–2 m from the reef with the buddy 1–2 m further outward. On fringing reefs, the pair contoured along the reef edge with the same spacing. The target depth matched the average depth used on wrecks (~25 m) where feasible; if not, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed.

Divers maintained a position of 1-2 m away from each other at all times, and maintained a consistent swimming pace while ensuring they stayed level with their buddy. The observation field included all fish present within a 10-metre radius in the 180-degree field of view ahead, covering the left, forward, and right directions, and extending vertically to form a 180-degree three-dimensional arc above and below the divers. A 10 m inclusion radius was selected to capture transient, mid-water higher trophic level predators (hereafter HTLPs) after multiple pilot surveys. Observer distance estimation was trained prior to fieldwork and reinforced during the sampling period using fixed range cues and team cross-checks. Surveys proceeded only when horizontal visibility was ≥ 3 m, and visibility was logged for quality control. Divers swam at a practiced pace of ~ 12 m min^{-1} for eight minutes, covering ~ 96 m per survey; the effective swath area was $\sim 1,920$ m² (distance $\times 2 \times 10$ m inclusion radius). This refines the 1,200 m² figure previously reported by Baillie et al. (2024).

All observed fish on the species list within the designated radius were recorded on waterproof slates in-situ. Counts of individuals by listed species were recorded for each survey, and zeros were retained so that presence and absence were represented.

2.3. Data Analysis

Data were analysed using Bayesian multivariate regression models, fit in R version 4.5.0 (2025-04-11) using the *brms* package (Bürkner 2017). Bayesian models were estimated using Markov chain Monte Carlo (MCMC) sampling implemented in Stan via *brms*. Data manipulation was

performed using the *tidyverse*, and visualisations were generated with *ggplot2* (Wickham et al. 2019). The species were first grouped into functional groups – grazers, invertivores, mesopredators, and higher trophic level predators (hereafter HTLPs) – based on trophic ecology (Table S2) for analysis of broader ecological patterns and to facilitate interpretation of community-level responses. Separate multivariate models were fitted to evaluate differences in functional group abundance across reef classifications and, independently, across geographic zones. Sites were classified by geographic zone as “nearshore” if located within 1 km of Koh Tao, “pelagic” if beyond 1 km, and “wreck” for intentionally deployed ships (Table S1). Zone was added to separate shoreline proximity from reef type when estimating classification effects. Subsequent species-specific models were used to examine finer-scale patterns within each functional group.

2.3.1. Functional group analysis

To examine differences in the abundance of these functional groups across reef classifications, a Bayesian multivariate regression model was used. The model specified the joint response as:

$$mvbind(Grazer, Invertivore, Mesopredator, HTLP) \sim Classification + (1 | Month_{year}) + (1 + \text{Site})$$

allowing inference across groups while accounting for potential correlations in abundance patterns. Classification refers to the site type, either shipwreck, fringing reef or pelagic pinnacle. Shipwrecks were set as the reference group for reef classification.

Prior to model fitting, exploratory analyses were conducted to examine survey effort, distribution of counts, and the composition of functional groups across sites. Zero-inflation rates were calculated for each species, and species with infrequent but sometimes extreme counts (e.g.

209 *Sphyræna* spp., *Gymnothorax* spp., *Tetraodon* and *Diodon* spp., all rays) were excluded due to
 210 their disproportionate influence on group-level patterns (<2% of observations removed). Mobile
 211 predators were retained in the analyses (e.g., *Caranx* spp.); only episodic schooling pulses of
 212 *Sphyræna* spp. that disproportionately influenced group-level estimates were excluded from
 213 model fitting and summarized descriptively. Visibility was not modelled because low-visibility
 214 (<3m) dives were excluded, remaining variation was moderate and comparable across sites and
 215 periods by design, and random effects for site and month-year absorbed residual heterogeneity.
 216 Quality control identified systematic undercounting in surveys conducted by one observer; those
 217 surveys were excluded. Functional group stability was assessed using coefficients of variation,
 218 and proportional group composition was assessed by site to evaluate consistency across reef
 219 types. These diagnostics informed data filtering and supported the robustness of the functional
 220 group classification. Down-weighting via transformation was considered but not implemented. A
 221 sensitivity check that retained *Sphyræna* under capped counts did not change qualitative higher-
 222 trophic-level predator trends. Zero-inflation was not evident in the remaining functional group
 223 data, while overdispersion was moderate, supporting the use of a negative binomial distribution.
 224 Several candidate models were evaluated, including those with and without zero-inflation terms
 225 and with random intercepts for site and month-year to account for repeated surveys and seasonal
 226 variation. Model performance was compared using leave-one-out cross-validation (LOO-CV;
 227 Vehtari et al. 2017). The best-supported model included a negative binomial likelihood with
 228 random intercepts for site and month-year, capturing both overdispersion and hierarchical
 229 structure in the data. The response for each group g , at site i , survey j , was modelled as:

$$Y_{ijg} \sim \text{NegBinom}(\mu_{ijg}, \phi_g)$$

$$\log(\mu_{ijg}) = \alpha_g + \beta_{g1} * \text{Fringing}_j + \beta_{g2} * \text{Pinnacle}_j + \mu_{ig} + v_{mg}$$

$$\mu_{ig} \sim N(0, \sigma^2_{site, g}), \nu_{mg} \sim N(0, \sigma^2_{month_{year}, g})$$

Here, Y_{ijg} represents the count of fish in group g at survey j within site i , μ_{ijg} is the expected count, ϕ_g is the dispersion parameter, μ_{ig} is the random site effect, and ν_{mg} is the random month-year effect. Models excluding site-level effects, month-year effects, or including zero-inflation performed substantially worse, and estimated zero-inflation probabilities were near zero across all responses. The final model was fitted with four chains of 4000 iterations each (1000 warmup), using control parameters `adapt_delta = 0.90` and `max_treedepth = 20`. Default weakly informative priors were applied, as no custom priors were specified. This included default priors on fixed effects, intercepts, and group-level standard deviations, as implemented in brms (Bürkner 2017). Posterior predictive checks were conducted using `pp_check()` to confirm adequate model fit. All parameters demonstrated convergence ($\hat{R} = 1.00$), with high effective sample sizes, supporting confidence in the posterior estimates.

Inferences were based on posterior distributions of fixed effects. Predicted mean abundances and proportional group compositions were visualised using faceted bar plots derived from population-level predictions via `conditional_effects()`. Differences across reef classifications were calculated by combining intercepts with classification-level effects on the log scale, following:

$$\log(\mu) = \text{Intercept} + \text{Classification}_{\text{effect}}$$

Posterior distributions of fixed effects were extracted using `spread_draws()` to estimate log-scale differences in functional group abundance across reef classifications. Differences were visualised using `stat_halfeye()`, which displays the full posterior distribution with overlaid medians and 95% credible intervals.

254 2.3.2. Zone-based analysis

255 A separate multivariate model was fitted to examine differences in functional group abundance
 256 across geographic zones. This analysis was conducted to assess whether spatial context,
 257 specifically, proximity to shore versus offshore exposure, also structured fish assemblages,
 258 independent of reef type. The response structure and model family remained identical to the
 259 classification-based model, with the predictor variable replaced by zone (nearshore, pelagic, or
 260 shipwreck):

261 $\text{mvbind}(\text{Grazer}, \text{Invertivore}, \text{Mesopredator}, \text{HTLP}) \sim \text{Zone} + (1 + \text{Month}_{\text{year}}) + (1 + \text{Site})$

262 $Y_{ijg} \sim \text{NegBinom}(\mu_{ijg}, \phi_g)$

263 $\log(\mu_{ijg}) = \alpha_g + \beta_{g1} * \text{Nearshore}_j + \beta_{g2} * \text{Pelagic}_j + \mu_{ig} + v_{mg}$
 264 $\mu_{ig} \sim N(0, \sigma^2_{\text{site}, g}), v_{ig} \sim N(0, \sigma^2_{\text{month}_{\text{year}}, g})$

265
 266 Where Y_{ijg} represents the count of fish in group g at survey j within site i , μ_{ijg} is the expected
 267 count, ϕ_g is the dispersion parameter, μ_{ig} is the random site effect, and v_{mg} is the random month-
 268 year effect. Shipwrecks were retained as the reference level for interpretability. Model fitting and
 269 diagnostics followed the same procedures, including the use of a negative binomial likelihood,
 270 site-level random intercepts, and evaluation via LOO-CV. Model diagnostics indicated good
 271 convergence and posterior sampling behaviour. LOO-CV showed minimal difference in
 272 expected predictive performance compared to the classification model ($\Delta\text{elpd} = -0.1$), indicating
 273 that reef type likely explains functional group composition slightly more effectively than spatial
 274 zone. Predicted mean abundances and differences across reef zones were extracted and
 275 visualised using the same procedure as the functional group model. Specifically, posterior draws
 276 were obtained via `spread_draws()`, and zone-level effects were visualised using

277 `stat_halfeye()` with 95% credible intervals, showing log-scale abundance differences
 278 relative to shipwrecks.

279 2.3.3. Species-specific analysis

280 Species-specific models were fitted to investigate fine-scale differences in abundance across reef
 281 classifications. Only species with at least 10 non-zero observations were included to ensure
 282 sufficient data for estimation. Survey-level data were aggregated and pivoted to a wide format,
 283 and a multivariate negative binomial model was fitted using the *brms* package, with abundance
 284 modelled as a function of reef classification. The response was specified as a multivariate bind of
 285 all included species, enabling joint estimation while accounting for overdispersion structured as:

286
$$mvbind(\text{Species}_1, \text{Species}_2, \dots, \text{Species}_n) \sim \text{Classification}$$

287 The response for species s at survey j was modelled as:

$$Y_{js} \sim \text{NegBinom}(\mu_{js}, \phi_s)$$

$$\log(\mu_{js}) = \alpha_s + \beta_{s1} * \text{Fringing}_j + \beta_{s2} * \text{Pinnacle}_j$$

290 Here, Y_{js} represents the count of species s in survey j , μ_{js} is the expected count, ϕ_s is the species-
 291 specific dispersion parameter, and shipwrecks were treated as the reference category. Site and
 292 month-year were not included as a random effect due to limited data and resulting model
 293 instability. With 18 taxa across 12 sites, the model became overparameterised and failed to
 294 converge reliably. Excluding random intercepts improved stability while retaining the ability to
 295 detect habitat-based trends. Model fitting used four chains of 4000 iterations each (1000
 296 warmup) with `adapt_delta = 0.90` and `max_treedepth = 20`. Species-level predictions
 297 were generated using `conditional_effects()` with population-level estimates. Differences
 298 in predicted abundance across reef types were visualised using faceted bar plots with 95%
 299 credible intervals.

3. RESULTS

A total of 350 fish surveys were conducted from March 2019 to June 2025 across 14 unique reef sites, comprising 5 fringing reefs, 6 pelagic pinnacles, and 3 shipwrecks (Table S1). Survey frequency varied among sites and years, with lower effort in 2020–22 as a result of COVID and monthly revisits from 2023 onward outside the monsoon period (Nov-Dec). Repeated surveys were conducted at each site, with 150 surveys on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Each survey covered an estimated area of $\sim 1,900 \text{ m}^2$, yielding a cumulative surveyed area of ~ 42 hectares. Across all sites, 162,894 individual fish were recorded. Mean ($\pm$ SD) fish density was highest on pinnacles (668 ± 928 fish per $1,900 \text{ m}^2$), followed by shipwrecks (348 ± 396) and fringing reefs (345 ± 220), indicating elevated average fish abundance at pinnacles, while shipwrecks and fringing reefs supported comparable mean densities.

3.1. Functional groups

Predicted fish abundance varied across reef types and functional groups (Figure 2, Table 1). Grazers and mesopredators were the most abundant groups across all classifications, followed by invertivores and higher trophic level predators (HTLPs). Posterior predictions indicated that grazer abundance was highest at fringing reefs (median: 100.7 fish per $1,900 \text{ m}^2$; 95% Credible Interval: [56.2, 179.0]), slightly lower at pinnacles (98.4; [64.6, 148.4]), and substantially lower at shipwrecks (47.8; [28.1, 83.4]). Invertivores showed comparable predicted abundance at pinnacles (58.7; [35.1, 100.5]) and fringing reefs (54.4; [24.2-120.7]) with lower estimates at shipwrecks (28.7; [28.1-83.4]). Mesopredator abundance was highest at pinnacles (134.8; [59.1, 311.3]), with lower values at fringing reefs (69.7; [182, 250.3]) and shipwrecks (41.7; [11.0,

150.2]). HTLPs had the highest predicted abundance at pinnacles (26.2; [13.4, 51.6]) and shipwrecks (24.6; [9.0, 70.2]) and lower values at fringing reefs (9.1; [3.0, 26.6]).

Posterior distributions indicated differences in functional group abundance across reef types, with varying levels of statistical support (Figure 3, Table 2). Grazers were more abundant at fringing reefs than shipwrecks, with a median effect of 0.75 (95% CrI: [0.03, 1.46]) and a 98% posterior probability that abundance was higher at fringing reefs. Pinnacles also supported higher grazer abundance (0.72; [0.14, 1.31]), with 99% posterior support for a positive effect.

Invertivores followed a similar pattern: abundance was higher at fringing reefs (0.63; [-0.48, 1.75]), and pinnacles (0.71; [-0.19, 1.61]), compared to shipwrecks, with posterior probabilities of 88% and 95%, respectively. Mesopredators were more abundant at pinnacles than shipwrecks (1.19; [-0.35, 2.70]), with 94% support for a positive effect, and may have also been more abundant at fringing reefs (0.51; [-1.34, 2.38]), though posterior support was weaker (71%). For higher trophic level predators (HTLPs), abundance was lower at fringing reefs compared to shipwrecks (-1.00; [-2.51, 0.47]), with a 92% probability of a negative effect. Pinnacles also supported higher HTLP abundance than fringing reefs (1.06; [-0.18, 2.34]), with 96% support. In contrast, the difference between pinnacles and shipwrecks was negligible (0.07; [-1.17, 1.25]), with posterior support evenly split (55%).

3.2. Geographic zones

While reef classification explained most of the variation in functional group composition, a secondary analysis evaluated whether geographic zone, defined by proximity to shore, also structured fish assemblages (Table 3, Figure 4). Grazers were more abundant in both nearshore (median effect: 0.79; 95% CrI: [0.18, 1.38]) and pelagic zones (0.75; [0.03, 1.49]) compared to

shipwrecks, with a 99% and 98% posterior probability, respectively, that abundance was higher in natural zones. Invertivores were also more abundant nearshore (0.80; [0.03, 1.54]), with a 98% probability that abundance exceeded that on shipwrecks, while the contrast between pelagic sites and shipwrecks was less clear (0.32; [-0.62, 1.21]), supported by a posterior probability of 77%. Mesopredators were somewhat more abundant nearshore (0.90; [-0.59, 2.38]) and in pelagic zones (0.79; [-0.97, 2.59]), with 89% and 82% posterior support, respectively, that abundance was higher than on shipwrecks. For higher trophic level predators (HTLPs), abundance was slightly lower at nearshore reefs relative to shipwrecks (-0.53; [-1.77, 0.72], 19 % support for a positive effect) and only marginally higher at pelagic sites (0.08; [-1.39, 1.56], 54 % support). Pelagic reefs may support somewhat higher HTLP abundance than nearshore sites (0.61; [-0.60, 1.81], 85 % support), though overall differences across zones were inconsistent.

3.3. Species-level contrasts

Posterior contrasts for individual species revealed consistent patterns in the direction and strength of habitat associations, particularly distinguishing shipwrecks from natural reef types. Species with the most pronounced or ecologically relevant differences among habitats are highlighted below; full posterior estimates and contrast summaries for all modelled taxa are provided in Tables S3, and S4, and Figures 6–7.

Among mesopredators and high trophic level predators, several species displayed distinct habitat associations. *Lutjanus* spp. <30 cm (small snappers; mesopredators) showed similar predicted abundance at pinnacles and shipwrecks, with a negligible median difference and low posterior support for any effect (-0.05; 95% CrI: [-0.51, 0.38]; 59%), but were substantially less abundant at fringing reefs compared to shipwrecks (-1.29; [-1.75, -0.84]; 100%). *Lutjanus* spp. >30 cm

Fish on Shipwrecks vs. Pinnacles

(large snappers; mesopredators) were more abundant at shipwrecks than pinnacles (−1.06; 95% CrI: [−1.44, −0.69]; 100%) and fringing reefs (−2.85; [−3.28, −2.44]; 100%). *Caranx* spp. (trevallies; high trophic level predators) were more abundant at pinnacles compared to shipwrecks (0.85; 95% CrI: [0.55, 1.15]; 100%) and more abundant at shipwrecks relative to fringing reefs (−0.54; [−0.86, −0.23]; 100%). *Diagramma/ Plectorhinchus* spp. (sweetlips; invertivores) were less abundant at both fringing reefs (−1.38; 95% CrI: [−1.79, −0.98]; 100%) and pinnacles (−0.65; [−1.05, −0.27]; 100%) compared to shipwrecks. *Lethrinus* spp. (emperors; mesopredators) showed substantially higher predicted abundance at pinnacles relative to shipwrecks (0.89; 95% CrI: [0.57, 1.19]; 100%), while differences between fringing reefs and shipwrecks were minor and weakly supported (0.13; [−0.20, 0.44]; 78%). *Epinephelus/ Plectropomus* spp. >30cm (large groupers; high trophic level predators) were more abundant at pinnacles relative to shipwrecks (0.37; 95% CrI: [0.14, 0.60]; 100%), with slightly higher but weakly supported differences at fringing reefs compared to shipwrecks (−0.07; [−0.32, 0.16]; 72%).

In contrast, several grazer and invertivore taxa were consistently lower on shipwrecks relative to natural reefs. *Pomacanthus* spp. (angelfish; invertivores) showed higher predicted abundance at both fringing reefs (0.94; 95% CrI: [0.69, 1.20]; 100%) and pinnacles (1.17; [0.92, 1.41]; 100%) compared to shipwrecks, with minimally elevated abundance at pinnacles relative to fringing reefs (0.23; [0.07, 0.39]; 99.8%). *Labroides dimidiatus* (cleaner wrasse; invertivores) also had higher predicted abundance at fringing reefs (2.29; 95% CrI: [1.99, 2.59]; 100%) and pinnacles (2.27; [1.98, 2.57]; 100%) compared to shipwrecks. *Epibulus insidiator* (slingjaw wrasse; invertivores) showed a similar pattern, with higher abundance at fringing reefs (3.64; [3.12, 4.24]; 100%) and pinnacles (1.86; [1.33, 2.46]; 100%) relative to shipwrecks. *Scarus* spp.

(parrotfish; grazers) had higher predicted abundance at fringing reefs (1.87; [1.68, 2.06]; 100%) and pinnacles (0.95; [0.77, 1.14]; 100%) compared to shipwrecks. *Chaetodon* spp. (butterflyfish; grazers) followed a similar pattern, with elevated abundance at fringing reefs (1.35; [1.06, 1.64]; 100%) and pinnacles (0.65; [0.36, 0.94]; 100%) relative to shipwrecks. Unlike other grazer taxa, *Siganus* spp. (rabbitfish; grazers) exhibited minimal variation, with slightly higher abundance on pinnacles (0.37; [0.15, 0.58]; 100%) and no strong difference between fringing reefs and shipwrecks (−0.07; [−0.29, 0.15]; 72% probability of a negative effect).

Across species, predatory taxa often reached comparable or higher abundance on shipwrecks compared with pinnacles, whereas reef-specialist grazers and invertivores were consistently lower on artificial structures (Figure 7). The consistently lower abundance of key grazers and invertivores indicates that meaningful compositional differences persist between artificial and natural reef assemblages.

4. DISCUSSION

This study evaluated fish assemblage structure across shipwrecks, pinnacles, and fringing reef environments to determine whether artificial reefs replicate natural community patterns. Based on their vertical relief and relative isolation, shipwrecks were expected to support assemblages more similar to pinnacles than found on fringing reefs. Results suggest that shipwreck assemblages closely resembled natural reef pinnacles in their support of mesopredators and higher trophic level predators, but diverged markedly in their low representation of grazers and invertivores. While several predatory and mobile species were abundant on both pinnacles and shipwrecks, many wrasses, parrotfishes, and small groupers were consistently underrepresented on artificial structures. These patterns highlight the functional selectivity of artificial reefs and

their potential to replicate certain ecological roles while failing to support others, reinforcing the need for functionally driven artificial reef designs and deployment strategies, and for natural reef conservation to remain a priority in Thailand (Paxton et al. 2020a, Baillie et al. 2024). Key outcomes are examined in terms of predator convergence, divergence in lower trophic guilds, and implications for artificial reef design and ecological function.

4.1. Shipwrecks resemble pinnacles in meso- and higher trophic level predators

Predator-dominated assemblages on shipwrecks closely mirrored those observed on pelagic pinnacles, aligning with the initial prediction. High trophic level predator (HTLP) abundance was particularly elevated on shipwrecks, and HTLPs were consistently more abundant at both shipwreck and pinnacle sites than on fringing reefs. This convergence is likely driven by shared physical features: vertical relief, isolation, and strong hydrodynamic flow are known to promote predator aggregation on both pinnacles and artificial reefs (Rilov & Benayahu 2000, Galbraith 2021, Galbraith et al. 2021, 2022, 2023, Pondella et al. 2022, Cresswell et al. 2023). Depth can influence predator distributions, but within shallow to upper-mesophotic zones its effect is often weaker than reef profile and structural relief. Studies report gradual declines in predator richness with depth and limited depth specialism across the first 100 m where structure remains a primary driver (Jankowski et al. 2015, Cresswell et al. 2025). Vertical relief has been identified as a key predictor of predator density on artificial structures, including shipwrecks and oil platforms (Paxton et al. 2020b, Harvey et al. 2021, Sibley et al. 2023, Schram et al. 2024). These features may enhance foraging efficiency and attract mobile predators from surrounding areas, potentially having implications for fish diversity at adjacent reef sites.

These findings are consistent with broader patterns observed across artificial reef systems, which frequently support dense assemblages of schooling pelagic and mobile mid-to-high trophic level predators (Streich et al. 2017, Schulze et al. 2020, Sibley et al. 2023). In this study, predators such as *Lutjanus*, *Epinephelus* / *Plectropomus*, *Caranx*, and *Diagramma* / *Plectorhinchus* spp. were particularly abundant on shipwrecks, echoing patterns reported in the Gulf of Thailand (Harvey et al. 2021, Baillie et al. 2024). In contrast, Paxton et al. (2020b) found that artificial reefs in the USA supported higher densities of transient species such as *Caranx*, but not resident predators like *Lutjanus* or *Epinephelus*, highlighting potential regional differences in species responses to reef structure. The strong resemblance between shipwreck and pinnacle predator assemblages observed in the current study suggests that ship-based artificial reefs may function more like pinnacle reef systems than fringing reefs in terms of predatory fish structure, providing vital insights for managers into how reef profile and spatial isolation influence community composition and can be used to guide artificial reef design, placement and ecological objectives.

4.2. Divergence in lower trophic guilds

Despite strong similarities in predator assemblages, shipwrecks diverged markedly from both natural reef types in their support of lower trophic guilds. Grazers and invertivores were consistently underrepresented, with higher densities of *Scarus*, *Chaetodon*, *Pomacanthus*, and various labrids (wrasses) observed on fringing and pinnacle reefs. These patterns are consistent with broader trends across tropical artificial reef systems, where small-bodied benthic feeders and cryptic invertivores are often scarce (Mattos & Yeemin 2020, Koeck et al. 2014, Hickman et al. 2024, Schram et al. 2024). Notably, *Labroides* and *Hemigymnus* spp. have also been reported as underrepresented on shipwrecks relative to natural pinnacles within the same region (Baillie et al. 2024), reinforcing the likelihood that artificial reefs tend to support distinct assemblages

454 lacking key functional groups associated with benthic foraging. Limited habitat complexity and
 455 resource availability likely constrain these groups (Fowler & Booth 2012, Simon et al. 2013).
 456 Although older or deteriorated shipwrecks can develop finer refugia and turf algal cover (Paxton
 457 et al. 2023), the relatively new metal structures surveyed here have not yet reached such stages of
 458 colonisation (Simon et al. 2013, Mattos & Yeemin 2020, Monchanin et al. 2021, 2025).

459 Structural complexity, rather than depth, appears to drive these patterns. Sites occurred at similar
 460 depths (< 30 m), where diversity and herbivore biomass typically remain stable and major
 461 declines occur only beyond 30–40 m when coral cover and fine-scale complexity decrease
 462 (Stefanoudis et al. 2019, Pinheiro et al. 2023). Nearby pinnacles at comparable depths still
 463 supported higher densities of grazers and invertivores, indicating that structural relief and
 464 microhabitat availability, rather than depth, were the primary drivers (Paxton et al. 2020b,
 465 Leitner et al. 2021, Galbraith 2021, Galbraith et al. 2021). Analyses by geographic zone
 466 (proximity to shore) confirmed that this divergence persisted, indicating that physical setting
 467 alone cannot explain the consistent underrepresentation of lower trophic groups on shipwrecks.
 468 Strong currents around wrecks may further limit benthic foraging and favour species that feed
 469 higher in the water column (Leitner et al. 2021, Galbraith 2021, Galbraith et al. 2021).

470 *Siganus* spp. were notable for their consistent abundance across reef types, diverging from the
 471 patterns observed for other grazer taxa. This may reflect species-level variation in ecological
 472 traits. Some of the *Siganus* species recorded in this study primarily feed on turf algae, but others
 473 surveyed, such as *Siganus javus*, are known to exhibit more flexible or planktivorous feeding
 474 strategies (Streit et al. 2015, Gress et al. 2023), potentially allowing persistence on shipwrecks
 475 where benthic forage is limited. Recent observations indicate that this flexibility may represent
 476 ecological opportunism rather than resource limitation, as *S. javus* has been proposed to feed

opportunistically on epibiotic algae and zooplankton from coral polyps (Gress et al. 2023). Their presence on structurally complex artificial reefs has also been reported in other parts of Thailand (Mattos & Yeemin 2020), suggesting a degree of dietary or habitat plasticity not shared by other herbivorous taxa. Similar trends have been reported in the eastern Mediterranean, where planktivores dominate shipwreck assemblages (Sensurat-Genc et al. 2022), implying that the structural features of wrecks may consistently favour planktivorous species.

Depth-associated shifts toward planktivory are well documented at greater depths, but the similarity in depths among our sites suggests that the wreck–pinnacle contrast observed here is better explained by structure and flow (Jankowski et al. 2015, Stefanoudis et al. 2019). Although other planktivorous taxa such as pomacentrids were not surveyed here, their dominance elsewhere highlights their potential importance in shaping trophic dynamics on artificial reefs. Incorporating this group through stationary point count surveys is a focus of ongoing work at these sites, which will allow a more complete assessment of community structure and trophic balance across reef types. Results here highlight that most low-trophic, benthic-feeding species remain functionally excluded from shipwreck communities, and that addressing habitat and resource limitations in artificial reef design may help support these groups, though it remains uncertain whether such changes would fully replicate natural reef assemblages.

4.3. Functional limitations of ARs as ecosystem surrogates

While overall fish density on shipwrecks was high, comparable to fringing reefs and second only to pinnacles, it was driven primarily by mesopredators and higher trophic level predators, with grazers and invertivores consistently underrepresented. Similar patterns have been documented elsewhere: artificial reefs often match or exceed natural reefs in total fish abundance but differ

markedly in functional composition, with consistently reduced species diversity and richness (Fowler & Booth, 2012, Paxton et al. 2020a). Assemblages dominated by predators and lacking lower trophic groups may reflect a combination of habitat features, species-specific responses, and top-down effects, where predator presence suppresses prey guilds (Boaden and Kingsford, 2015). Artificial reef fish communities are thus compositionally novel and shaped by deployment context, structural design, and regional species pools (Folpp et al. 2013, Schulze et al. 2020).

Coral surveys at the same reef sites included in this study found that communities on artificial structures remain distinct from those on natural reefs, showing partial convergence with pinnacle assemblages over time (Monchanin et al. 2021, 2025). While shipwrecks may offer fisheries value and act as de facto refuges under high exploitation pressure (as they serve as physical barriers to bottom-trawling practices (Hickman et al. 2024)), they appear to fall short of replicating the full ecological function of natural reefs. These findings underscore the need for functionally driven reef design and context-aware management goals when deploying artificial structures.

4.4. Design and management implications

Recognising these structural and ecological mismatches is essential for guiding the design of artificial reefs that support a broader range of functional groups. Artificial reefs can contribute to marine conservation and fisheries management (Fowler & Booth 2012, Paxton et al. 2020a, Pondella et al 2022, Sibley et al. 2023), but their ecological function depends heavily on how they are designed and where they are deployed (Streich et al. 2017, Paxton et al. 2020a, Baillie et al. 2024). Within this depth range, design variables controlling microhabitat and refuge space exert greater influence on functional outcomes than depth itself (Jankowski et al. 2015, Pinheiro

et al. 2023). Structures providing vertical relief are effective at attracting predators, yet replicating the full ecological breadth of natural reefs requires fine-scale complexity: branching substrates, cavities, and textured surfaces that support turf algae and invertebrate prey (Koeck et al. 2014, Mattos & Yeemin 2020). Modular and internally complex designs promote colonisation by benthic specialists more effectively than uniform hulls or concrete blocks (Jaxion-Harm et al. 2018, Monchanin et al. 2021, 2025, Baillie et al. 2024). Incorporating varied surface materials and internal cavities, often absent from metal shipwrecks, can enhance habitat heterogeneity and functional diversity.

The surrounding environment also shapes reef performance. Hydrodynamic conditions, such as current strength and direction, can influence the foraging efficiency of pelagic feeders and may enhance the persistence of planktivores and filter feeders (Galbraith et al. 2023, Mattos et al. 2023). Site context matters as much as structure: while deploying artificial reefs near natural reefs may enhance coral recruitment (Monchanin et al. 2021, Monchanin et al. 2025), it may not necessarily ensure ecological similarity or functional equivalence in fish assemblages (Simon et al. 2013, Galbraith, 2021). Emerging evidence suggests that artificial reefs placed near natural reefs may displace fish and fragment populations rather than enhance overall biodiversity (Layman & Allegier 2020, Medeiros et al. 2022), a central issue to the attractor-producer debate in the artificial reef literature (see Paxton et al. 2023 review). Although displacement can heighten exploitation risk when fish move away from protected areas, other studies report minimal effects of proximity (Ross et al. 2016, Sensurat-Genc et al. 2022). Even where displacement occurs, it may not be ecologically detrimental in all contexts. In heavily fished zones, shipwrecks can nonetheless act as de facto refuges, increasing biomass and structural complexity relative to trawled habitats (Hickman et al. 2024). These patterns emphasise spatial

544 planning for artificial reefs that promotes functional complementarity with natural reefs, while
545 mitigating anthropogenic impacts in degraded seascapes.

546 Broader management strategies should ensure that artificial reefs are implemented alongside
547 natural reef conservation to sustain ecosystem function. Shipwrecks and similar structures can
548 contribute to fisheries management, particularly by supporting larger predators (Harvey et al.
549 2021), but they are not ecological replacements for natural reefs and must be integrated within
550 comprehensive reef management frameworks that prioritise ecosystem conservation (Streich et
551 al. 2017, Pondella et al. 2022, Sibley et al. 2023). Furthermore, given that nearshore shipwrecks
552 have been associated with reduced community heterogeneity and limited ecological benefits
553 (Medeiros et al. 2022), offshore deployments may offer greater functional value. Deploying
554 metal shipwrecks as artificial reefs in offshore settings that mimic the vertical structure of natural
555 pinnacles could enhance support for vulnerable and commercially important species. Achieving
556 trophic balance requires purpose-built artificial reefs designed to support a range of functional
557 groups, not just HTLP species (Paxton et al. 2020a, Pondella et al. 2022). Structures should be
558 tailored to ecological objectives, incorporating habitat complexity for lower trophic guilds and
559 sited based on hydrodynamic conditions and connectivity. Evaluations must go beyond fish
560 abundance to consider trophic structure, mutualist presence, and long-term community
561 composition. As artificial reefs tend to form novel assemblages (Fowler & Booth 2012, Folpp et
562 al. 2013) functional outcomes, not just biomass metrics, should guide design, deployment, and
563 monitoring (Paxton et al. 2020a, Baillie et al. 2024, Scram et al. 2024). Management frameworks
564 and permitting processes should reflect this by linking artificial reef approval to clearly defined
565 ecological functions (Pondella et al. 2022) and ensuring alignment with regional reef
566 conservation priorities. Ultimately, artificial reefs can complement conservation goals, yet the

protection and restoration of natural reefs remain fundamental to sustaining biodiversity and resilience.

Future research should extend beyond fish assemblages to include benthic and fouling communities that influence habitat development and successional dynamics (Fowler & Booth 2012, Galbraith et al. 2021, 2022). Long-term monitoring of colonisation and successional trajectories as done in Monchanin et al. (2021, 2025) will help capture shifts in community composition and ecosystem function (Folpp et al. 2013). Comparative analyses of modular and monolithic designs can identify strategies that enhance habitat suitability for lower trophic guilds (Mattos & Yeemin 2020). Incorporating hydrodynamic modelling could improve understanding of spatial patterns in fish abundance and diversity (Galbraith et al. 2023), while behavioural tracking tools, such as acoustic telemetry, can clarify patterns of fish movement, site fidelity, and habitat use. Seasonal monitoring and assessments of invasive species and habitat connectivity will also be critical for managing long-term reef health (Jaxion-Harm et al. 2018, Schulze et al. 2020). Continued emphasis on context-specific design, ecological function, and long-term monitoring will ensure that artificial reefs contribute effectively to ecosystem resilience, biodiversity conservation, and sustainable reef management under accelerating environmental change.

5. CONCLUSION

This study is the first to compare shipwrecks, pelagic pinnacles, and fringing reefs in terms of fish assemblage structure. Shipwrecks mirrored pinnacles in supporting mesopredators and higher trophic level predators but consistently lacked grazers and invertivores compared to both natural reef types, resulting in a skewed trophic balance. These patterns likely reflect structural

589 and ecological limitations, though site-level variability and seasonal effects may also play a role.
590 Future research using long-term monitoring, fine-scale habitat surveys, and behavioural data will
591 help clarify the drivers of these differences. Effective deployment of artificial reefs should
592 incorporate fine-scale complexity to support lower trophic guilds, and site selection should
593 consider hydrodynamics, connectivity, and spatial placement relative to natural reefs. In
594 Thailand and other tropical regions, high-relief offshore structures may better meet objectives for
595 vulnerable and commercially important species. Artificial reefs can complement, but not replace,
596 the protection and restoration of natural reefs, and their success depends on clear ecological
597 goals, functional design, and integration within broader conservation strategies.

598 **6. TABLES**

599 Table 1 Predicted mean abundances (with 95% credible intervals) of four functional fish groups
 600 across three reef classifications: shipwreck, fringing reef, and pelagic pinnacle. Estimates are
 601 derived from a multivariate negative binomial model with reef classification as a fixed effect and
 602 site and month-year as random intercepts. Values reflect the expected total count per survey for
 603 each functional group.

Functional Group	Shipwreck	Fringing	Pinnacle
Grazer	47.8 [28.1–83.4]	100.7 [56.2–179.0]	98.4 [64.6–148.4]
Invertivore	28.7 [13.1–63.9]	54.5 [24.2–120.7]	58.7 [35.1–100.5]
Mesopredator	41.7 [11.0–150.2]	69.7 [18.2–250.3]	134.8 [59.1–311.3]
HTLP	24.6 [9.0–70.2]	9.1 [3.0–26.6]	26.2 [13.4–51.6]

604 Table 2 Posterior summaries of functional group abundance contrasts across zones from the
 605 across reef classifications (shipwreck, fringing reef, pinnacle). Values are derived from a
 606 Bayesian multivariate negative binomial model. The table reports the posterior median, 95%
 607 credible interval (CI), and the proportion of posterior samples supporting positive or negative
 608 differences. Estimates are derived from posterior draws of the reef-classification model, with
 609 shipwrecks set as the reference level.

Functional Group	Comparison	Median	95% CI	P($\beta > 0$)	P($\beta < 0$)
Grazer	Fringing – Shipwreck	0.745	[0.029, 1.460]	0.978	0.022
	Pinnacle – Shipwreck	0.717	[0.144, 1.310]	0.991	0.009
	Pinnacle – Fringing	–0.031	[–0.642, 0.589]	0.457	0.543
Invertivore	Fringing – Shipwreck	0.628	[–0.476, 1.750]	0.880	0.120
	Pinnacle – Shipwreck	0.711	[–0.188, 1.610]	0.945	0.055
	Pinnacle – Fringing	0.077	[–0.857, 0.994]	0.574	0.426
Mesopredator	Fringing – Shipwreck	0.514	[–1.340, 2.380]	0.714	0.286
	Pinnacle – Shipwreck	1.190	[–0.345, 2.700]	0.940	0.060
	Pinnacle – Fringing	0.668	[–0.865, 2.200]	0.820	0.180
HTLP	Fringing – Shipwreck	–0.997	[–2.510, 0.466]	0.077	0.923
	Pinnacle – Shipwreck	0.066	[–1.170, 1.250]	0.551	0.449
	Pinnacle – Fringing	1.060	[–0.175, 2.340]	0.956	0.044

Fish on Shipwrecks vs. Pinnacles

610 Table 3 Posterior summaries of functional group abundance contrasts across zones from the
611 Bayesian multivariate model. The table reports the posterior median, 95% credible interval (CI),
612 and the proportion of posterior samples supporting positive or negative differences. Estimates are
613 derived from posterior draws of the zone-level model, with shipwrecks set as the reference level.
614

Functional Group	Comparison	Median	95% CI	P($\beta > 0$)	P($\beta < 0$)
Grazer	Nearshore – Wreck	0.793	[0.182, 1.380]	0.994	0.006
	Pelagic – Wreck	0.747	[0.033, 1.490]	0.980	0.020
	Pelagic – Nearshore	-0.033	[-0.636, 0.584]	0.453	0.547
Invertivore	Nearshore – Wreck	0.804	[0.030, 1.540]	0.978	0.022
	Pelagic – Wreck	0.322	[-0.618, 1.210]	0.767	0.233
	Pelagic – Nearshore	-0.479	[-1.240, 0.277]	0.099	0.901
Mesopredator	Nearshore – Wreck	0.897	[-0.588, 2.380]	0.891	0.109
	Pelagic – Wreck	0.790	[-0.971, 2.590]	0.824	0.176
	Pelagic – Nearshore	-0.119	[-1.580, 1.390]	0.436	0.564
HTLP	Nearshore – Wreck	-0.525	[-1.770, 0.720]	0.186	0.814
	Pelagic – Wreck	0.082	[-1.390, 1.560]	0.541	0.459
	Pelagic – Nearshore	0.605	[-0.595, 1.810]	0.852	0.148

615

7. FIGURES

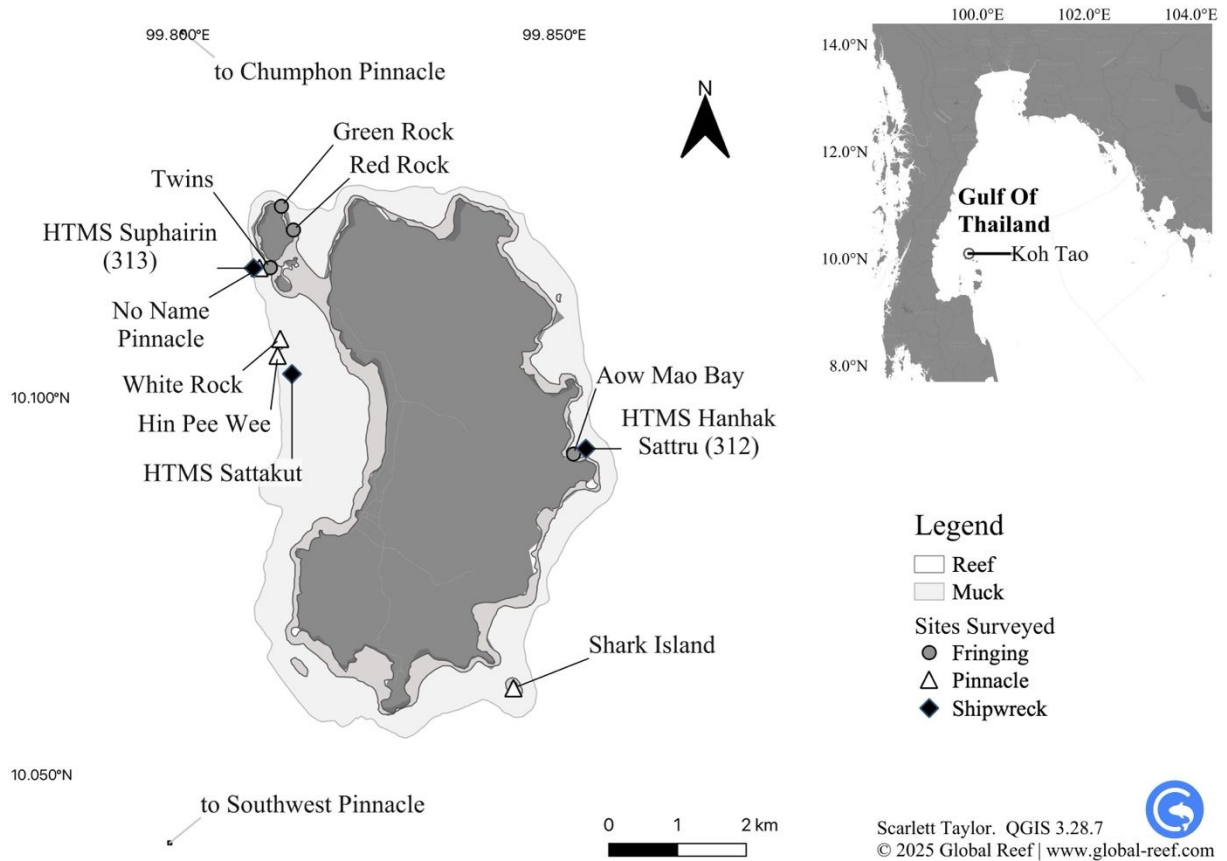


Figure 1 Dive sites surveyed around Koh Tao in the Gulf of Thailand (10.10980°N, 99.81180°E).

Fish on Shipwrecks vs. Pinnacles

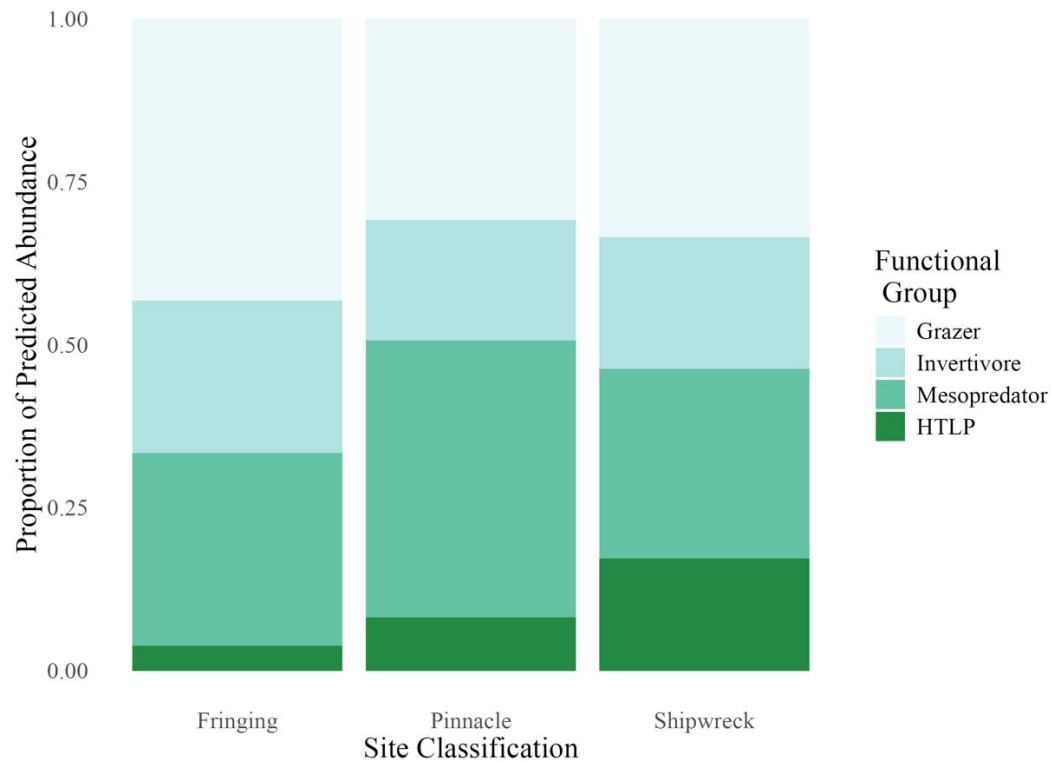
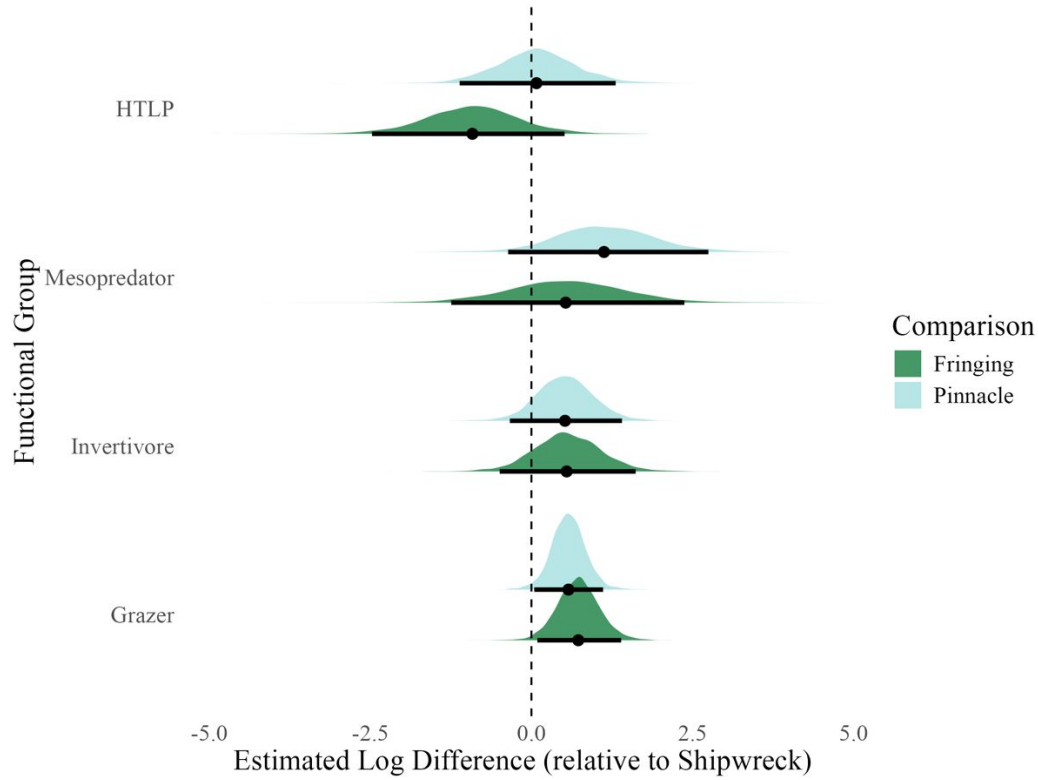


Figure 2 Predicted mean abundances of four functional fish groups—grazers, invertivores, mesopredators, and higher trophic level predators (HTLPs)—across three reef classifications: fringing reefs, pelagic pinnacles, and shipwrecks. Each panel displays the posterior mean abundance for one functional group based on multivariate negative binomial models.

Fish on Shipwrecks vs. Pinnacles



625

626 Figure 3 Posterior distributions show estimated log-scale differences in abundance for each
 627 functional group on fringing reefs (dark green) and pinnacles (light blue), relative to shipwrecks.
 628 Densities represent posterior uncertainty, with medians (dots) and 95% credible intervals (thick
 629 bars) overlaid. Zero indicates no difference from shipwrecks; positive values denote higher
 630 predicted abundance.

Fish on Shipwrecks vs. Pinnacles

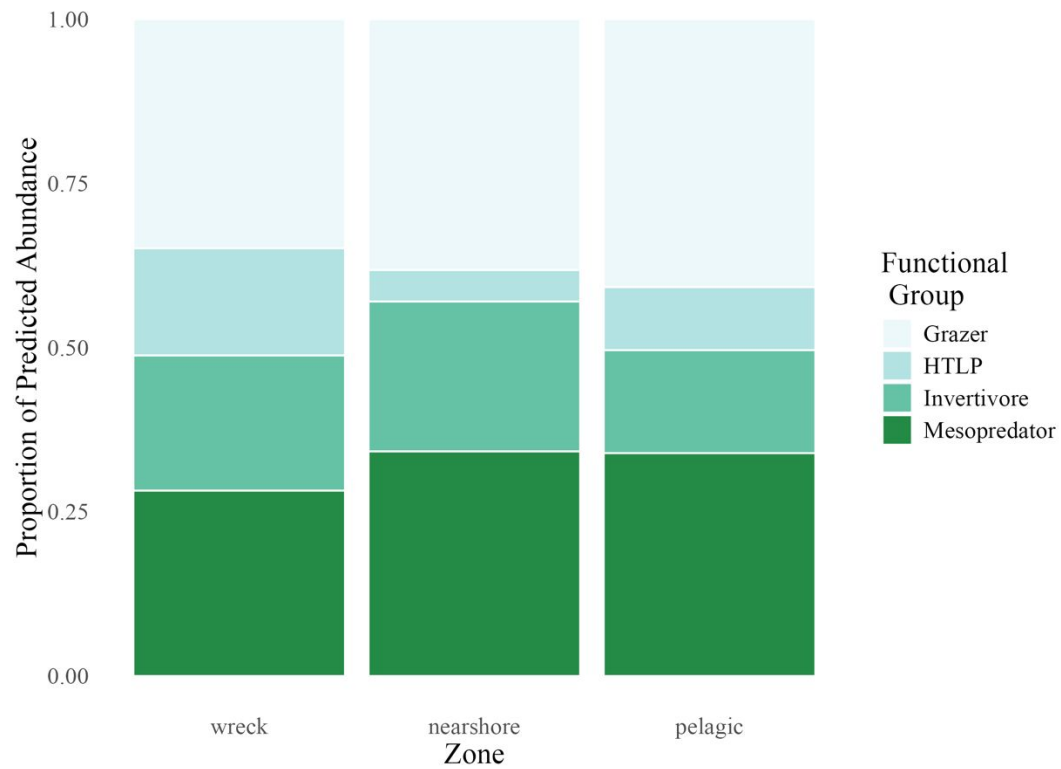


Figure 4 Predicted proportional composition of four functional fish groups across three reef zones (defined by proximity to shore): wreck, nearshore, and pelagic. Each bar represents the relative contribution of grazers, invertivores, mesopredators, and higher trophic level predators (HTLPs) to the total fish count within each zone. Proportions are normalised to 1.0 for comparison across zones.

Fish on Shipwrecks vs. Pinnacles

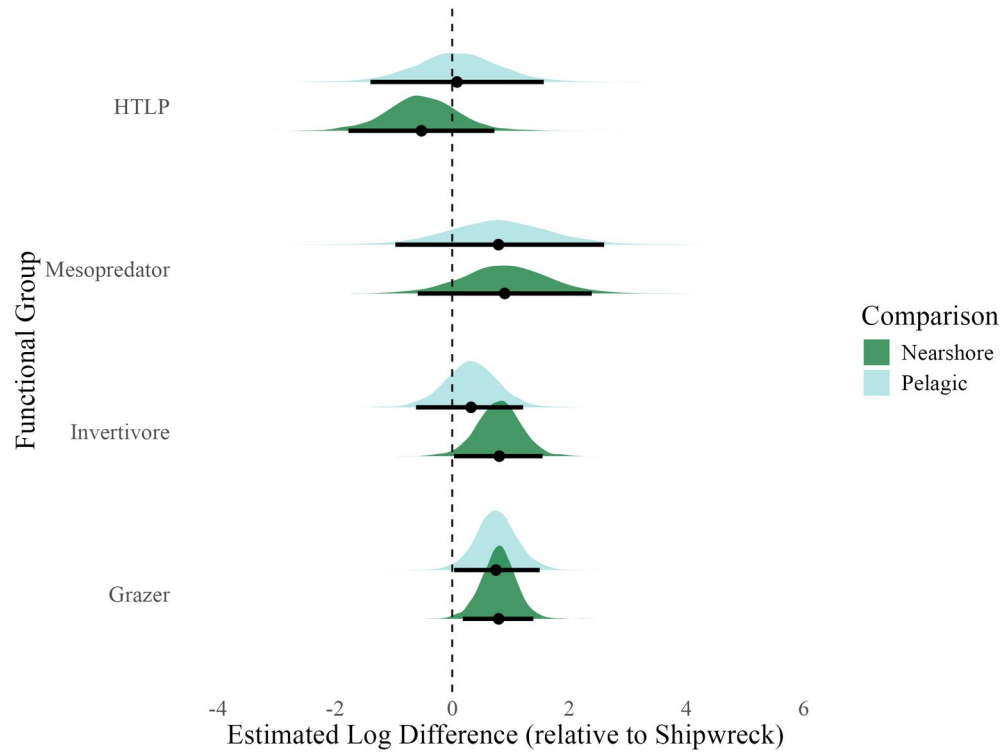


Figure 5 Posterior distributions show estimated log-scale differences in abundance for each functional group in nearshore (dark green) and pelagic (light blue) zones, relative to shipwrecks. Densities represent posterior uncertainty, with medians (dots) and 95% credible intervals (thick bars) overlaid. Zero indicates no difference from shipwrecks; positive values denote higher predicted abundance.

Fish on Shipwrecks vs. Pinnacles

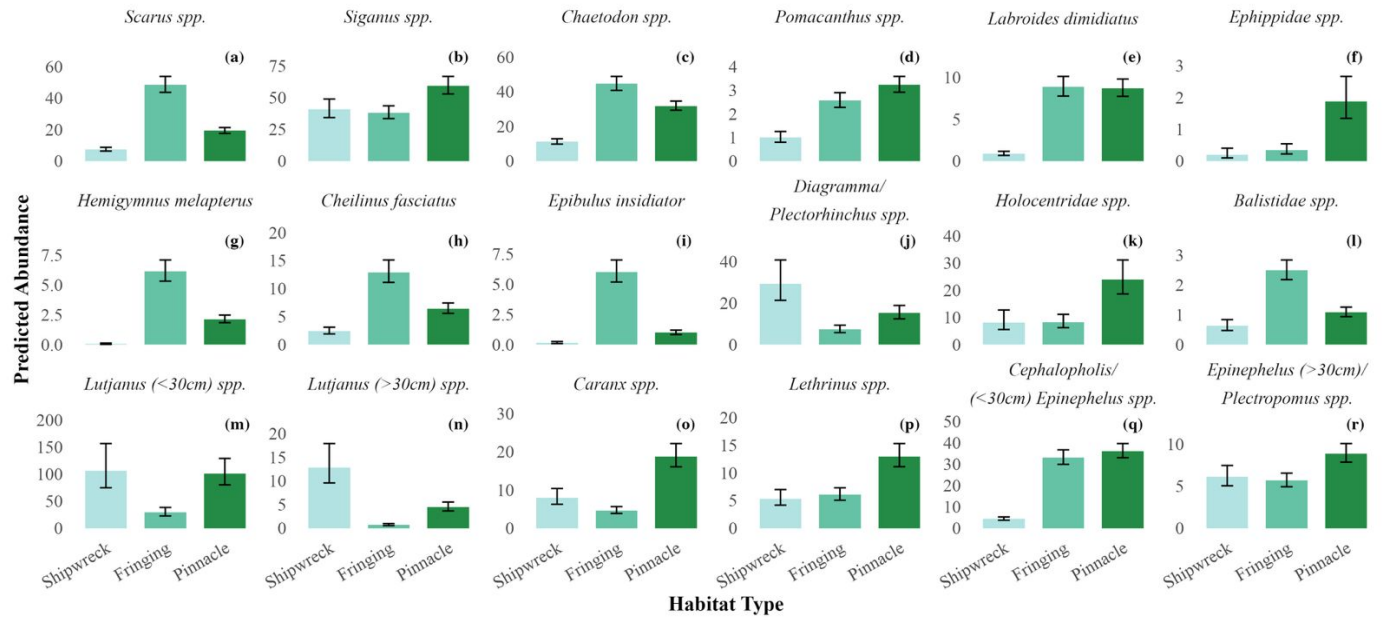


Figure 6 Predicted mean abundances of 18 reef fish taxa (a through r) across three reef classifications: fringing reef, pinnacle, and shipwreck. Each panel represents one species, with bars showing the posterior mean abundance from multivariate negative binomial models. Error bars represent Bayesian 95% credible intervals. Bars are coloured by reef classification to facilitate comparison of relative abundance patterns across reef types.

Fish on Shipwrecks vs. Pinnacles

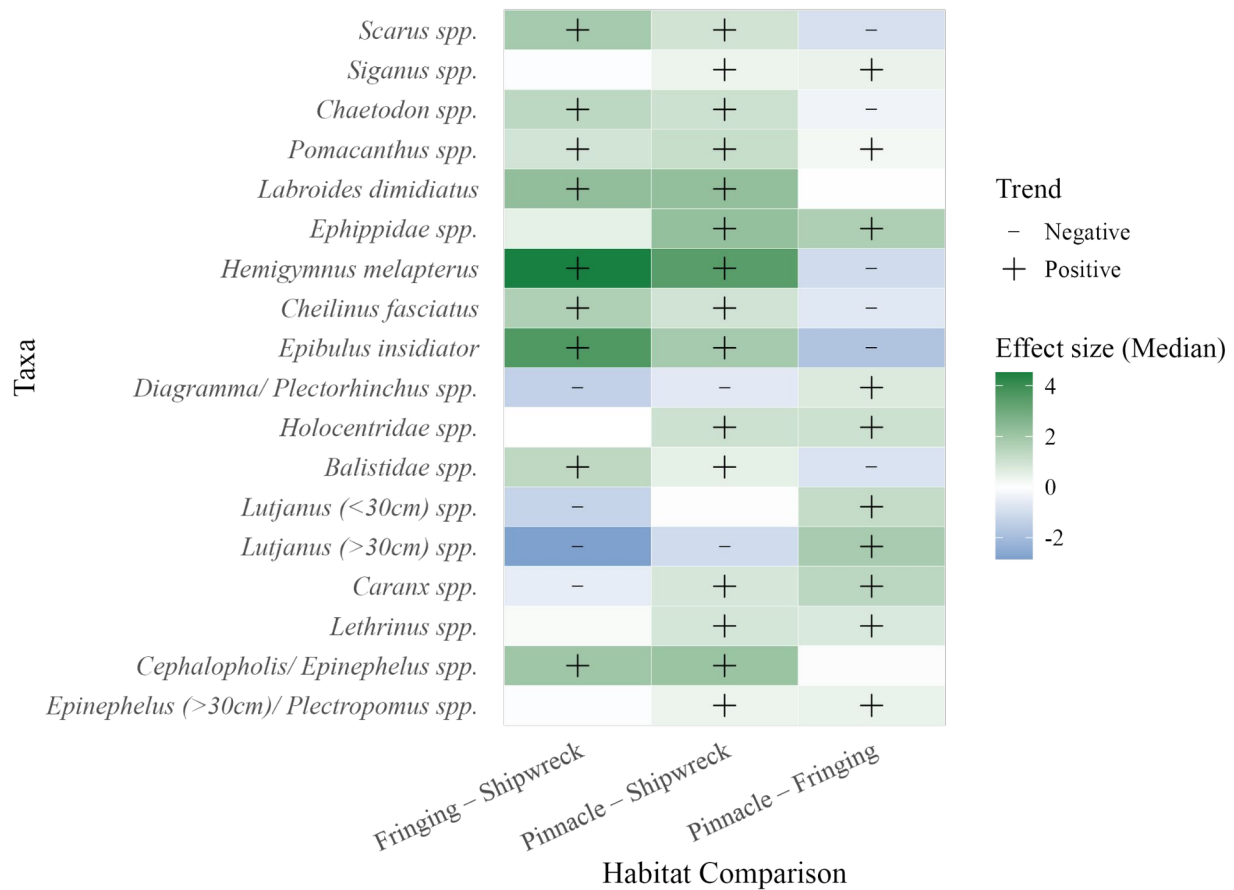


Figure 7 Species-level estimates of fish abundance differences across habitat types, relative to shipwrecks, from a Bayesian multivariate model. Each tile represents the median posterior effect size for a given species-habitat comparison (Fringing or Pinnacle reef vs. Shipwreck baseline). Positive values indicate higher predicted abundance on natural reefs relative to shipwrecks, and negative values indicate the opposite. Black symbols denote species with strong support for directional effects (posterior probability > 0.95 or < 0.05); plus signs indicate positive effects, and minus signs indicate negative effects.

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9. FUNDING

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10. CONFLICT OF INTERESTS

Authors of the current study declare no conflict of interest.

11. ETHICAL APPROVAL

This study did not require formal ethical approval, and data collection followed standard scientific ethical guidelines. All necessary permits for observational research were secured.

12. DATA AVAILABILITY

All data and code used in this study are openly available through Zenodo from <https://doi.org/10.5281/zenodo.16784382> , archived at the time of submission.

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Response to Reviews (1) — Manuscript: Shipwrecks can mirror predator assemblages of pelagic pinnacles but may lack the trophic balance of natural reef environments (MEPS-2025-08-016)

Authors: Scarlett R. Taylor, Gavin Miller, Piers Baillie

Date: 2025-11-03

Cover note to the Editor

Dear Prof. Peck,

Thank you for coordinating the constructive reviews of our manuscript. We have revised the paper substantially and address each point raised by the reviewers. Below we quote each comment in italics followed by our response and the exact locations of manuscript changes (Line numbers refer to the manuscript showing edits).

Summary of major revisions:

- Expanded methods for trophic-group assignment, survey depth, geographic zones, and sampling design including pseudoreplication safeguards.
- Added site characteristics table (depth, zone, vertical relief), clarified survey timing, and refined path description.
- Streamlined text by ~25% to remove repetition and tightened or caveated claims where appropriate.
- Updated literature to include suggested recent work on pinnacles, depth effects, and artificial reef design and socioeconomics.

We thank the reviewers sincerely for their constructive comments, insights and their time.

Best,

Scarlett R. Taylor (corresponding author) on behalf of all authors

Global Reef
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Editor-in-Chief comments

EIC-01 — Editor-in-Chief comment

Original comment:

[Scope & tone] Streamline repetitive sections by ~25% without loss of content.

Response:

Thank you for this comment. The manuscript has been carefully revised to reduce redundancy and improve readability while preserving all essential information. Repetitive phrasing and overlapping explanations were condensed especially across the Introduction and Discussion, resulting in a streamlined narrative without loss of content or clarity.

Action taken:

Revised wording throughout the document to make the text more concise.

EIC-02 — Editor-in-Chief comment

Original comment:

[Methods – trophic groups] Provide more detail on how fish trophic groups were assigned.

Response:

Thank you for this comment. Additional detail has been provided in Section 2.2 to clarify how trophic groups were assigned. Survey species were classified into four functional groups (grazers, invertivores, mesopredators, and higher trophic level predators) using a predefined scheme based on established ecological roles of coral-reef fishes. Assignments followed the precedent in Baillie et al. (2024) in the same study system, and were informed by trophic parameters and feeding ecology reported in regional literature and FishBase. The mapping of each species to its functional group is shown in Table S2. This clarification ensures transparency and consistency in how species were categorized for analysis.

Action taken:

Methods text updated at L207-214:

“Survey species were assigned to these four functional groups using precedent consistent with Baillie et al. (2024); the full mapping is provided in Table S2. The list was derived from pilot surveys and regional checklists and was held constant across sites and months to standardize counts, reduce identification error, and focus on conspicuous

taxa that are reliably detected within a 10 m inclusion radius (Baillie et al. 2024). Pomacentrids were excluded as their small size and high abundance make them poorly suited to timed-swim methods; accurate assessment would require stationary point counts, which were beyond the scope of this study.”

EIC-03 — Editor-in-Chief comment

Original comment:

[Methods – depth] Clarify the role of reef depth and where surveys occurred.

Response:

Thank you for this comment. All three reviewers noted this as well. We have added substantial detail in the methods section on site characteristics (for both the shipwrecks and the natural reef sites), time frame and frequency of sampling effort, and included a new table (Table S1) with per-site counts, first and last survey dates, and additional characteristics (reef type classifications, geographical zones, survey depths) for all the sites.

Action taken:

Added text to Methods Section 2.1: Study Sites at L185-192 to explain shipwreck characteristics. Revised Methods Section 2.1: Study Sites at L196-204 to add detail about frequency of surveys, effort and time frame of study. Added to Results text at L367-369 to clarify survey frequency and sampling effort. Added Table S1 - with per-site counts, first and last survey dates, and additional characteristics (reef type classifications, geographical zones, survey depths) for all the dive sites.

EIC-04 — Editor-in-Chief comment

Original comment:

[Design – geography] Address potential relationship between reef type and geographic zones.

Response:

Thank you for this comment. Additional clarification has been provided to address the relationship between reef type and geographic zone. Although reef classifications and geographic zones were partially correlated, since fringing reefs generally occurred nearshore and pinnacles were often offshore, the overlap was not complete. Several pinnacles (e.g., Shark Island, No Name Pinnacle, and Twins Pinnacles) were located within 1 km of shore, while others (e.g., Chumphon, White Rock, and Southwest Pinnacle) occurred in pelagic positions beyond 1 km. To explicitly evaluate potential geographic effects independent of reef type, a separate multivariate model was fitted

with zone (nearshore, pelagic, or wreck) as the fixed effect rather than reef classification. This analysis helped confirm that the main assemblage patterns were driven primarily by reef classification rather than proximity to shore. The relationship between these variables has been clarified in the Methods, and Table S1 was added to summarize site characteristics, including reef type, geographic zone, and depth.

Action taken:

Clarified relationship between reef type and geographic zone in the Methods at L255-257 and Discussion at L553-556; noted that an additional zone-based model was fitted to evaluate spatial effects independently; Table S1 added to include reef classification, geographic zone, and depth.

EIC-05 — Editor-in-Chief comment

Original comment:

[Design – pseudoreplication] Address potential pseudoreplication with repeated sampling design.

Response:

Thank you for this comment. Both Reviewers 1 and 3 raised valuable points regarding potential pseudoreplication and temporal structure.

While *Site* was already included as a random effect in all models to account for repeated sampling at fixed locations, we have now incorporated a temporal random effect to address non-independence through time. We tested models with *Year* and with *Month–Year* as temporal random effects; the Month–Year structure provided a substantially better fit ($\Delta\text{elpd} \approx 30$) and was retained in the final analysis. This revision ensures that both spatial and temporal dependence are explicitly modelled, preventing pseudoreplication and improving overall model performance.

Action taken:

Methods (2.31, 2.3.2) and Results (3.1, 3.2) sections updated to describe inclusion of Month–Year as a temporal random effect, following model comparison with Year. Figures 2-5 and tables 1-3 updated with revised outputs; overall trends and interpretations remain consistent with the original analysis.

Reviewer 1

R1-01 — Reviewer 1 comment

Original comment:

[Intro] L82 - When introducing pinnacles here, I think a little more context would be good for readers not familiar with these reef geomorphologies, particularly in the context of seamounts which are similar but more well known . Perhaps change to “Isolated pelagic pinnacles (steep sided bathymetric features similar to seamounts) are often exposed to strong hydrodynamic flow and tend to support high densities of mesopredators and larger predators such as Caranx, Lethrinus, Lutjanus, and Epinephelus spp. (Letessier et al. 2019, Galbraith et al. 2022a, 2022b, Cresswell et al. 2023, Galbraith et al. 2023, Mattos et al. 2023).

Response:

Thank you for this comment. We have taken your excellent suggestion and updated the introduction text. We placed it earlier than suggested (in the first paragraph of the introduction, rather than in P6).

Action taken:

Revised at L49 to clarify reef geomorphology.

R1-02 — Reviewer 1 comment

Original comment:

[Intro] L84 - If citing Galbraith et al. 2022 here, suggest Galbraith, G. F., Cresswell, B. J., McCormick, M. I., Bridge, T. C., & Jones, G. P. (2022). Contrasting hydrodynamic regimes of submerged pinnacle and emergent coral reefs. Plos one, 17(8), e0273092 as the peer reviewed reference over a report. Or should this be Galbraith et al. 2022a, 2022b and 2023? Citations here don't match references.

Response:

Thank you for this comment. We have added this reference to our paper and replaced most mentions of the report with the peer-reviewed paper. We have checked the Galbraith et al. references throughout the document to make sure they are correct.

Action taken:

Checked and clarified references throughout the text.

Added citation: Galbraith, G. F., Cresswell, B. J., McCormick, M. I., Bridge, T. C., & Jones, G. P. (2022). Contrasting hydrodynamic regimes of submerged pinnacle and emergent coral reefs. Plos one, 17(8), e0273092

R1-03 — Reviewer 1 comment

Original comment:

[Intro] L87 - Suggest citing Leitner et al. 2021 for explicit reference to pinnacles here.

Response:

Thank you for this comment. We have added the reference to Leitner et al. 2021.

Action taken:

Added citation at Introduction L105 and L111.

Leitner, A. B., Friedrich, T., Kelley, C. D., Travis, S., Partridge, D., Powell, B., & Drazen, J. C. (2021). Biogeophysical influence of large-scale bathymetric habitat types on mesophotic and upper bathyal demersal fish assemblages: a Hawaiian case study. Marine Ecology Progress Series, 659, 219–236. <https://doi.org/10.3354/MEPS13581>

R1-04 — Reviewer 1 comment

Original comment:

[Intro] L91 - Suggest adding Galbraith et al. 2021 as well as the Galbraith 2021 thesis as a peer reviewed reference.

Response:

Thank you for this comment. We have added the Galbraith et al. 2021 reference throughout the text.

Action taken:

Added references to Galbraith et al. 2021 throughout.

Galbraith, G. F., Cresswell, B. J., McCormick, M. I., Bridge, T. C., & Jones, G. P. (2021). High diversity, abundance and distinct fish assemblages on submerged coral reef pinnacles compared to shallow emergent reefs. Coral Reefs, 40(2), 335-354.

R1-05 — Reviewer 1 comment

Original comment:

[Intro] L92 – Suggest adding Done 1983 as a seminal reference here

Response:

Thank you for this comment. We have added Done 1983.

Action taken:

Added Done 1983 as a seminal reference at Introduction L117.

Done TJ. (1983). Coral zonation, its nature and significance. Australian Institute of Marine Science. Pages: 107-147.

R1-06 — Reviewer 1 comment

Original comment:

[Intro/AR context] L101 - Suggest adding Gress et al. 2025 as a very recent addition to the literature around shipwrecks and fringing reefs, which explicitly examines how differences in benthic habitat affect fish assemblages and habitat functions.

Response:

Thank you for this comment. We have added the reference to Gress et al. 2025

Action taken:

Add Gress et al. (2025) at Introduction L130.

Gress, E., Bairos-Novak, K. R., Bridge, T. C., & Galbraith, G. F. (2025). Neglected Biodiversity of Fish Assemblages Associated with Antipatharia (Black Corals) on Tropical Shallow Reef Ecosystems. Ecology and Evolution, 15(8), e72015.

R1-07 — Reviewer 1 comment

Original comment:

[Significance] L116 – I agree, but it's also important for fundamental marine ecology and understanding how different reef geomorphologies support different marine biodiversity patterns, processes and functions. As shallow and coastal reefs become increasingly degraded and are closer to human influence than offshore pinnacles and reefs, these more isolated features may become important bastions of resilience in the future. Consider expanding to include this point here, or maybe after L129 where you make the point to emphasise the value of this study in the broader context of coral reef decline.

Response:

Thank you for this insightful comment. We have expanded the paragraph to emphasise the broader ecological relevance of this work, highlighting the importance of understanding how different reef geomorphologies support biodiversity and may contribute to resilience as shallow coastal reefs decline.

Action taken:

Revised Introduction at L153 to include additional context on the ecological and conservation significance of reef geomorphology:

“Characterising such patterns is critical to (...) and understanding how reef geomorphology shapes biodiversity, trophic structure, and ecological function. As shallow and coastal reefs continue to degrade under increasing human pressure, isolated offshore features such as pinnacles and shipwrecks may become increasingly important refuges for reef-associated communities and potential sources of ecological resilience.”

R1-08 — Reviewer 1 comment

Original comment:

[Methods – study period] L138 – Great survey effort! But what was the actual study period, from 2011 when HMS Sattakut was deployed to 2023 when the other two were scuppered as artificial reefs? Once this is defined could you clarify the description of the repeated surveys? You state that each site was surveyed repeatedly over the study period but this could be daily, monthly, annually if you have a 12 year study period – maybe best to say “multiple times” and provide the n for surveys on each site?

Response:

Thank you for this comment. Reviewer 3 also pointed this out (R3-02) so we have added detail about the frequency of surveys, effort and time frame of study in both the methods and results sections. We have also added an additional Table (now Table S1) to clarify site characteristics and survey effort.

Action taken:

Revised Methods Section 2.1: Study Sites at L196-204 to add detail about frequency of surveys, effort and time frame of study:

“Each site was surveyed multiple times from March 2019 to June 2025. Surveys were planned at least monthly outside the November–December monsoon pause, though frequency varied by site and year, with lower effort in 2020–22. In total, 350 fish surveys were completed, with 150 on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at predetermined locations within each reef type to

ensure consistency across dives. Per-site counts, first and last survey dates, and median intervals are provided in Table S1.”

Added to Results text at L367-369 to clarify survey frequency and sampling effort: “Survey frequency varied among sites and years, with lower effort in 2020–22 as a result of COVID and monthly revisits from 2023 onward outside the monsoon period (Nov-Dec). “

Added Table S1 - with per-site counts, first and last survey dates, and additional characteristics (reef type classifications, geographical zones, survey depths) for all the dive sites.

R1-09 — Reviewer 1 comment

Original comment:

[Methods – wording] L144 – insert “a” before the word “standardised

Response:

Thank you for this comment. We have amended our wording.

Action taken:

Revised at L206.

R1-10 — Reviewer 1 comment

Original comment:

[Methods – taxa list] L145 – Was there a reason that pomacentrids were omitted from the species list? Given the trophic focus of this paper it seems strange to not include this key trophic group (mostly planktivores).

Response:

Thank you for this comment. Pomacentrids were excluded because their high abundance, small body size, and use of fine-scale microhabitats make them poorly suited to the timed-swim method employed in this study. Accurately quantifying this group would require an additional protocol such as stationary point counts, which was beyond the scope of our design and would have reduced replication for other functional groups. The survey list therefore focused on medium- to large-bodied species that can be reliably detected within a timed-swim framework. However, we recognise their ecological importance, and have incorporated stationary point counts targeting

pomacentrids in an ongoing/upcoming complementary study comparing fish production and attraction on artificial reefs in the same study system.

Action taken:

Updated Methods 2.2. at L215 to clarify rationale for exclusion:

“Pomacentrids were excluded as their small size and high abundance make them poorly suited to timed-swim methods; accurate assessment would require stationary point counts, which were beyond the scope of this study.”

R1-11 — Reviewer 1 comment

Original comment:

[Methods – depth & area] L144 – 162 – Great detail in these survey methods but no mention of depth? What depths were the surveys conducted at and did these differ between sites? Also L158 – what was the area/distance covered by each survey and did these differ between sites? Was it the same divers swimming a practiced speed each survey, based on the pilot timed-swims?

Response:

Thank you for this comment. Survey paths followed the structural contour from a fixed start point at a fixed target depth; because surveys were time-standardized, the end point varied. Buddy positioning sampled both the structure and adjacent water column and was applied consistently across reef types. The target depth matched the wreck surveys (~25 m) where feasible; otherwise, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed. Average survey depths are reported in Table 1.

The distance covered per survey was ~96 m (8 min at ~12 m·min⁻¹), and the corresponding swath area was ~1,920 m², calculated as distance × 2 × 10 m inclusion radius. A practiced pace (~12 m·min⁻¹) was maintained across surveys following pilot timed-swims. The earlier estimate of 1,200 m² reported in Baillie et al. (2024) is refined here to 1,920 m² to reflect the full inclusion radius used.

Action taken:

Methods 2.2 updated at L221-228 to add:

“From the fixed start point, the transect path followed the structural contour at the target depth; the end point varied because surveys were time-standardized. On shipwrecks, one diver maintained a position ~1 m above the hull edge while the buddy remained 1–2 m into open water at the same depth. On pinnacles, one diver remained 1–2 m from the reef with the buddy 1–2 m further outward. On fringing reefs, the pair contoured along the reef edge with the same spacing. The target depth matched the average depth used

on wrecks (~25 m) where feasible; if not, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed.”

Methods 2.2 updated at L237-240 to add:

“Divers swam at a practiced pace of ~12 m min⁻¹ for eight minutes, covering ~96 m per survey; the effective swath area was ~1,920 m² (distance × 2 × 10 m inclusion radius). This refines the 1,200 m² figure previously reported by Baillie et al. (2024).”

Results updated to reflect the refined survey area (~1,920 m²) at L374, L379, and L384.

R1-12 — Reviewer 1 comment

Original comment:

[Methods – phrasing] L 160 – suggest changing to “All observed fish on the species list within the designated radius were recorded on waterproof slates in-situ”.

Response:

Thank you for this comment. We have added the reference to in-situ.

Action taken:

Revised by adding “in-situ” at L244.

R1-13 — Reviewer 1 comment

Original comment:

[Methods – remove detail] L 161 – Remove unnecessary detail here about the separate section of the slate.

Response:

Thank you for this comment. We have removed that line.

Action taken:

Revised by removal of “each new survey was recorded in a separate section of the slate to maintain data integrity” at L245-247.

R1-14 — Reviewer 1 comment

Original comment:

[Methods – geography var] L172 - What are these geographic zones? Please clarify or introduce this variable to be tested.

Response:

Thank you for this comment. We have introduced and clarified this in the methods.

Action taken:

Updated Methods 2.3 at L258-260:

“Sites were classified by geographic zone as “nearshore” if located within 1 km of Koh Tao, “pelagic” if offshore pinnacles beyond 1 km, and “wreck” for intentionally deployed ships. Zone was added to separate shoreline proximity from reef type when estimating classification effects.”

R1-15 — Reviewer 1 comment**Original comment:**

[Stats – random effects] L178 – Is there a random effect for site in this equation? Given that surveys seem likely to have been conducted over multiple years should perhaps a random effect for survey_year be included?

Response:

Thank you for this comment. Yes, a random effect for Site was included in all models to account for repeated surveys at the same locations.

In response to this and related comments, we also tested temporal random effects to address potential non-independence across survey periods. Models incorporating Month–Year as a temporal random effect performed best ($\Delta\text{elpd} \approx 30$ relative to Year-only models) and were retained in the final analysis. This structure accounts for both spatial and temporal dependence while avoiding pseudoreplication.

Action taken:

We re-ran the analysis to incorporate temporal random effects, testing both Year and Month–Year structures. The Month–Year model was adopted and the Methods (2.3) section updated accordingly. Functional group results (3.1, 3.2) were revised slightly, though overall trends and interpretations remain consistent with the original analysis. Tables 1-3 and Figures 2-5 updated.

R1-16 — Reviewer 1 comment**Original comment:**

[Results – site size] L269 - I’m again curious as to the size of the sites here. How big are the pinnacle sites? From the methods the survey duration seems to have been governed by the size of the smallest wreck to enable a full lap.

Response:

Thank you for this comment. Site size does vary among reef types, with natural reefs (particularly pinnacles) generally larger and more spatially complex than shipwrecks. Survey duration was standardised to eight minutes to maintain consistency across sites and was determined based on the approximate time required to complete a full circuit of the smallest wreck while remaining within no-decompression limits. This ensured comparable survey effort and detection probability across reef types despite differences in total site area.

Action taken:

Added Table S1: Site Characteristics with per-site counts, first and last survey dates, depth of sites.

R1-17 — Reviewer 1 comment**Original comment:**

[Results – geography] L300 – Sorry, I'm still not clear what these geographic zones are? Wrecks are clear but does the nearshore/pelagic classification mean that you have some pinnacles in both nearshore and pelagic positions?

Response:

Thank you for this comment. We have clarified the definition of geographic zones to make the classification scheme explicit. “Nearshore” sites refer to reefs located within 1 km of Koh Tao, “pelagic” sites are those situated beyond 1 km, and “wreck” refers to intentionally deployed ships. This zoning framework separates shoreline proximity from reef type, as both natural pinnacles and fringing reefs occur across the nearshore–pelagic gradient.

Action taken:

Updated Methods 2.3 at L258-260:

“Sites were classified by geographic zone as “nearshore” if located within 1 km of Koh Tao, “pelagic” if offshore pinnacles beyond 1 km, and “wreck” for intentionally deployed ships. Zone was added to separate shoreline proximity from reef type when estimating classification effects.”

R1-18 — Reviewer 1 comment**Original comment:**

[Discussion – depth] Discussion section general comments – This section is well rounded, interesting and well written. I do find it a bit strange that there is little to no mention of depth though (briefly L432-424)? I'm not sure if the sites were surveyed at different depths but I think there should be something a little earlier in the discussion that touches on the influence of depth on fish community assemblages as it really is a key variable. I'm not concerned that this will affect the analysis as site was included in the model as a random effect but I do think depth related effects should be acknowledged. See the following recent studies and suggest including a few sentences that briefly cover the effects of depth (or sometimes lack of), particularly on predators: [suggested citations: Jankowski et al. 2015; Stefanoudis et al. 2019; Pinheiro et al. 2023; Cresswell et al. 2025]

Response:

Thank you for this suggestion. We have added new text throughout the Discussion summarizing the suggested recent studies on depth-related variation in reef fish communities at Sections 4.1, 4.2, and 4.4 (Jankowski et al. 2015, Stefanoudis et al. 2019, Pinheiro et al. 2023, Cresswell et al. 2025). These works collectively show that while depth can influence community composition—particularly beyond 30 m—habitat complexity and reef profile are often stronger determinants within shallow and upper-mesophotic zones. Because all sites in this study occurred at similar depths (< 30 m), depth is unlikely to explain the observed divergence, which instead reflects structural and functional differences among reef types.

Action taken:

Added to Discussion 4.1 at L503-506:

“Depth can influence predator distributions, but within shallow to upper-mesophotic zones its effect is often weaker than reef profile and structural relief. Studies report gradual declines in predator richness with depth and limited depth specialism across the first 100 m where structure remains a primary driver (Jankowski et al. 2015; Cresswell et al. 2025).”

Added to Discussion 4.2 at L544-556:

“Structural complexity, rather than depth, appears to drive these patterns. Sites occurred at similar depths (< 30 m), where diversity and herbivore biomass typically remain stable and major declines occur only beyond 30–40 m when coral cover and fine-scale complexity decrease (Stefanoudis et al. 2019, Pinheiro et al. 2023). Nearby pinnacles at comparable depths still supported higher densities of grazers and invertivores, indicating that structural relief and microhabitat availability, rather than depth, were the primary drivers (Paxton et al. 2020b, Leitner et al. 2021, Galbraith 2021, Galbraith et al. 2021).”

Added to Discussion 4.2 at L580-582:

“Depth-associated shifts toward planktivory are well documented at greater depths, but the similarity in depths among our sites suggests that the wreck–pinnacle contrast

observed here is better explained by structure and flow than by depth alone (Jankowski et al. 2015, Stefanoudis et al. 2019).”

Added to Discussion 4.4 at L617-619:

“Within this depth range, design variables that control microhabitat and refuge space will have greater leverage on functional outcomes than depth itself (Jankowski et al. 2015, Pinheiro et al. 2023).”

Added citations:

Cresswell, B. J., Galbraith, G. F., Barnett, A., Harrison, H. B., Jones, G. P., McClure, E. C., Quimpo, T. J. R., & Hoey, A. S. (2025). Limited Evidence for Depth Specialism in Isolated Seamount Reef Predators. *Ecology and Evolution*, 15(9), e72044.
<https://doi.org/10.1002/ECE3.72044>

Jankowski, M. W., Gardiner, N. R., & Jones, G. P. (2015). Depth and reef profile: Effects on the distribution and abundance of coral reef fishes. *Environmental Biology of Fishes*, 98(5), 1373–1386. <https://doi.org/10.1007/S10641-014-0365-1>
Pinheiro, H. T., MacDonald, C., Quimbayo, J. P., Shepherd, B., Phelps, T. A., Loss, A. C., Teixeira, J. B., & Rocha, L. A. (2023). Assembly rules of coral reef fish communities along the depth gradient. *Current Biology: CB*, 33(8), 1421-1430.e4.
<https://doi.org/10.1016/J.CUB.2023.02.040>

Stefanoudis, P. v., Gress, E., Pitt, J. M., Smith, S. R., Kincaid, T., Rivers, M., Andradi-Brown, D. A., Rowlands, G., Woodall, L. C., & Rogers, A. D. (2019). Depth-dependent structuring of reef fish assemblages from the shallows to the rariphotic zone. *Frontiers in Marine Science*, 6(JUN), 461131. <https://doi.org/10.3389/FMARS.2019.00307>

R1-19 — Reviewer 1 comment

Original comment:

[Discussion – terminology] L384 – By 'vertical extent' do you mean depth? Clarify.

Response:

Thank you for this comment. We have amended the wording to make it clearer that we are referring to the vertical relief provided by tall artificial reefs.

Action taken:

Amended Discussion text at L506: from “vertical extent” to “vertical relief”.

R1-20 — Reviewer 1 comment

Original comment:

[Citations] L387 – Again suggest Galbraith et al. 2021 as a better reference than thesis alone.

Response:

Thank you for this comment. We have amended the references.

Action taken:

Added references to Galbraith et al. 2021 throughout.

R1-21 — Reviewer 1 comment

Original comment:

[Discussion – Siganids] L430 – Super interesting! This is an excellent addition to the ecology of Siganids. Please see Gress et al. 2023 for interest and consider citing alongside Streit et al. 2015 as a more contemporary reference.

Response:

Thank you for this comment. The paper you suggested was an excellent read, both for interest, and for adding more context to *S. javus*' feeding behaviours. We've added the citation to this paper as well as lightly edited the discussion here to mention it.

Action taken:

Added to Discussion 4.2 at L569-571:

"Recent observations indicate that this flexibility may represent ecological opportunism rather than resource limitation, as *S. javus* has been proposed to feed opportunistically on epibiotic algae and zooplankton from coral polyps (Gress et al. 2023)."

R1-22 — Reviewer 1 comment

Original comment:

[Scope/Discussion] L437 – I'm not suggesting any form of reanalysis at all as the data collected are the data collected, but it would be good to acknowledge that this study didn't include planktivores and maybe provide a reason why in the methods? Perhaps point to this as a direction for future studies? I think you've got a really neat study system out there around Koh Tao with heaps of potential for further work at these sites. It would be great to see full fish assemblage surveys conducted as part of a long-term monitoring project so that data would be comparable to other global LTMP datasets for spatial and temporal comparisons.

Response:

Thank you for this helpful suggestion. We have added text to the Discussion acknowledging the omission of planktivorous Pomacentrids and noting that this group is the focus of ongoing surveys at the same sites. These data will allow future analyses to integrate the full fish assemblage for comparison with global long-term monitoring programs.

Action taken:

Added text in Discussion Section 4.2 at L582-587 to note the exclusion of planktivores and future monitoring directions:

“Although other planktivorous taxa such as pomacentrids were not surveyed here, their dominance elsewhere highlights their potential importance in shaping trophic dynamics on artificial reefs. Incorporating this group through stationary point count surveys is a focus of ongoing work at these sites, which will allow a more complete assessment of community structure and trophic balance across reef types.”

R1-23 — Reviewer 1 comment

Original comment:

[Citations] L530 – Galbraith et al. 2022 would be a better reference here

Response:

Thank you for this comment. We have added the reference to Galbraith et al. 2022.

Action taken:

Added reference to Galbraith et al. 2022 in Discussion section 4.4 at L697.

R1-24 — Reviewer 1 comment

Original comment:

[References] L666 – Please check this reference is complete/correct. “v.” ?

Response:

Thank you for this comment. We have checked the citation and it was a formatting issue for a middle initial from the reference software used (Mendeley).

Action taken:

Revised at L868.

Reviewer 2

R2-01 — Reviewer 2 comment

Original comment:

[Methods – trophic assignment] 1) How were the survey species assigned to functional groups? Was this done using Fishbase trophic parameters, for instance, or another way?

Response:

Thank you for this comment. Survey species were assigned to four functional groups using a predefined scheme reflecting widely accepted ecological roles on coral reefs. This is consistent with the precedent in Baillie et al. (2024), who used a predetermined list representing multiple functional groups in timed-swim surveys; the full mapping is shown in Table S2.

Action taken:

Methods 2.2 text updated at L210-212 to add:

“Survey species were assigned to these four functional groups using precedent consistent with Baillie et al. (2024); the full mapping is provided in Table S2.”

R2-02 — Reviewer 2 comment

Original comment:

[Methods – species list] 2) Why was a predetermined species list used? I don't fully understand the rationale for this. I'm not suggesting that the authors resurvey but instead more fully explain why this decision was made for the study.

Response:

Thank you for this comment. We have amended the wording in section 2.2 to make this clearer.

A predetermined species list was used to standardize timed-swim counts across sites and months, focus effort on conspicuous taxa that span key functional groups, and reduce identification error under a 10 m inclusion radius. The list was derived from pilot surveys and regional checklists, then held constant to support comparability through time. This approach follows published precedent for our survey design (Baillie et al. 2024; adapted from Hill and Wilkinson 2004), and it fits the study's aim to compare assemblage structure across reef types rather than to census all taxa.

Action taken:

Methods 2.2 text updated at L212-215 to add:

“The list was derived from pilot surveys and regional checklists and was held constant across sites and months to standardize counts, reduce identification error, and focus on conspicuous taxa that are reliably detected within a 10 m inclusion radius (Baillie et al. 2024). “

R2-03 — Reviewer 2 comment

Original comment:

[Design – geography vs type] 3) Is the geographic zone fully different from the reef type? For example, I would assume that pinnacles are typically offshore and fringing reefs are typically nearshore. I may be wrong, though! If reef type and geographic zone / distance from shore are highly correlated, then I wonder about the value added of doing the geographic zone analysis. It would be helpful to include a site table that specifies characteristics like geographic zone, depth, etc. (see #4 below)..

Response:

Thank you for this comment. We agree that reef type and geographic zone are often correlated, as fringing reefs tend to occur nearshore and pinnacles are commonly offshore. However, in our study these variables were not perfectly aligned: several pinnacles (e.g., Shark Island, White Rock, and Twins Pinnacles) are located within 1 km of shore, while others (e.g., Chumphon, No Name Pinnacle, Southwest) occur in pelagic positions beyond 1 km. To account for this overlap, geographic zone was included as an independent variable to separate shoreline proximity from reef classification when evaluating assemblage patterns. We have clarified this distinction in the Methods and added a table summarising site characteristics, including reef type, zone, and depth.

Action taken:

Clarified distinction between reef type and geographic zone in the Methods 2.3 at L258 and added Table S1: Site Characteristics, which lists per-site depth, geographic zone, and reef classification.

R2-04a — Reviewer 2 comment

Original comment:

[Study sites – depth] a. Can the authors please provide information on reef depth (e.g., maybe add a table on site characteristics)?

Response:

Thank you for this comment. We have added information on reef depth and other site-level characteristics to provide clearer context for the study locations. Depth varied slightly among sites but remained within the shallow to upper-mesophotic range (< 30 m) across all reef types.

Action taken:

Methods 2.2 text updated at L223-228 to add:

“On shipwrecks, one diver maintained a position ~1 m above the hull edge while the buddy remained 1–2 m into open water at the same depth. On pinnacles, one diver remained 1–2 m from the reef with the buddy 1–2 m further outward. On fringing reefs, the pair contoured along the reef edge with the same spacing. The target depth matched the average depth used on wrecks (~25 m) where feasible; if not, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed.”

Added Table S1: Site Characteristics with per-site counts, first and last survey dates, depth of sites.

R2-04b — Reviewer 2 comment

Original comment:

[Sampling design] b. Were reefs surveyed in a particular season / time period or at fairly regular intervals across the year of the study?

Response:

Thank you for this comment. We have clarified details about sampling effort in both the methods and results, and added a table of site characteristics (Table S1). We added detail about frequency of surveys, effort and time frame of study.

Action taken:

Methods 2.3 text updated at L196-201 to add:

“Each site was surveyed multiple times from March 2019 to June 2025. Surveys were planned at least monthly outside the November–December monsoon pause, though frequency varied by site and year, with lower effort in 2020–22. In total, 350 fish surveys were completed, with 150 on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at predetermined locations within each reef type to ensure consistency across dives. Per-site counts, first and last survey dates, and median intervals are provided in Table S1.”

Results updated at L370-372 to add:

“Survey frequency varied among sites and years, with lower effort in 2020–22 as a result of COVID and monthly revisits from 2023 onward outside the monsoon period (Nov-Dec). “

Added Table S1: Site Characteristics with per-site counts, first and last survey dates, depth of sites.

R2-04c — Reviewer 2 comment

Original comment:

[Habitat metrics] c. Were measures of reef complexity or vertical relief obtained? I ask this since the article discusses reef complexity and vertical relief as likely drivers of the patterns in fish communities. If these data exist, it would be interesting to see if these factors indeed predict fish community outcomes.

Response:

Thank you for this comment. Measures of reef complexity and vertical relief were not collected as part of this dataset, as our focus was on community-level fish responses across established site types. While no current projects are quantifying these metrics at our sites, we agree that incorporating direct habitat measures would strengthen future analyses and plan to include them in subsequent monitoring efforts.

Action taken:

NA.

R2-04d — Reviewer 2 comment

Original comment:

[Transect path] d. I know that each reef survey had the same starting point, but did the surveys also have the same end point? I'd like to see a bit more information on how the survey path was decided upon while underwater, if appropriate. And, were the decisions on where to survey similar for artificial vs. natural reefs? It seems that maybe the divers swam around the edge of the ships. Did they also swim around the edge of pinnacles? If so, how was this done on fringing reefs?

Response:

(Same as R1-11)

Thank you for this comment. Reviewer 1 also asked for a similar revision in comment R1-11. Survey paths followed the structural contour from a fixed start point at a fixed target depth; because surveys were time-standardized, the end point varied. Buddy positioning sampled both the structure and adjacent water column and was applied consistently across reef types. The target depth matched the wreck surveys (~25 m) where feasible; otherwise, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed. Average survey depths are reported in Table 1.

The distance covered per survey was ~96 m (8 min at ~12 m·min⁻¹), and the corresponding swath area was ~1,920 m², calculated as distance × 2 × 10 m inclusion radius. A practiced pace (~12 m·min⁻¹) was maintained across surveys following pilot timed-swims. The earlier estimate of 1,200 m² reported in Baillie et al. (2024) is refined here to 1,920 m² to reflect the full inclusion radius used.

Action taken:

Methods 2.2 updated at L221-228 to add:

“From the fixed start point, the transect path followed the structural contour at the target depth; the end point varied because surveys were time-standardized. On shipwrecks, one diver maintained a position ~1 m above the hull edge while the buddy remained 1–2 m into open water at the same depth. On pinnacles, one diver remained 1–2 m from the reef with the buddy 1–2 m further outward. On fringing reefs, the pair contoured along the reef edge with the same spacing. The target depth matched the average depth used on wrecks (~25 m) where feasible; if not, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed.”

Methods 2.2 updated at L237-240 to add:

“Divers swam at a practiced pace of ~12 m min⁻¹ for eight minutes, covering ~96 m per survey; the effective swath area was ~1,920 m² (distance × 2 × 10 m inclusion radius). This value refines the 1,200 m² area previously reported by Baillie et al. (2024).”

Results updated to reflect the refined survey area (~1,920 m²) at L374, L379, and L384.

R2-05 — Reviewer 2 comment

Original comment:

[Terminology] Line 50: I think it's OK to use "shipwrecks," but you may want to consider using "ships" instead (and throughout) since shipwrecks typically mean those that wrecked during storms, war, etc., and the study sites that you used were intentionally deployed ships.

Response:

Thank you for this comment. We appreciate the suggestion and agree that “shipwrecks” can also refer to unintentionally sunk vessels. However, the term is also widely used in the artificial reef literature to describe intentionally deployed ships functioning as artificial reefs. To remain consistent with established usage and with terminology applied by local management authorities, we have retained the term “shipwrecks” throughout the manuscript. We have clarified in the introduction that these were intentionally deployed ships.

Action taken:

Addition of “intentional” and “intentionally deployed” at various points (L61, L73, L133, etc.) throughout the introduction to make it clear that the shipwrecks in this study were intentionally deployed for benefits to stakeholders.

R2-06 — Reviewer 2 comment

Original comment:

[Framing] Line 60: “balanced trophic assemblages” implies that natural reefs (even degraded reefs) always have balanced trophic assemblages; consider clarifying

Response:

Thank you for this comment. We agree that the phrase “balanced trophic assemblages” could imply that natural reefs are always in equilibrium, which was not the intended meaning. We have revised the sentence to clarify that the comparison refers to the representation of trophic groups typically observed on natural reefs, rather than an assumption of balance.

Action taken:

Revised Introduction text at L76 to clarify phrasing:
replaced “balanced trophic assemblages” with “trophic structures comparable to those of natural reefs”

R2-07 — Reviewer 2 comment

Original comment:

[Clarity] Line 65-66: I don’t follow what you mean by metrics obscuring differences in communities and functions. Please clarify.

Response:

Thank you for this comment. We have clarified that abundance and richness metrics, while commonly used, can mask important differences in the composition and functional structure of fish communities. Artificial reefs may appear similar to natural reefs when measured only by total abundance or species count, yet differ substantially in trophic makeup and ecological roles.

Action taken:

Revised Introduction text at L82 to clarify the intended meaning; (changes in bold):
“Where evaluations exist, artificial reefs often support higher fish abundance and species richness than nearby natural reefs (Fowler & Booth 2012, Folpp et al. 2013, Schulze et

al. 2020); **however, these metrics (total abundance, richness)** can obscure differences in community and trophic structure (Ross et al. 2016, Paxton et al. 2020a, Paxton et al. 2020b, Paxton et al. 2023).”

R2-08 — Reviewer 2 comment

Original comment:

[Context] Line 108: Consider tempering this statement, as artificial reefs have also been compared to sedimentary habitats, seagrass meadows, etc.

Response:

Thank you for this comment. We agree that artificial reefs have been compared to a variety of habitats, including sedimentary environments and seagrass meadows, in other contexts. We have revised the sentence to clarify that our distinction refers specifically to studies comparing artificial reefs to different types of coral reef habitats, rather than to all possible habitat comparisons.

Action taken:

Revised Introduction text at L136-140:

“Unlike most artificial coral reef studies limited to fringing-reef comparisons, this study tests shipwreck similarity across two contrasting natural reef types: fringing reefs representing shallow coastal systems and pelagic pinnacles representing deep, high-relief offshore reefs, to determine which they more closely resemble.”

R2-09 — Reviewer 2 comment

Original comment:

[Filtering] Line 188-189: Was additional data filtering conducted besides removing rare species.

Response:

Thank you for this comment. Yes. In addition to excluding the specified episodic taxa, two filters were applied: (i) surveys with horizontal visibility <3 m were not included, and (ii) a quality-control review identified systematic undercounting in surveys by one observer; those surveys were excluded. No other filters were applied.

Action taken:

Methods 2.2 updated at L236-237 to add:

“Surveys proceeded only when horizontal visibility was ≥ 3 m, and visibility was logged for quality control.”

Methods 2.3.1 updated at L278-280 to add:

“Visibility was not modelled because low-visibility (<3m) dives were excluded, remaining variation was moderate and comparable across sites and periods by design, and random effects for site and month-year absorbed residual heterogeneity.”

Methods 2.3 updated at L281-282 to add:

“Quality control identified systematic undercounting in surveys conducted by one observer; those surveys were excluded. No additional filtering was performed.”

R2-10 — Reviewer 2 comment

Original comment:

[Results wording] Line 272-273: Based on the values provided, it doesn't seem that shipwrecks have elevated average abundance of fish compared to fringing reefs. The values on these two reef types seem pretty similar, especially with the error provided. Consider rephrasing.

Response:

Thank you for this comment. We agree that the difference between shipwrecks and fringing reefs was small and not clearly distinguishable given the variation observed. We have rephrased this sentence to emphasise that only pinnacles showed a clear increase in fish abundance.

Action taken:

Revised Results S3 at L378-379 to clarify interpretation:

Replaced “and, to a lesser extent, shipwrecks relative to fringing reefs”, with “(...) while shipwrecks and fringing reefs supported comparable mean densities.”

R2-11 — Reviewer 2 comment

Original comment:

[Terminology] Line 381: “large predators” – Weren't the HTLP larger predators anyways? Or, are you saying some of the HTLP are larger predators but not others? It's possible that I'm misunderstanding.

Response:

Thank you for this comment. We amended the wording; it was an attempt to vary the wording for interest, but we see how it could lead to confusion.

Action taken:

Revised Discussion Section 4.2 at L498:

Replaced “larger predators” with “HTLPs”

R2-12 — Reviewer 2 comment

Original comment:

[Implications] Line 401-402: Suggest explaining why it would be the case that these findings would provide insights for managers.

Response:

Thank you for this comment. Additional explanation has been provided to clarify the management relevance of these findings. The resemblance between shipwreck and pinnacle predator assemblages indicates that high-relief artificial structures may function ecologically as predator-aggregating habitats, similar to natural pinnacles. This understanding informs site selection and design strategies by identifying where such reefs are likely to promote predator-dominated communities and where additional habitat complexity would be required to support lower trophic guilds.

Action taken:

Expanded point at Discussion 4.1 at L523-525:

“(…) providing vital insights for managers into how reef profile and spatial isolation influence community composition and can be used to guide artificial reef design, placement and ecological objectives.”

R2-13 — Reviewer 2 comment

Original comment:

[Caveat] Line 419: Consider caveating the statement that shipwrecks don't provide fine-scale refugia or cryptic habitats; especially in shipwrecks that have deteriorated or sunk during war events, we typically see that they do have a lot of structural complexity that provides microhabitats and refugia.

Response:

Thank you for this comment. We agree that the degree of structural complexity varies among shipwrecks, especially between older or deteriorated wrecks and newer metal structures. We have clarified that the shipwrecks surveyed in this study are relatively recent metal deployments, which have not yet undergone sufficient deterioration to develop extensive fine-scale refugia or cryptic microhabitats comparable to natural reefs

Action taken:

Revised at Discussion 4.2 at L538-540 to include context on the age and material of shipwrecks surveyed:

“Although older or deteriorated shipwrecks can develop finer refugia and turf algal cover (Paxton et al. 2023), the relatively new metal structures surveyed here have not yet reached such stages of colonisation (Simon et al. 2013, Mattos & Yeemin 2020, Monchanin et al. 2021, 2025).

R2-14 — Reviewer 2 comment

Original comment:

[Hypothesis strength] Line 439-440: Consider making the later part of this statement more hypothetical, as I don't think we know that if habitat limitation was addressed that the benthic species on shipwrecks would be similar to those on naturally occurring reefs.

Response:

Thank you for this comment. We agree that this statement could be interpreted too definitively. We have revised it to more cautiously suggest that addressing habitat and resource limitations may help improve representation of benthic-feeding species, without implying that such changes would necessarily recreate natural reef assemblages.

Action taken:

Revised Discussion 4.2 at L588-591 to frame the statement more hypothetically: “These patterns (...) and that addressing habitat and resource limitations in artificial reef design may help support these groups, though it remains uncertain whether such changes would fully replicate natural reef assemblages.”

R2-15 — Reviewer 2 comment

Original comment:

[Style] Line 498-499: Nit picky but “ensure” is used twice; I think this sentence could be made a bit more concise.

Response:

Thank you for this comment. We have amended our wording.

Action taken:

Revised wording at L644 to remove duplicated word and make more concise: “Broader management strategies should ensure that artificial reefs are implemented alongside natural reef conservation to sustain ecosystem function.”

R2-16 — Reviewer 2 comment

Original comment:

[AR design nuance] Line 506: “Positioning...vertical structure” – But even if the artificial reef is in deep water, it may not mimic the structure of pinnacles depending on what type / design of artificial reef is deployed...

Response:

Thank you for this comment. We have amended our wording to specify that we are talking about metal shipwrecks deployed for the purpose of artificial reefs.

Action taken:

Changed “positioning artificial reefs (...)” to “*deploying metal shipwrecks as artificial reefs* in offshore settings (...)” at Discussion L677-678.

R2-17 — Reviewer 2 comment

Original comment:

[Redundancy] Line 533-534: “Overall, this...reefs” – this sentence seems slightly out of place and is redundant with content elsewhere; consider removing..

Response:

Thank you for this comment. We have removed this line.

Action taken:

Removed Discussion L711-712.

Reviewer 3

R3-01 — Reviewer 3 comment

Original comment:

[Study sites – structure] In the section on study sites, I would recommend that you provide some information on the size and structural complexity of the shipwrecks.

Response:

Thank you for this comment. We have added detail about each of the shipwrecks in the methods section.

Action taken:

Added text to Methods Section 2.1: Study Sites at L185-192 to explain shipwreck characteristics:

“At 48m in length and 7m in width, *HTMS Sattakut* sits on the sandy sea floor in the west of Koh Tao (10.10169°N, 99.81537°E) at a maximum depth of 30m and provides 12m of vertical relief from the base of the hull to the highest point. *HTMS Suphairin* and *HTMS Hanhak-Sattru* were deployed in the north-west (10.1166°N, 99.80993°E), at a maximum depth of 30m, and north-east (10.09239°N, 99.85292°E), at a maximum depth of 31m, of the island respectively. *HTMS Suphairin* and *HTMS Hanhak-Sattru* have identical measurements at 45m in length and 7m in width and provide approximately 16m of vertical relief from hull base to highest point.”

R3-02 — Reviewer 3 comment

Original comment:

[Sampling frame] In your methods you state that you had “14 unique reef sites, comprising 5 fringing reefs, 6 pelagic pinnacles, and 3 shipwrecks. Repeated surveys were conducted at each site, with 150 surveys on fringing reefs, 130 on pinnacles, and 70 on shipwrecks”.

Can you please add the time frame over which surveys were conducted?

Response:

Thank you for this comment. Reviewer 1 also pointed this out (R1-08). We have added detail about the sampling effort in both the methods and results sections, and created a new table (Table S1) with per-site counts, first and last survey dates, and additional characteristics (reef type classifications, geographical zones, survey depths) for all the sites in the interest of transparency and reproducibility.

Action taken:

Revised Methods Section 2.1: Study Sites at L196-204 to add detail about frequency of surveys, effort and time frame of study:

“Each site was surveyed multiple times from March 2019 to June 2025. Surveys were planned at least monthly outside the November–December monsoon pause, though frequency varied by site and year, with lower effort in 2020–22. In total, 350 fish surveys were completed, with 150 on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at predetermined locations within each reef type to ensure consistency across dives. Per-site counts, first and last survey dates, and median intervals are provided in Table S1.”

Added to Results text at L367-369 to clarify survey frequency and sampling effort:

“Survey frequency varied among sites and years, with lower effort in 2020–22 as a result of COVID and monthly revisits from 2023 onward outside the monsoon period (Nov-Dec). “

Added Table S1 - with per-site counts, first and last survey dates, and additional characteristics (reef type classifications, geographical zones, survey depths) for all the dive sites.

R3-03 — Reviewer 3 comment

Original comment:

[Design – pseudoreplication] My concern is that each site was surveyed repeatedly over the study period, and I question whether your sampling is pseudo-replicated?

Please explain why your samples aren't pseudo replicated and how you have dealt with time which could/should be incorporated into your data analysis.

Response:

Thank you for this important comment. Another reviewer also pointed this out. We have clarified that repeated surveys at each site were accounted for statistically by including Site as a random effect, which controls for non-independence among observations from the same location.

To address temporal autocorrelation and potential seasonal effects, we have now updated the analysis to additionally incorporate Month–Year as a temporal random effect. This structure accounts for repeated measures through time and prevents pseudoreplication by modelling temporal and spatial dependence explicitly. We also tested alternative models with Year alone, but the Month–Year random effect provided superior fit ($\Delta\text{elpd} \approx 30$) and captured finer-scale variation.

Action taken:

We re-ran the analysis to incorporate temporal structure, testing models with Year and Month–Year as random effects. The Month–Year model provided a better fit and was adopted in the final analysis. The Methods section has been updated to reflect this change, and results for the functional group analysis were revised accordingly. Although numerical estimates changed slightly, the overall trends and interpretations remain consistent with the original findings.

R3-04 — Reviewer 3 comment

Original comment:

[Methods – precedent] Your sampling method is unusual, and it would be good if you could provide a reference for who has used this type of approach before.

Response:

Thank you for this comment. Precedent for this approach was cited by adding Baillie et al. (2024) and Hill and Wilkinson (2004).

Action taken:

Citations inserted at L207; Hill and Wilkinson (2004) added to the reference list.

Hill, J., and Wilkinson, C. (2004) Methods for Ecological Monitoring of Coral Reefs: A Resource for Managers. Version 1. Australian Institute for Marine Science. ISBN 0 642 322 376.

R3-05 — Reviewer 3 comment

Original comment:

[Methods – detectability] I recommend that you need to explain the probability of detecting small fish at a distance of 10 metres and how you ensured that you weren't over or undercounting fish by getting your distance wrong. Divers are notoriously poor at estimating distance underwater.

Response:

Thank you for this comment. We have clarified our reasoning in the methods text.

A 10 m inclusion radius was selected to capture transient, mid-water higher trophic level predators that were missed at smaller radii in pilot work. Detectability of very small taxa was de-emphasized by not targeting such families (e.g. Pomacentridae) in the species list. To reduce distance-related error, observer distance estimation was trained and periodically calibrated with fixed range cues, and a standard survey arc and pace were used. A visibility threshold of ≥ 3 m was applied, and visibility was logged for quality control. Visibility was not modelled, as dives below the threshold were excluded and remaining variation was moderate and comparable across sites and periods; random effects for site and date captured the residual heterogeneity.

Action taken:

Methods 2.2 updated at L233-237 to add:

“A 10 m inclusion radius was selected to capture transient, mid-water higher trophic level predators (hereafter HTLPs). Observer distance estimation was trained prior to fieldwork and reinforced during the sampling period using fixed range cues and team cross-checks. Surveys proceeded only when horizontal visibility was ≥ 3 m, and visibility was logged for quality control.”

Exploratory analysis note added at L278-280:

“Visibility was not modelled because dives with visibility < 3 m were excluded, remaining variation was moderate and comparable across sites and periods, and random effects for site and date absorbed residual heterogeneity.”

R3-06 — Reviewer 3 comment**Original comment:**

[Methods – Data description] Line 161. I suggest in the methods you state what data was collected. Absence presence, count, length? Obviously count from the data analysis, but please state it.

Response:

Thank you for this comment. We have specified more clearly what data was collected.

Action taken:

Methods text at L244 updated to add:

“Counts of individuals by listed species were recorded for each survey, and zeros were retained so that presence and absence were represented.”

R3-07 — Reviewer 3 comment**Original comment:**

[Methods - Data treatment] Line 185. You state that you have removed schooling species from your analysis. Given schooling species are typical of high relief structures (both in Thailand and other locations) why not down weight those species by transforming the data? Those schooling species are important.

Response:

Thank you for this comment. Schooling species were acknowledged as important on high-relief structures. Down-weighting via transformation was considered but not implemented. Mobile predators were retained generally (e.g., *Caranx* spp.). Only episodic *Sphyrna* pulses that dominated group estimates were excluded from model fitting and reported descriptively ($< 2\%$ of observations). An exploratory sensitivity check

that retained *Sphyraena* with capped counts produced unchanged higher-trophic-level predator trends. Barracudas will be examined in future work.

Action taken:

Methods 2.3.1 at L275-276 updated to add:

“Mobile predators were retained in the analyses (e.g., *Caranx* spp.); only episodic schooling pulses that disproportionately influenced group-level estimates were excluded from model fitting and summarized descriptively, chiefly *Sphyraena* spp.”

Methods 2.3.1. at L285-287 updated to add:

“Down-weighting via transformation was considered but not implemented. A sensitivity check that retained *Sphyraena* under capped counts did not change qualitative higher-trophic-level predator trends.”

R3-08 — Reviewer 3 comment

Original comment:

[Citations – Design of ARs] You are missing one reference which I think will really help your discussion and introduction which I have found fundamental.

Pondella, D.J., Claisse, J.T., Williams, C.M., 2022. Theory, practice, and design criteria for utilizing artificial reefs to increase production of marine fishes. Frontiers in Marine Science 9. <https://doi.org/10.3389/fmars.2022.983253>

Response:

Thank you for this comment. We have added references to Pondella et al. (2022) throughout the text.

Action taken:

Added citation:

Pondella, D.J., Claisse, J.T., Williams, C.M., 2022. Theory, practice, and design criteria for utilizing artificial reefs to increase production of marine fishes. Frontiers in Marine Science 9. <https://doi.org/10.3389/fmars.2022.983253>

R3-09 — Reviewer 3 comment

Original comment:

[Citations – socioeconomics] Similarly, artificial reefs/shipwrecks are being placed primarily to enhance the social and economic benefits to stakeholders. This particularly the case in Thailand and one of the main reasons DMCR are creating artificial reefs. I consider you add a sentence or paragraph to the introduction. The reference below will aide you.

Elrick-Barr, Carmen E., et al. "Man-made structures in the marine environment: A review of stakeholders' social and economic values and perceptions." Environmental Science & Policy 129 (2022): 12-18.

Response:

Thank you for this comment. This is thoughtful context for this work, and strong for us to keep in mind for future artificial reef studies here in Thailand. We have added various references to this paper throughout the article.

Action taken:

Updated introduction text at L62-66 to add:

“Intentional shipwreck deployment in Thailand is managed under the national Artificial Reef Program (since 1978) by the Thai Department of Marine and Coastal Resources (DMCR) to reduce diver pressure on natural reefs and enhance fisheries, reflecting the high social and economic value placed on artificial reefs by recreational users and fishers (Yeemin et al. 2006, Mattos & Yeemin 2020, Harvey et al. 2021, Elrick-Barr et al. 2022, Baillie et al. 2024).”

Updated methods text in section 2.1 at L186-188:

“All three were intentionally sunk by the DMCR to enhance social and economic benefits to key stakeholders, in this case, divers and recreation fishers (Elrick-Barr et al. 2022).”

Added citation:

Elrick-Barr, C. E., Zimmerhackel, J. S., Hill, G., Clifton, J., Ackermann, F., Burton, M., & Harvey, E. S. (2022). Man-made structures in the marine environment: A review of stakeholders' social and economic values and perceptions. *Environmental Science and Policy*, 129, 12–18. <https://doi.org/10.1016/j.envsci.2021.12.006>

We trust that these revisions fully address the reviewers' and editorial concerns and have significantly improved the clarity, methodological transparency, and interpretive strength of the manuscript.

Scarlett R. Taylor, on behalf of all authors.

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Shipwrecks can mirror predator assemblages of pelagic pinnacles but may lack the trophic balance of natural reef environments.

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~~Scarlett R. Taylor~~**S.R.T.**: Methodology, Investigation, Software, Formal Analysis, Visualisation, Writing - Original draft preparation, Writing - Reviewing and Editing. ~~Gavin Miller~~**G.M.**: Conceptualisation, Methodology, Investigation, Writing - Original draft preparation, Writing - Reviewing and Editing. ~~Piers Baillie~~**P.B.**: Methodology, Investigation, Project Administration, Supervision, Writing - Reviewing and Editing.

13 **ABSTRACT**

14 Artificial reefs are widely deployed to enhance fish populations, yet their ecological equivalence
15 to natural reefs remains uncertain, partly because comparisons are typically limited to fringing
16 coral reefs. Artificial reefs are widely deployed to enhance fish populations, yet their capacity to
17 replicate the ecological roles of natural reef systems remains uncertain, and comparisons are
18 often limited to nearby fringing reefs. Such assessments may overlook how structural features
19 and varying reef morphologies influence ecological outcomes. Fish assemblages were compared
20 across shipwrecks, pelagic pinnacles, and fringing reefs around an island in the Gulf of Thailand
21 to evaluate how artificial structures of differing relief align with natural community and trophic
22 patterns. This study compared fish assemblages across shipwrecks, pelagic pinnacles, and
23 fringing reefs in the Gulf of Thailand to evaluate how artificial structures align with natural
24 patterns of community structure and trophic composition. Natural reef sites were selected to
25 represent contrasting structural profiles, with pinnacles characterized by high vertical relief and
26 fringing reefs by lower relief slopes. This design enabled assessment of whether high relief
27 artificial structures, such as shipwrecks, more closely reflect fish communities found on similarly
28 structured natural reefs. Bayesian multivariate models applied to 350 ~~underwater~~ surveys from
29 14 sites showed that shipwrecks resembled pelagic pinnacles in ~~supporting~~ mesopredators and
30 higher trophic level predators, but had consistently lower representation of grazers and
31 invertivores relative to both natural reef types. This functional divergence likely reflects reduced
32 benthic complexity and forage availability; shipwrecks may support predator biomass but not
33 fully function as ecological analogues of natural reefs. ~~These results indicate functional~~
34 ~~divergence likely driven by reduced benthic complexity and limited forage availability. While~~
35 ~~shipwrecks may support fisheries species and predator biomass, they do not fully function as~~

Fish on Shipwrecks vs. Pinnacles

36 ~~ecological analogues of natural reefs.~~ Findings highlight the importance of aligning artificial reef
37 design with ~~defined~~ ecological objectives and considering structural context in management
38 strategies across tropical reef systems worldwide.

39 **Keywords:** artificial reefs, fish assemblages, trophic structure, Bayesian ~~inference, marine~~
40 ~~conservation, coral reefs~~

1. INTRODUCTION

~~Artificial reef performance is typically evaluated by comparison with nearby fringing coral reef systems, although such comparisons may overlook structural influences on ecological outcomes. Artificial reef performance is typically evaluated by comparison with nearby fringing reef systems, yet these assessments may overlook how structural features influence ecological outcomes.~~ Fringing coral reefs, which dominate nearshore tropical coastlines, are generally low in vertical relief and often embedded in sediment-rich, human-impacted environments (Bell & Galzin 1984, Letessier et al. 2019). In contrast, many artificial reefs, particularly shipwrecks, share greater physical similarity with ~~offshore pelagic~~ pinnacles (steep-sided bathymetric features similar to seamounts), including ~~steep high vertical relief~~ vertical profiles, current exposure, and spatial isolation (Letessier et al. 2019, Paxton et al. 2023~~0b~~, Galbraith et al. 2022). These ~~structural~~ features influence foraging opportunities, predator access, and planktonic input, ~~potentially~~ shaping fish community composition and trophic balance (White et al. 2007, Galbraith et al. 2023). Despite these parallels, few studies have examined whether shipwrecks more closely resemble the ecological characteristics of pinnacles, align instead with fringing reef communities, or represent a functionally novel assemblage.

Artificial reefs are deployed worldwide to mitigate habitat loss and enhance fish populations (Rilov & Benayahu 2000, Paxton et al. 2020a, Harvey et al. 2021, Pondella et al. 2022, Elrick-Barr et al. 2022, Baillie et al. 2024). Among the various designs, shipwrecks are frequently used due to their complex structure, tourism appeal, and availability through both intentional deployment and maritime accidents (Mattos & Yeemin 2020, Medeiros et al. 2022, Hickman et al. 2024). Intentional shipwreck deployment in Thailand is managed under the national Artificial Reef Program (since 1978) by the Thai Department of Marine and Coastal Resources (DMCR) to

~~reduce diver pressure on natural reefs and enhance fisheries, reflecting the high social and economic value placed on artificial reefs by recreational users and fishers (Yeemin et al. 2006, Mattos & Yeemin 2020, Harvey et al. 2021, Elrick-Barr et al. 2022, Baillie et al. 2024). In Thailand, shipwreck deployment has been widespread since the launch of the national Artificial Reef Program in 1978, aimed at reducing diver pressure on natural reefs and supporting fisheries enhancement (Yeemin et al. 2006, Mattos & Yeemin 2020, Harvey et al. 2021, Baillie et al. 2024).~~

The Gulf of Thailand contains three primary reef habitat types relevant to this study: nearshore fringing reefs that slope gradually toward sandy bottoms (Yeemin et al. 2009), deeper offshore pinnacles with substantial vertical relief, and a growing number of intentionally deployed artificial structures such as shipwrecks (Mattos et al. 2023). Despite the prevalence of these artificial reefs, ~~relatively~~ few studies have assessed their ecological functioning, particularly their capacity to support ~~balanced trophic assemblages~~ trophic structures comparable to those of natural reefs and replicate the ecological roles of natural reef systems (Fowler & Booth 2012, Simon et al. 2013, Streich et al. 2017, Medeiros et al. 2022, Paxton et al. 2023, Schram et al. 2024).

Where evaluations exist, artificial reefs often support higher fish abundance and species richness than nearby natural reefs (Fowler & Booth 2012, Folpp et al. 2013, Schulze et al. 2020); however, these metrics (total abundance, richness) can obscure differences in community and trophic structure (Ross et al. 2016, Paxton et al. 2020a, Paxton et al. 2020b, Paxton et al. 2023).
~~Where evaluations do exist, artificial reefs, particularly shipwrecks, often support high total fish abundance and species richness compared to nearby natural reefs (Fowler & Booth 2012, Folpp et al. 2013, Schulze et al. 2020). However, these metrics can obscure differences in community~~

~~composition and ecological function (Ross et al. 2016, Paxton et al. 2020a, Paxton et al. 2020b, Paxton et al. 2023).~~ Predator biomass is frequently elevated, while lower trophic guilds ~~such as (grazers, s and invertivores)~~ tend to be underrepresented on artificial reefs (Fowler & Booth 2012, Schram et al. 2024). This imbalance may limit processes such as algal control, nutrient cycling, and trophic transfer, ~~indicating that high fish abundance alone does not equate to ecological equivalence with natural reefs, showing that high abundance alone does not imply ecological equivalence.~~

Such divergences are influenced by multiple ecological and structural factors, including vertical relief, habitat complexity, reef age, and proximity to natural systems (Rilov & Benayahu 2000, Simon et al. 2013, Streich et al. 2017, Paxton et al. 2023). ~~Shipwrecks tend to be high relief and spatially isolated,~~ High relief, isolated shipwrecks favouring aggregation of upper trophic level species while offering limited support for lower trophic guilds (Harvey et al. 2021, Medeiros et al. 2022, Schram et al. 2024). Over time, these dynamics may produce assemblages that diverge functionally and compositionally from adjacent natural reefs (Paxton et al. 2020b, ~~Paxton et al. 2023,~~ Sibley et al. 2023). Distinguishing ~~superficial~~ species-level similarity from broader ecological performance is ~~therefore~~ essential when evaluating artificial reef outcomes (Simon et al. 2013, Medeiros et al. 2022, Schram et al. 2024).

Variation among natural reef types provides important context for interpreting artificial reef performance. Isolated pelagic pinnacles, ~~s,~~ often exposed to strong hydrodynamic flow (Leitner et al. 2021), tend to support high densities of mesopredators and larger predators such as *Caranx*, *Lethrinus*, *Lutjanus*, and *Epinephelus* spp. (Letessier et al. 2019, Galbraith et al. 2022, Cresswell et al. 2023, Galbraith et al. 2023a, ~~Galbraith et al. 2023b,~~ Mattos et al. 2023). These physical features enhance planktonic input and nutrient delivery, increasing productivity at lower trophic

levels and facilitating biomass accumulation at higher trophic levels (White et al. 2007, [Leitner et al. 2021](#), Galbraith et al. 2022, 2023). [High relief and structural complexity also provide refuge and foraging habitat across trophic groups \(Letessier et al. 2019, Galbraith et al. 2021, 2022, 2023, Pondella et al. 2022\).](#) ~~Pinnacles also exhibit substantial vertical relief and structural complexity, which provide refuge and foraging opportunities across multiple trophic guilds (Galbraith et al. 2023, Letessier et al. 2019).~~

[In contrast, fringing reefs and other nearshore systems are more strongly shaped by local benthic variation, hydrodynamic exposure, and human disturbance \(Done 1983, Bell & Galzin 1984\).](#) ~~In contrast, fringing reefs and other emergent coastal systems are shaped more strongly by local variation in benthic composition, hydrodynamic exposure, and human disturbance (Bell & Galzin, 1984, Letessier et al. 2019, Galbraith 2021).~~ These environments ~~tend to~~ [typically](#) support ~~more trophically balanced assemblages, with a greater~~ [higher](#) relative abundance of grazers and invertivores that ~~play key roles in~~ [maintain](#) reef resilience and benthic regulation (Koeck et al. 2014, Boaden & Kingsford 2015, Streit et al. 2015, Schram et al. 2024). Comparisons between artificial reefs and adjacent natural reefs, including fringing systems, have consistently shown that shipwrecks support higher densities of mid- to higher-trophic fish, ~~such as *Lutjanus*, *Sphyræna*, *Plectorhynchus*, and *Epinephelus* spp.~~, but reduced densities of lower trophic guilds (Fowler & Booth 2012, Paxton et al. 2020a, Medeiros et al. 2022, Baillie et al. 2024, Schram et al. 2024). These patterns likely reflect limited food availability for grazers and invertivores on artificial substrates, as well as differences in benthic microhabitats and structural complexity (Rilov & Benayahu, 2000, Schram et al. 2024, [Gress et al. 2025](#)).

This study builds on previous assessments of fish assemblages in the Gulf of Thailand (Mattos & Yeemin 2020, Harvey et al. 2021, Mattos et al. 2023, Baillie et al. 2024), [to evaluate whether](#)

intentionally deployed shipwrecks replicate the community structure and trophic composition of
natural reef analogues or represent a functionally novel assemblage~~with the aim of evaluating~~
~~whether shipwrecks replicate the community structure and trophic composition of natural reef~~
~~analogues or represent a functionally novel assemblage type.~~ Unlike most artificial coral reef
studies limited to fringing-reef comparisons, this study tests shipwreck similarity across two
contrasting natural reef types: fringing reefs representing shallow coastal systems and pelagic
pinnacles representing deep, high-relief offshore reefs, to determine which they more closely
resemble. ~~Unlike most prior studies, which have compared artificial reefs only to nearby fringing~~
~~reefs, this work explicitly tests shipwreck similarity against two contrasting natural reef types:~~
~~fringing reefs and pelagic pinnacles, to determine which they more closely resemble.~~ Given their
structural similarity to pinnacles, shipwrecks were hypothesised to support predator-dominated
assemblages more closely aligned with pinnacles than with fringing reefs, though limited food
availability and reduced benthic complexity may constrain lower trophic guilds such as grazers
and invertivores compared to both natural reef types (Fowler & Booth 2012, Simon et al. 2013,
Schram et al. 2024). ~~Given their structural similarity to pelagic pinnacles, shipwrecks are~~
~~hypothesised to support predator-dominated assemblages more closely aligned with pinnacle~~
~~reefs than with fringing systems. However, artificial reefs may lack the food availability, benthic~~
~~complexity, or microhabitats required to sustain lower trophic guilds such as herbivores and~~
~~invertivores (Fowler & Booth 2012, Simon et al. 2013, Schram et al. 2024). These divergences~~
~~may reflect differences in habitat structure, trophic filtering, or broader ecological constraints.~~
Characterising such patterns is critical to evaluating the ecological role of artificial reefs in
tropical marine systems and understanding how reef geomorphology shapes biodiversity, trophic
structure, and ecological function. As shallow and coastal reefs continue to degrade under

increasing human pressure, isolated offshore features such as pinnacles and shipwrecks may become increasingly important refuges for reef-associated communities and potential sources of ecological resilience. Characterising such patterns is critical to evaluating the ecological role of artificial reefs in tropical marine systems. By comparing functional group and species-level patterns across reef types, and relating these to structural features such as vertical relief and complexity, the study aims to clarify the ecological significance of shipwrecks and inform context-specific deployment strategies in Thailand and other tropical regions pursuing artificial reef development (Paxton et al. 2020a, Baillie et al. 2024).

~~By explicitly comparing fish assemblages across shipwrecks, pinnacles, and fringing reefs, this study evaluates whether shipwrecks more closely resemble a particular type of natural reef or support distinct assemblages. Patterns in functional group composition, including the relative abundance of grazers, invertivores, mesopredators, and higher trophic level predators, are used alongside species-specific responses to assess ecological similarity and divergence. Key design features such as vertical relief, surface complexity, and orientation are considered in interpreting assemblage patterns, given their known influence on artificial reef performance (Rilov & Benayahu 2000, Paxton et al. 2020a). These findings contribute to a broader understanding of how artificial reefs function across structurally varied reef contexts and are intended to inform ecologically grounded deployment strategies not only in Thailand, where shipwreck use continues to expand, but in other tropical regions pursuing artificial reef development to support reef fisheries and conservation goals (Paxton et al. 2020a, Baillie et al. 2024).~~

2. MATERIALS AND METHODS

2.1. Study Sites

Surveys were conducted across three reef categories: fringing reefs (n = 5 sites), pelagic pinnacles (n = 6 sites), and artificial reefs (shipwrecks; n = 3 sites), totalling 14 individual sites (Figure 1, [Table S1](#)). All sites were located around Koh Tao (10.10980° N, 99.81180° E) in the Gulf of Thailand, the largest semi-enclosed area in the central Indo-Pacific (Mattos et al. 2023). Koh Tao lies approximately 65 km offshore and provides access to a range of reef structures for comparative study.

The three shipwrecks studied were decommissioned Royal Thai Navy vessels: the landing craft *HTMS Sattakut*, which was deployed in June 2011, and the two fast-attack vessels *HTMS Supharin* and *HTMS Hanhak Sattru*, both deployed in September 2023. All three were intentionally sunk by the DMCR to enhance social and economic benefits to key stakeholders, in this case divers and recreational fishers (Elrick-Barr et al. 2022). At 48m in length and 7m in width, *HTMS Sattakut* sits on sandy sea floor in the west of Koh Tao (10.10169°N, 99.81537°E) at a maximum depth of 30m and provides 12m of vertical relief from the base of the hull to the highest point. *HTMS Suphairin* and *HTMS Hanhak-Sattru* were deployed on sand in the north-west (10.1166°N, 99.80993°E), at a maximum depth of 30m, and north-east (10.09239°N, 99.85292°E), at a maximum depth of 31m, of the island respectively. *HTMS Suphairin* and *HTMS Hanhak-Sattru* have identical measurements at 45m in length and 7m in width, and provide approximately 16 m of vertical relief from the hull base to the highest point.

Each site was surveyed multiple times from March 2019 to June 2025. Surveys were planned at least monthly outside the November–December monsoon pause, though frequency varied by site

and year, with lower effort in 2020–22. In total, 350 fish surveys were completed, with 150 on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at predetermined locations within each reef type to ensure consistency across dives. Per-site counts, first and last survey dates, and median intervals are provided in Table S1. ~~Each site was surveyed repeatedly over the study period, yielding a total of 350 fish surveys: 150 on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Repeated surveys were performed at predetermined locations within each reef type to ensure consistency across dives.~~

2.2. Survey Protocol

All surveys followed a standardised underwater census (UVC) methodology. Divers conducted eight minute timed-swim surveys (used in Baillie et al. 2024; adapted from Hill and Wilkinson 2004) using a predetermined survey species list (Table S2), which encompasses multiple functional groups (grazers, invertivores, mesopredators and higher trophic level predators) and ensures consistency in species identification and data collection. Survey species were assigned to these four functional groups using precedent consistent with Baillie et al. (2024); the full mapping is provided in Table S2. The list was derived from pilot surveys and regional checklists and was held constant across sites and months to standardize counts, reduce identification error, and focus on conspicuous taxa that are reliably detected within a 10 m inclusion radius (Baillie et al. 2024). Pomacentrids were excluded as their small size and high abundance make them poorly suited to timed-swim methods; accurate assessment would require stationary point counts, which were beyond the scope of this study. Eight minutes was selected as the survey duration based on multiple pilot surveys conducted on the *HTMS Sattakut*, where one full circuit of the wreck could be completed without overlap, minimizing double counting.

Fish on Shipwrecks vs. Pinnacles

Each survey commenced from the same designated starting point on every dive site. The survey team consisted of two divers working in a buddy team on SCUBA. From the fixed start point, the transect path followed the structural contour at the target depth; the end point varied because surveys were time standardised. On shipwrecks, one diver maintained a position ~1 m above the hull edge while the buddy remained 1–2 m into open water at the same depth. On pinnacles, one diver remained 1–2 m from the reef with the buddy 1–2 m further outward. On fringing reefs, the pair contoured along the reef edge with the same spacing. The target depth matched the average depth used on wrecks (~25 m) where feasible; if not, the deepest contiguous reef ledge was followed while remaining ~1 m above the seabed.

Divers maintained a position of 1-2 m away from each other at all times, and maintained a consistent swimming pace while ensuring they stayed level with their buddy. The observation field included all fish present within a 10-metre radius in the 180-degree field of view ahead, covering the left, forward, and right directions, and extending vertically to form a 180-degree three-dimensional arc above and below the divers. A 10 m inclusion radius was selected to capture transient, mid-water higher trophic level predators (hereafter HTLPs) after multiple pilot surveys. Observer distance estimation was trained prior to fieldwork and reinforced during the sampling period using fixed range cues and team cross-checks. Surveys proceeded only when horizontal visibility was ≥ 3 m, and visibility was logged for quality control. Divers swam at a practiced pace of ~ 12 m min^{-1} for eight minutes, covering ~ 96 m per survey; the effective swath area was $\sim 1,920$ m^2 (distance $\times 2 \times 10$ m inclusion radius). This refines the $1,200$ m^2 figure previously reported by Baillie et al. (2024). ~~Divers swam slowly to maximise observation time and reduce disturbance to the fish. The estimated area covered during each survey was calculated~~

~~by measuring the distance covered during multiple pilot timed swim surveys on the HTMS Sattakut.~~

All observed fish on the species list within the designated radius were recorded on waterproof slates in-situ. Counts of individuals by listed species were recorded for each survey, and zeros were retained so that presence and absence were represented. Data were recorded in real-time during the survey. Each new survey was recorded in a separate section of the slate to maintain data integrity.

2.3. Data Analysis

Data were analysed using Bayesian multivariate regression models, fit in R version 4.5.0 (2025-04-11) using the *brms* package (Bürkner 2017). Bayesian models were estimated using Markov chain Monte Carlo (MCMC) sampling implemented in Stan via *brms*. Data manipulation was performed using the *tidyverse*, and visualisations were generated with *ggplot2* (Wickham et al. 2019). The species were first grouped into functional groups – grazers, invertivores, mesopredators, and higher trophic level predators (hereafter HTLPs) – based on trophic ecology (Table S2~~1~~) for analysis of broader ecological patterns and to facilitate interpretation of community-level responses. Separate multivariate models were fitted to evaluate differences in functional group abundance across reef classifications and, independently, across geographic zones. Sites were classified by geographic zone as “nearshore” if located within 1 km of Koh Tao, “pelagic” if beyond 1 km, and “wreck” for intentionally deployed ships (Table S1). Zone was added to separate shoreline proximity from reef type when estimating classification effects. Subsequent species-specific models were used to examine finer-scale patterns within each functional group.

2.3.1. Functional group analysis

To examine differences in the abundance of these functional groups across reef classifications, a Bayesian multivariate regression model was used. The model specified the joint response as:

$$\begin{aligned} & \text{mvbind}(\text{Grazer}, \text{Invertivore}, \text{Mesopredator}, \text{HTLP}) \\ & \sim \text{Classification} + (1 \mid \text{Month_Year}) + (1 + \mid \text{Site}) \end{aligned}$$

allowing inference across groups while accounting for potential correlations in abundance patterns. Classification refers to the site type, either shipwreck, fringing reef or pelagic pinnacle. Shipwrecks were set as the reference group for reef classification.

Prior to model fitting, exploratory analyses were conducted to examine survey effort, distribution of counts, and the composition of functional groups across sites. Zero-inflation rates were calculated for each species, and species with infrequent but sometimes extreme counts (e.g. *Sphyræna* spp., *Gymnothorax* spp., *Tetraodon* and *Diodon* spp., all rays) were excluded due to their disproportionate influence on group-level patterns (<2% of observations removed). Mobile predators were retained in the analyses (e.g., *Caranx* spp.); only episodic schooling pulses of *Sphyræna* spp. that disproportionately influenced group-level estimates were excluded from model fitting and summarized descriptively. Visibility was not modelled because low-visibility (<3m) dives were excluded, remaining variation was moderate and comparable across sites and periods by design, and random effects for site and month-year absorbed residual heterogeneity. Quality control identified systematic undercounting in surveys conducted by one observer; those surveys were excluded. Functional group stability was assessed using coefficients of variation, and proportional group composition was assessed by site to evaluate consistency across reef types. These diagnostics informed data filtering and supported the robustness of the functional

group classification. Down-weighting via transformation was considered but not implemented. A sensitivity check that retained *Sphyræna* under capped counts did not change qualitative higher-trophic-level predator trends. Zero-inflation was not evident in the remaining functional group data, while overdispersion was moderate, supporting the use of a negative binomial distribution.

Several candidate models were evaluated, including those with and without zero-inflation terms and with random intercepts for site and month-year to account for repeated surveys and seasonal variation. ~~and with or without site-level random intercepts to account for repeated surveys.~~

Model performance was compared using leave-one-out cross-validation (LOO-CV; Vehtari et al. 2017). The best-supported model included a negative binomial likelihood with random intercepts ~~by~~ for site and month-year, capturing both overdispersion and hierarchical structure in the data.

The response for each group g , at site i , survey j , was modelled as:

$$Y_{ijg} \sim \text{NegBinom}(\mu_{ijg}, \phi_g)$$

$$\log(\mu_{ijg}) = \alpha_g + \beta_{g1} * \text{Fringing}_j + \beta_{g2} * \text{Pinnacle}_j + \mu_{ig} + v_{mg}$$

$$\mu_{ig} \sim N(0, \sigma^2_{\text{site},g}), \quad v_{mg} \sim N(0, \sigma^2_{\text{month_year},g})$$

Here, Y_{ijg} represents the count of fish in group g at survey j within site i , μ_{ijg} is the expected count, ϕ_g is the dispersion parameter, μ_{ig} is the random site effect, and v_{mg} is the random

month-year effect. ~~and μ_{ig} is the random site effect.~~ Models excluding site-level effects, month-year effects, or including zero-inflation performed substantially worse, and estimated zero-

inflation probabilities were near zero across all responses. The final model was fitted with four

chains of 4000 iterations each (1000 warmup), using control parameters `adapt_delta = 0.90`

and `max_treedepth = 20`. Default weakly informative priors were applied, as no custom

priors were specified. This included default priors on fixed effects, intercepts, and group-level

standard deviations, as implemented in brms (Bürkner 2017). Posterior predictive checks were conducted using `pp_check()` to confirm adequate model fit. All parameters demonstrated convergence ($\hat{R} = 1.00$), with high effective sample sizes, supporting confidence in the posterior estimates.

Inferences were based on posterior distributions of fixed effects. Predicted mean abundances and proportional group compositions were visualised using faceted bar plots derived from population-level predictions via `conditional_effects()`. Differences across reef classifications were calculated by combining intercepts with classification-level effects on the log scale, following:

$$\log(\mu) = \text{Intercept} + \text{Classification_effect}$$

Posterior distributions of fixed effects were extracted using `spread_draws()` to estimate log-scale differences in functional group abundance across reef classifications. Differences were visualised using `stat_halfeye()`, which displays the full posterior distribution with overlaid medians and 95% credible intervals.

2.3.2. Zone-based analysis

A separate multivariate model was fitted to examine differences in functional group abundance across geographic zones. This analysis was conducted to assess whether spatial context, specifically, proximity to shore versus offshore exposure, also structured fish assemblages, independent of reef type. The response structure and model family remained identical to the classification-based model, with the predictor variable replaced by zone (nearshore, pelagic, or shipwreck):

mvbind(Grazer, Invertivore, Mesopredator, HTLP)

$\sim \text{Zone} + (1 + | \text{Month_Year}) + (1 + | \text{Site})$

$Y_{ijg} \sim \text{NegBinom}(\mu_{ijg}, \phi_g)$

$\log(\mu_{ijg}) = \alpha_g + \beta_{g1} * \text{Nearshore}_j + \beta_{g2} * \text{Pelagic}_j + \mu_{ig} + v_{mg}$

$\mu_{ig} \sim N(0, \sigma^2_{\text{site},g}), \quad v_{ig} \sim N(0, \sigma^2_{\text{month_year},g})$

Where Y_{ijg} represents the count of fish in group g at survey j within site i , μ_{ijg} is the expected count, ϕ_g is the dispersion parameter, ~~and~~ μ_{ig} is the random site effect, and v_{mg} is the random month-year effect. Shipwrecks were retained as the reference level for interpretability. Model fitting and diagnostics followed the same procedures, including the use of a negative binomial likelihood, site-level random intercepts, and evaluation via LOO-CV. Model diagnostics indicated good convergence and posterior sampling behaviour. LOO-CV showed minimal difference in expected predictive performance compared to the classification model ($\Delta\text{elpd} = -0.1$), indicating that reef type likely explains functional group composition slightly more effectively than spatial zone. Predicted mean abundances and differences across reef zones were extracted and visualised using the same procedure as the functional group model. Specifically, posterior draws were obtained via `spread_draws()`, and zone-level effects were visualised using `stat_halfeye()` with 95% credible intervals, showing log-scale abundance differences relative to shipwrecks.

2.3.3. Species-specific analysis

Species-specific models were fitted to investigate fine-scale differences in abundance across reef classifications. Only species with at least 10 non-zero observations were included to ensure sufficient data for estimation. Survey-level data were aggregated and pivoted to a wide format, and a multivariate negative binomial model was fitted using the *brms* package, with abundance

modelled as a function of reef classification. The response was specified as a multivariate bind of all included species, enabling joint estimation while accounting for overdispersion structured as:

$$mvbind(Species_1, Species_2, \dots, Species_n) \sim Classification$$

The response for species s at survey j was modelled as:

$$Y_{js} \sim NegBinom(\mu_{js}, \phi_s)$$
$$\log(\mu_{js}) = \alpha_s + \beta_{s1} * Fringing_j + \beta_{s2} * Pinnacle_j$$

Here, Y_{js} represents the count of species s in survey j , μ_{js} is the expected count, ϕ_s is the species-specific dispersion parameter, and shipwrecks were treated as the reference category.

Site ~~and month-year were~~ was not included as a random effect due to limited data and resulting model instability. With 18 taxa across 12 sites, the model became overparameterised and failed to converge reliably. Excluding random intercepts improved stability while retaining the ability to detect habitat-based trends. Model fitting used four chains of 4000 iterations each (1000 warmup) with `adapt_delta = 0.90` and `max_treedepth = 20`. Species-level predictions were generated using `conditional_effects()` with population-level estimates. Differences in predicted abundance across reef types were visualised using faceted bar plots with 95% credible intervals.

3. RESULTS

A total of 350 fish surveys were conducted from March 2019 to June 2025 across 14 unique reef sites, comprising 5 fringing reefs, 6 pelagic pinnacles, and 3 shipwrecks (Table S1). Survey frequency varied among sites and years, with lower effort in 2020–22 as a result of COVID and monthly revisits from 2023 onward outside the monsoon period (Nov-Dec). Repeated surveys were conducted at each site, with 150 surveys on fringing reefs, 130 on pinnacles, and 70 on shipwrecks. Each survey covered an estimated area of $\sim 1,9200 \text{ m}^2$, yielding a cumulative

surveyed area of ~42 hectares. Across all sites, 162,894 individual fish were recorded. Mean ($\pm$ SD) fish density was highest on pinnacles (668 ± 928 fish per $1,9200 \text{ m}^2$), followed by shipwrecks (348 ± 396) and fringing reefs (345 ± 220), indicating elevated average fish abundance at pinnacles, while shipwrecks and fringing reefs supported comparable mean densities. ~~and, to a lesser extent, shipwrecks relative to fringing reefs.~~

3.1. Functional groups

Predicted fish abundance varied across reef types and functional groups (Figure 2, Table 1). Grazers and mesopredators were the most abundant groups across all classifications, followed by invertivores and higher trophic level predators (HTLPs). Posterior predictions indicated that grazer abundance was highest at fringing reefs (median: ~~127.9~~100.7 fish per $1,9200 \text{ m}^2$; 95% Credible Interval: [~~91.56.2~~, ~~179.0484~~]), slightly lower at pinnacles (~~105.9~~98.4; [~~78.64.6~~, ~~148.45~~]), and substantially lower at shipwrecks (~~59.747.8~~; [~~38.28.1~~, ~~83.494~~]). Invertivores showed comparable predicted abundance at pinnacles (~~56.0~~58.7; [~~35.134~~, ~~100.594~~]) and fringing reefs (~~54.4~~; [~~24.2-120.7~~])(~~55.9~~; [~~32~~, ~~97~~]), with lower estimates at shipwrecks (~~33.0~~; [~~16~~, ~~66~~])28.7; [~~28.1-83.4~~]). Mesopredator abundance was highest at pinnacles (~~1340.85~~; [~~59.16~~, ~~311.302~~]), with lower values at fringing reefs (~~68.769.7~~; [~~18226~~, ~~250.3472~~]) and shipwrecks (~~43.541.7~~; [~~11.043~~, ~~150.240~~]). HTLPs had the highest predicted abundance at pinnacles (26.2; [13.4, 51.6]) and shipwrecks (24.6; [9.0, 70.2] and lower values at fringing reefs (9.1; [3.0, 26.6]).~~lowest predicted abundance at fringing reefs (11.0; [5, 23]), with similar values at pinnacles (26.4; [13, 51]) and shipwrecks (25.6; [10, 68]).~~

Posterior distributions indicated differences in functional group abundance across reef types, with varying levels of statistical support (Figure 3, Table 2). Grazers were more abundant at

fringing reefs than shipwrecks, with a median effect of 0.75 (95% CrI: [0.03, 1.46])~~0.76 (95% CrI: [0.20, 1.34])~~ and a 98% posterior probability that abundance was higher at fringing reefs. Pinnacles also supported higher grazer abundance (0.72; [0.14, 1.31])~~(0.57; [0.02, 1.13])~~, with 99% posterior support for a positive effect. Invertivores followed a similar pattern: abundance was higher at fringing reefs (0.63; [-0.48, 1.75])~~(0.52; [-0.36, 1.42])~~ and pinnacles (0.71; [-0.19, 1.61])~~(0.52; [-0.34, 1.42])~~ compared to shipwrecks, with posterior probabilities of 88% and 95%, respectively. Mesopredators were more abundant at pinnacles than shipwrecks (1.19; [-0.35, 2.70])~~(1.10; [-0.35, 2.58])~~, with 94% support for a positive effect, and may have also been more abundant at fringing reefs (0.51; [-1.34, 2.38])~~(0.45; [-1.07, 1.99])~~, though posterior support was weaker (71%). For higher trophic level predators (HTLPs), abundance was lower at fringing reefs compared to shipwrecks (-1.00; [-2.51, 0.47])~~(-0.86; [-2.10, 0.40])~~, with a 92% probability of a negative effect. Pinnacles also supported higher HTLP abundance than fringing reefs (1.06; [-0.18, 2.34])~~(0.88; [-0.14, 1.86])~~, with 96% support. In contrast, the difference between pinnacles and shipwrecks was negligible (0.072; [-1.17, 1.4625]), with posterior support evenly split (55%).

3.2. Geographic zones

While reef classification explained most of the variation in functional group composition, a secondary analysis evaluated whether geographic zone, defined by proximity to shore, also structured fish assemblages (Table 3, Figure 4). Grazers were more abundant in both nearshore (median effect: 0.79; 95% CrI: [0.18, 1.38])~~0.67; 95% CrI: [0.12, 1.28])~~ and pelagic zones (0.75; [0.03, 1.49])~~(0.62; [-0.09, 1.32])~~ compared to shipwrecks, with a 99% and 98% posterior probability, respectively, that abundance was higher in natural zones. Invertivores were also more abundant nearshore (0.80; [0.03, 1.54])~~(0.67; [-0.08, 1.44])~~, with a 98% probability that

abundance exceeded that on shipwrecks, while the contrast between pelagic sites and shipwrecks was less clear $(0.32; [-0.62, 1.21])$, $(0.12; [-0.77, 1.05])$, supported by a posterior probability of ~~61.77%~~ 64.77%. Mesopredators were somewhat more abundant nearshore $(0.90; [-0.59, 2.38])$ ~~$(0.84; [-0.68, 2.29])$~~ and in pelagic zones $(0.79; [-0.97, 2.59])$, ~~$(0.68; [-1.17, 2.49])$~~ , with 89.4% and 78.82% posterior support, respectively, that abundance was higher than on shipwrecks. For higher trophic level predators (HTLPs), abundance was slightly lower at nearshore reefs relative to shipwrecks $(-0.53; [-1.77, 0.72])$, 19 % support for a positive effect) and only marginally higher at pelagic sites $(0.08; [-1.39, 1.56])$, 54 % support). Pelagic reefs may support somewhat higher HTLP abundance than nearshore sites $(0.61; [-0.60, 1.81])$, 85 % support), though overall differences across zones were inconsistent. For HTLPs, pelagic zones were estimated to support slightly higher abundance than both nearshore reefs $(0.57; [-0.68, 1.84])$ and shipwrecks $(0.07; [-1.45, 1.60])$, though posterior support was weak in both cases (57% and 54%, respectively), suggesting no consistent difference in HTLP abundance across zones.

3.3. Species-level contrasts

Posterior contrasts for individual species revealed consistent patterns in the direction and strength of habitat associations, particularly distinguishing shipwrecks from natural reef types. Species with the most pronounced or ecologically relevant differences among habitats are highlighted below; full posterior estimates and contrast summaries for all modelled taxa are provided in Tables S32, and S43, and Figures 6–7.

Among mesopredators and high trophic level predators, several species displayed distinct habitat associations. *Lutjanus* spp. <30 cm (small snappers; mesopredators) showed similar predicted abundance at pinnacles and shipwrecks, with a negligible median difference and low posterior

support for any effect (-0.05 ; 95% CrI: $[-0.51, 0.38]$; 59%), but were substantially less abundant at fringing reefs compared to shipwrecks (-1.29 ; $[-1.75, -0.84]$; 100%). *Lutjanus* spp. >30 cm (large snappers; mesopredators) were more abundant at shipwrecks than pinnacles (-1.06 ; 95% CrI: $[-1.44, -0.69]$; 100%) and fringing reefs (-2.85 ; $[-3.28, -2.44]$; 100%). *Caranx* spp. (trevallies; high trophic level predators) were more abundant at pinnacles compared to shipwrecks (0.85 ; 95% CrI: $[0.55, 1.15]$; 100%) and more abundant at shipwrecks relative to fringing reefs (-0.54 ; $[-0.86, -0.23]$; 100%). *Diagramma/ Plectorhinchus* spp. (sweetlips; invertivores) were less abundant at both fringing reefs (-1.38 ; 95% CrI: $[-1.79, -0.98]$; 100%) and pinnacles (-0.65 ; $[-1.05, -0.27]$; 100%) compared to shipwrecks. *Lethrinus* spp. (emperors; mesopredators) showed substantially higher predicted abundance at pinnacles relative to shipwrecks (0.89 ; 95% CrI: $[0.57, 1.19]$; 100%), while differences between fringing reefs and shipwrecks were minor and weakly supported (0.13 ; $[-0.20, 0.44]$; 78%). *Epinephelus/ Plectropomus* spp. >30cm (large groupers; high trophic level predators) were more abundant at pinnacles relative to shipwrecks (0.37 ; 95% CrI: $[0.14, 0.60]$; 100%), with slightly higher but weakly supported differences at fringing reefs compared to shipwrecks (-0.07 ; $[-0.32, 0.16]$; 72%).

In contrast, several grazer and invertivore taxa were consistently lower on shipwrecks relative to natural reefs. *Pomacanthus* spp. (angelfish; invertivores) showed higher predicted abundance at both fringing reefs (0.94 ; 95% CrI: $[0.69, 1.20]$; 100%) and pinnacles (1.17 ; $[0.92, 1.41]$; 100%) compared to shipwrecks, with minimally elevated abundance at pinnacles relative to fringing reefs (0.23 ; $[0.07, 0.39]$; 99.8%). *Labroides dimidiatus* (cleaner wrasse; invertivores) also had higher predicted abundance at fringing reefs (2.29 ; 95% CrI: $[1.99, 2.59]$; 100%) and pinnacles (2.27 ; $[1.98, 2.57]$; 100%) compared to shipwrecks. *Epibulus insidiator* (slingjaw wrasse;

invertivores) showed a similar pattern, with higher abundance at fringing reefs (3.64; [3.12, 4.24]; 100%) and pinnacles (1.86; [1.33, 2.46]; 100%) relative to shipwrecks. *Scarus* spp. (parrotfish; grazers) had higher predicted abundance at fringing reefs (1.87; [1.68, 2.06]; 100%) and pinnacles (0.95; [0.77, 1.14]; 100%) compared to shipwrecks. *Chaetodon* spp. (butterflyfish; grazers) followed a similar pattern, with elevated abundance at fringing reefs (1.35; [1.06, 1.64]; 100%) and pinnacles (0.65; [0.36, 0.94]; 100%) relative to shipwrecks. Unlike other grazer taxa, *Siganus* spp. (rabbitfish; grazers) exhibited minimal variation, with slightly higher abundance on pinnacles (0.37; [0.15, 0.58]; 100%) and no strong difference between fringing reefs and shipwrecks (−0.07; [−0.29, 0.15]; 72% probability of a negative effect).

Across species, predatory taxa often reached comparable or higher abundance on shipwrecks compared with pinnacles, whereas reef-specialist grazers and invertivores were consistently lower on artificial structures (Figure 7). The consistently lower abundance of key grazers and invertivores indicates that meaningful compositional differences persist between artificial and natural reef assemblages.

4. DISCUSSION

This study evaluated fish assemblage structure across shipwrecks, pinnacles, and fringing reef environments to determine whether artificial reefs replicate natural community patterns. Based on their vertical relief and relative isolation, shipwrecks were expected to support assemblages more similar to pinnacles than found on fringing reefs. ~~By modelling both functional group and species-level patterns,~~ results suggest that shipwreck assemblages closely resembled natural reef pinnacles in their support of mesopredators and higher trophic level predators, but diverged markedly in their low representation of grazers and invertivores. While several predatory and

mobile species were abundant on both pinnacles and shipwrecks, many wrasses, parrotfishes, and small groupers were consistently underrepresented on artificial structures. These patterns highlight the functional selectivity of artificial reefs and their potential to replicate certain ecological roles while failing to support others, reinforcing the need for functionally driven artificial reef designs and deployment strategies, and for natural reef conservation to remain a priority in Thailand ([Paxton et al. 2020a](#), Baillie et al. 2024; ~~Paxton et al. 2020a~~). Key outcomes are examined in terms of predator convergence, divergence in lower trophic guilds, and implications for artificial reef design and ecological function.

4.1. Shipwrecks resemble pinnacles in meso- and higher trophic level predators

Predator-dominated assemblages on shipwrecks closely mirrored those observed on pelagic pinnacles, aligning with the initial prediction. High trophic level predator (HTLP) abundance was particularly elevated on shipwrecks, and ~~larger predators~~ HTLPs were consistently more abundant at both shipwreck and pinnacle sites than on fringing reefs. This convergence is likely driven by shared physical features: vertical relief, isolation, and strong hydrodynamic flow are known to promote predator aggregation on both pinnacles and artificial reefs (Rilov & Benayahu 2000, Galbraith 2021, Galbraith et al. [2021, 2022, 2023](#), [Pondella et al. 2022](#), Cresswell et al. 2023). [Depth can influence predator distributions, but within shallow to upper-mesophotic zones its effect is often weaker than reef profile and structural relief. Studies report gradual declines in predator richness with depth and limited depth specialism across the first 100 m where structure remains a primary driver \(Jankowski et al. 2015, Cresswell et al. 2025\).](#) Vertical ~~extent~~ relief has ~~also~~ been identified as a key predictor of predator density on artificial structures, including shipwrecks and oil platforms (Paxton et al. 2020b, Harvey et al. 2021, Sibley et al. 2023, Schram

et al. 2024). These features may enhance foraging efficiency and attract mobile predators from surrounding areas, potentially having implications for fish diversity at adjacent reef sites.

These findings are consistent with broader patterns observed across artificial reef systems, which frequently support dense assemblages of schooling pelagic and mobile mid-to-high trophic level predators (Streich et al. 2017, Schulze et al. 2020, Sibley et al. 2023). In this study, predators such as *Lutjanus*, *Epinephelus* / *Plectropomus*, *Caranx*, and *Diagramma* / *Plectorhinchus* spp. were particularly abundant on shipwrecks, echoing patterns reported in the Gulf of Thailand (Harvey et al. 2021, Baillie et al. 2024). In contrast, Paxton et al. (2020b) found that artificial reefs in the USA supported higher densities of transient species such as *Caranx*, but not resident predators like *Lutjanus* or *Epinephelus*, highlighting potential regional differences in species responses to reef structure. The strong resemblance between shipwreck and pinnacle predator assemblages observed in the current study suggests that ship-based artificial reefs may function more like pinnacle reef systems than fringing reefs in terms of predatory fish structure, potentially providing vital insights ~~into management decision making for artificial reef deployments in the region~~ for managers into how reef profile and spatial isolation influence community composition and can be used to guide artificial reef design, placement and ecological objectives.

4.2. Divergence in lower trophic guilds

Despite strong similarities in predator assemblages, shipwrecks diverged markedly from both natural reef types in their support of lower trophic guilds. Grazers and invertivores were consistently underrepresented, with higher densities of *Scarus*, *Chaetodon*, *Pomacanthus*, and various labrids (wrasses) observed on fringing and pinnacle reefs. These patterns ~~appear~~ are

consistent with broader trends across tropical artificial reef systems, where small-bodied benthic feeders and cryptic invertivores are often scarce (Mattos & Yeemin 2020, Koeck et al. 2014, Hickman et al. 2024, Schram et al. 2024). Notably, *Labroides* and *Hemigymnus* spp. have also been reported as underrepresented on shipwrecks relative to natural pinnacles within the same region (Baillie et al. 2024), reinforcing the likelihood that artificial reefs tend to support distinct assemblages lacking key functional groups associated with benthic foraging. Limited habitat complexity and resource availability likely constrain these groups (Fowler & Booth 2012, Simon et al. 2013). Although older or deteriorated shipwrecks can develop finer refugia and turf algal cover (Paxton et al. 2023), the relatively new metal structures surveyed here have not yet reached such stages of colonisation (Simon et al. 2013, Mattos & Yeemin 2020, Monchanin et al. 2021, 2025). These patterns suggest a degree of functional divergence from natural reef communities, driven by limitations in habitat complexity and resource availability on artificial structures (Fowler & Booth 2012, Simon et al. 2013, Schram et al. 2024).

Structural complexity, rather than depth, appears to drive these patterns. Several ecological and physical constraints likely contribute to this divergence. Structural complexity is likely insufficient to establish shipwrecks as effective ecological analogues of natural reefs; shipwrecks often lack fine-scale refugia, cryptic microhabitats, and turf algal cover needed by benthic feeders (Simon et al. 2013, Koeck et al. 2014, Mattos & Yeemin 2020, Monchanin et al. 2021, Monchanin et al. 2025). Differences in depth, location, and reef structure may also play a role. However, the nearby pinnacles surveyed at similar depths supported higher densities of grazers and invertivores. Sites occurred at similar depths (< 30 m), where diversity and herbivore biomass typically remain stable and major declines occur only beyond 30–40 m when coral cover and fine-scale complexity decrease (Stefanoudis et al. 2019, Pinheiro et al. 2023). Nearby

pinnacles at comparable depths still supported higher densities of grazers and invertivores, indicating that structural relief and microhabitat availability, rather than depth, were the primary drivers (Paxton et al. 2020b, Leitner et al. 2021, Galbraith 2021, Galbraith et al. 2021). Analyses by geographic zone (proximity to shore) confirmed that this divergence persisted, indicating that physical setting alone cannot explain the consistent underrepresentation of lower trophic groups on shipwrecks. Zone-based analyses confirmed that trophic divergence persisted, suggesting that physical setting alone cannot fully explain the consistent underrepresentation of lower trophic groups on shipwrecks. Strong currents around wrecks may further limit benthic foraging and favour species that feed higher in the water column (Leitner et al. 2021, Galbraith 2021, Galbraith et al. 2021).

Siganus spp. were notable for their consistent abundance across reef types, diverging from the patterns observed for other grazer taxa. This may reflect species-level variation in ecological traits. Some of the *Siganus* species recorded in this study primarily feed on turf algae, but others surveyed, such as *Siganus javus*, are known to exhibit more flexible or planktivorous feeding strategies (Streit et al. 2015, Gress et al. 2023), potentially allowing persistence on shipwrecks where benthic forage is limited. Recent observations indicate that this flexibility may represent ecological opportunism rather than resource limitation, as *S. javus* has been proposed to feed opportunistically on epibiotic algae and zooplankton from coral polyps (Gress et al. 2023). Their presence on structurally complex artificial reefs has also been reported in other parts of Thailand (Mattos & Yeemin 2020), suggesting a degree of dietary or habitat plasticity not shared by other herbivorous taxa. Similar trends have been reported in the eastern Mediterranean, where planktivores dominate shipwreck assemblages (Sensurat-Genc et al. 2022), implying that the structural features of wrecks may consistently favour planktivorous species. Additionally, a

~~dominance of planktivores has been observed on shipwrecks in other regions, including the eastern Mediterranean (Sensurat-Gene et al. 2022), suggesting that the structural features of wrecks may consistently favour planktivorous species.~~

Depth-associated shifts toward planktivory are well documented at greater depths, but the similarity in depths among our sites suggests that the wreck–pinnacle contrast observed here is better explained by structure and flow (Jankowski et al. 2015, Stefanoudis et al. 2019). Although other planktivorous taxa such as pomacentrids were not surveyed here, their dominance elsewhere highlights their potential importance in shaping trophic dynamics on artificial reefs. Incorporating this group through stationary point count surveys is a focus of ongoing work at these sites, which will allow a more complete assessment of community structure and trophic balance across reef types. ~~These patterns~~Results here highlight that most low-trophic, benthic-feeding species remain functionally excluded from shipwreck communities, -and that addressing habitat and resource limitations in artificial reef design may help support these groups, though it remains uncertain whether such changes would fully replicate natural reef assemblages. unless habitat and resource limitations are addressed explicitly in artificial reef design.

4.3. Functional limitations of ARs as ecosystem surrogates

While overall fish density on shipwrecks was high, comparable to fringing reefs and second only to pinnacles, it was driven primarily by mesopredators and higher trophic level predators, with grazers and invertivores consistently underrepresented. Similar patterns have been documented elsewhere: artificial reefs often match or exceed natural reefs in total fish abundance but differ markedly in functional composition, with consistently reduced species diversity and/ richness (Fowler & Booth, 2012, Paxton et al. 2020a). Assemblages dominated by predators and lacking

lower trophic groups may reflect a combination of habitat features, species-specific responses, and top-down effects, where predator presence suppresses prey guilds (Boaden and Kingsford, 2015). Artificial reef fish communities are thus compositionally novel and shaped by deployment context, structural design, and regional species pools (Folpp et al. 2013, Schulze et al. 2020). Coral surveys at the same reef sites included in this study found that communities on artificial structures remain distinct from those on natural reefs, showing partial convergence with pinnacle assemblages over time (Monchanin et al. 2021, ~~Monchanin et al.~~ 2025). While shipwrecks may offer fisheries value and act as de facto refuges under high exploitation pressure (as they serve as physical barriers to bottom-trawling practices ~~-(Hickman et al. 2024))~~, they appear to fall short of replicating the full ecological function of natural reefs. These findings underscore the need for functionally driven reef design and context-aware management goals when deploying artificial structures.

4.4. Design and management implications

Recognising these structural and ecological mismatches is essential for guiding the design of artificial reefs that support a broader range of functional groups. Artificial reefs can ~~play a valuable role in~~ contribute to marine conservation and fisheries management (Fowler & Booth 2012, Paxton et al. 2020a, Pondella et al 2022, Sibley et al. 2023), but their ecological function depends heavily on how they are designed and where they are deployed (Streich et al. 2017, Paxton et al. 2020a, Baillie et al. 2024). Within this depth range, design variables controlling microhabitat and refuge space exert greater influence on functional outcomes than depth itself (Jankowski et al. 2015, Pinheiro et al. 2023). Structures providing vertical relief are effective at attracting predators, yet replicating the full ecological breadth of natural reefs requires fine-scale

complexity: branching substrates, cavities, and textured surfaces that support turf algae and invertebrate prey (Koeck et al. 2014, Mattos & Yeemin 2020). Structures that offer vertical relief are highly effective at attracting predators, particularly mobile and pelagic species (Simon et al. 2013, Paxton et al. 2020b). However, replicating the full ecological breadth of natural reefs requires more than height alone (Monchanin et al. 2021, Schram et al. 2024, Monchanin et al. 2025). Small grazers and invertivores depend on benthic complexity: fine-scale refugia, branching substrates, crevices, and textured surfaces that support turf algae and invertebrate prey; features that are often lacking from artificial reef structures (Koeck et al. 2014; Mattos & Yeemin 2020). Modular and internally complex designs promote colonisation by benthic specialists more effectively than uniform hulls or concrete blocks. Studies show that modular units and internally complex structures can promote the colonisation of benthic specialists more effectively than uniform concrete blocks or ship hulls (Jaxion-Harm et al. 2018, Monchanin et al. 2021, Monchanin et al. 2025, Jaxion-Harm et al. 2018, Baillie et al. 2024). Incorporating varied surface materials and internal cavities, often absent from metal shipwrecks, can enhance habitat heterogeneity and functional diversity. Internal cavities and varied surface materials that shipwrecks may lack, can help to improve habitat heterogeneity and support functional diversity in fish assemblages.

The surrounding environment ~~may~~ also ~~strongly~~ shape reef performance. Hydrodynamic conditions, such as current strength and direction, can influence the foraging efficiency of pelagic feeders and may enhance the persistence of planktivores and filter feeders (Galbraith et al. 2023, Mattos et al. 2023). Site context matters as much as structure: while deploying artificial reefs near natural reefs may enhance coral recruitment (Monchanin et al. 2021, Monchanin et al. 2025), it may not necessarily ensure ecological similarity or functional equivalence in fish

assemblages (Simon et al. 2013, Galbraith, 2021). Emerging evidence suggests that artificial reefs placed near natural reefs may displace fish and fragment populations rather than enhance overall biodiversity (Layman & Allegier 2020, Medeiros et al. 2022, ~~Paxton et al. 2023~~). ~~This is), a -central issue to the attractor versus producer hypothesis that is subject to debate~~attractor-producer debate in the artificial reef literature (see Paxton et al. 2023 review). Although displacement can heighten exploitation risk when fish move away from protected areas, other studies report minimal effects of proximity (Ross et al. 2016, Sensurat-Genc et al. 2022). Such displacement can also increase vulnerability to exploitation, particularly when artificial habitats draw fish away from protected areas. However, other studies report limited effects of proximity to natural reefs on fish assemblages at shipwrecks (Ross et al. 2016, Sensurat-Genc et al. 2022). Even where displacement occurs, it may not be ecologically detrimental in all contexts. In heavily fished zones, shipwrecks can nonetheless act as de facto refuges, increasing biomass and structural complexity relative to trawled habitats (Hickman et al. 2024). ~~In areas with high fishing pressure, shipwrecks can act as de facto marine refuges, supporting greater fish biomass and more structured communities than surrounding trawled zones (Hickman et al. 2024). These patterns emphasise spatial planning for artificial reefs that promotes functional complementarity with natural reefs, while mitigating anthropogenic impacts in degraded seascapes. These findings highlight the importance of spatial planning that prioritises functional complementarity with natural reefs, while also recognising the potential of ARs to provide refuge and mitigate anthropogenic impacts in degraded seascapes.~~

Broader management strategies should ensure that artificial reefs are implemented alongside natural reef conservation to sustain ecosystem function. ~~Beyond site-level design, broader management strategies in the Gulf of Thailand should ensure that artificial reefs are paired with~~

~~natural reef conservation to ensure ecosystem function is maintained. Shipwrecks and similar structures can contribute to fisheries management, particularly by supporting larger predators (Harvey et al. 2021), but they are not ecological replacements for natural reefs and must be integrated within comprehensive reef management frameworks that prioritise ecosystem conservation (Streich et al. 2017, Pondella et al. 2022, Sibley et al. 2023). While shipwrecks and similar structures can support fisheries management, particularly with larger predators (Harvey et al. 2021), they are not ecological replacements for natural reefs and must be integrated within broader reef management strategies that prioritise ecosystem conservation (Streich et al. 2017, Sibley et al. 2023).~~ Furthermore, given that nearshore shipwrecks have been associated with reduced community heterogeneity and limited ecological benefits (Medeiros et al. 2022), offshore deployments may offer greater functional value. Deploying metal shipwrecks as artificial reefs ~~Positioning artificial reefs~~ in offshore settings that mimic the vertical structure of natural pinnacles could enhance support for vulnerable and commercially important species. Achieving trophic balance requires purpose-built artificial reefs designed to support a range of functional groups, not just HTLP species (Paxton et al. 2020a, Pondella et al. 2022). Structures should be tailored to ecological objectives, incorporating habitat complexity for lower trophic guilds and sited based on hydrodynamic conditions and connectivity. Evaluations must go beyond fish abundance to consider trophic structure, mutualist presence, and long-term community composition. As artificial reefs tend to form novel assemblages (Fowler & Booth 2012, Folpp et al. 2013) functional outcomes, not just biomass metrics, should guide design, deployment, and monitoring (Paxton et al. 2020a, Baillie et al. 2024, Scram et al. 2024). Management frameworks and permitting processes should reflect this by linking artificial reef approval to clearly defined ecological functions (Pondella et al. 2022) and ensuring alignment

with regional reef conservation priorities. Ultimately, artificial reefs can complement conservation goals, yet the protection and restoration of natural reefs remain fundamental to sustaining biodiversity and resilience.~~Ultimately, artificial reefs have the potential to complement conservation goals, but protecting and restoring natural reefs remains essential to sustaining biodiversity and ecosystem resilience in the region.~~

Future research should extend beyond fish assemblages to include benthic and fouling communities that influence habitat development and successional dynamics (Fowler & Booth 2012, Galbraith et al. 2021, 2022).~~Where this study focused on variance of fish assemblages across natural and artificial reef sites, future research should extend beyond fish to include benthic and fouling communities, which influence habitat development and substrate use over time (Fowler & Booth 2012).~~ Long-term monitoring of colonisation and successional trajectories as done in Monchanin et al. (2021, 2025) will help capture shifts in community composition and ecosystem function (Folpp et al. 2013). Comparative analyses of modular and monolithic designs can identify strategies that enhance habitat suitability for lower trophic guilds (Mattos & Yeemin 2020).~~Structural comparisons between modular and monolithic designs, particularly with respect to internal complexity, may reveal strategies for enhancing habitat suitability for lower trophic guilds (Mattos & Yeemin 2020).~~ Incorporating hydrodynamic modelling could improve understanding of spatial patterns in fish abundance and diversity (Galbraith et al. 2023), while behavioural tracking tools, such as acoustic telemetry, can clarify patterns of fish movement, site fidelity, and habitat use. Seasonal monitoring and assessments of invasive species and habitat connectivity will also be critical for managing long-term reef health (Jaxion-Harm et al. 2018, Schulze et al. 2020). Overall, this study provides a foundation for evaluating functional performance and outcomes of artificial reefs.~~Continued emphasis on context-specific design,~~

ecological function, and long-term monitoring will ensure that artificial reefs contribute effectively to ecosystem resilience, biodiversity conservation, and sustainable reef management under accelerating environmental change.~~Ongoing research should prioritise context-specific design, ecological function, and long-term monitoring to inform adaptive management and reef policy in tropical systems. Together, these efforts will help ensure that artificial reefs contribute meaningfully to ecosystem resilience, biodiversity conservation, and sustainable reef management in the face of accelerating environmental change.~~

5. CONCLUSION

This study is the first to compare shipwrecks, pelagic pinnacles, and fringing reefs in terms of fish assemblage structure. Shipwrecks mirrored pinnacles in supporting mesopredators and higher trophic level predators but consistently lacked grazers and invertivores compared to both natural reef types, resulting in a skewed trophic balance. These patterns likely reflect structural and ecological limitations, though site-level variability and seasonal effects may also play a role. Future research using long-term monitoring, fine-scale habitat surveys, and behavioural data will help clarify the drivers of these differences. Effective deployment ~~depends on more than high relief and isolation.~~ Designs of artificial reefs should incorporate fine-scale complexity to support lower trophic guilds, and site selection should consider hydrodynamics, connectivity, and spatial placement relative to natural reefs. In Thailand and other tropical regions, high-relief offshore structures may better meet objectives for vulnerable and commercially important species. Artificial reefs can complement, but not replace, the protection and restoration of natural reefs,

735 and their success depends on clear ecological goals, functional design, and integration within
736 broader conservation strategies.

6. TABLES

Table 1 Predicted mean abundances (with 95% credible intervals) of four functional fish groups across three reef classifications: shipwreck, fringing reef, and pelagic pinnacle. Estimates are derived from a multivariate negative binomial model with reef classification as a fixed effect and site and month-year as ~~a~~ random intercepts. Values reflect the expected total count per survey for each functional group.

Functional Group	Shipwreck	Fringing	Pinnacle
Grazer	47.8 [28.1–83.4] 59.7 [38.1–94.1]	100.7 [56.2–179.0] 127.9 [90.6–180.8]	98.4 [64.6–148.4] 105.9 [77.6–145.1]
Invertivore	28.7 [13.1–63.9] 33.0 [15.9–65.8]	54.5 [24.2–120.7] 55.9 [31.6–96.7]	58.7 [35.1–100.5] 56.0 [33.9–93.6]
Mesopredator	41.7 [11.0–150.2] 43.5 [13.2–140.1]	69.7 [18.2–250.3] 68.7 [26.2–171.9]	134.8 [59.1–311.3] 130.5 [56.6–301.7]
HTLP	24.6 [9.0–70.2] 25.6 [10.2–68.0]	9.1 [3.0–26.6] 11.0 [5.1–23.3]	26.2 [13.4–51.6] 26.4 [13.4–51.1]

Table 2 Posterior summaries of functional group abundance contrasts across zones from the across reef classifications (shipwreck, fringing reef, pinnacle). Values are derived from a Bayesian multivariate negative binomial model. The table reports the posterior median, 95% credible interval (CI), and the proportion of posterior samples supporting positive or negative differences. Estimates are derived from posterior draws of the reef-classification model, with shipwrecks set as the reference level.

Functional Group	Comparison	Median	95% CI	P($\beta > 0$)	P($\beta < 0$)
Grazer	Fringing – Shipwreck	0.7450 760	[0.029, 1.460] [0.197, 1.340]	0.9780 994	0.0220 006
	Pinnacle – Shipwreck	0.7170 574	[0.144, 1.310] [0.022, 1.130]	0.9910 979	0.0090 021
	Pinnacle – Fringing	–0.031 0.188	[–0.642, 0.589] [–0.657, 0.278]	0.4570 203	0.5430 797
Invertivore	Fringing – Shipwreck	0.6280 525	[–0.476, 1.750] [–0.359, 1.420]	0.8800 891	0.1200 109

Fish on Shipwrecks vs. Pinnacles

Mesopredator	Pinnacle – Shipwreck	0.7110 522	[-0.188, 1.610] [-0.337, 1.420]	0.9450 897	0.0550 103
	Pinnacle – Fringing	0.0770 006	[-0.857, 0.994] [-0.733, 0.738]	0.5740 507	0.4260 493
	Fringing – Shipwreck	0.5140 447	[-1.340, 2.380] [-1.070, 1.990]	0.7140 738	0.2860 262
	Pinnacle – Shipwreck	1.1901 100	[-0.345, 2.700] [-0.346, 2.580]	0.9400 939	0.0600 061
	Pinnacle – Fringing	0.6680 642	[-0.865, 2.200] [-0.616, 1.980]	0.8200 854	0.1800 146
	Fringing – Shipwreck	-0.997 0.855	[-2.510, 0.466] [-2.100, 0.396]	0.0770 079	0.9230 921
HTLP	Pinnacle – Shipwreck	0.0660 022	[-1.170, 1.250] [-1.170, 1.160]	0.5510 518	0.4490 482
	Pinnacle – Fringing	1.0600 880	[-0.175, 2.340] [-0.141, 1.860]	0.9560 960	0.0440 041

Table 3 Posterior summaries of functional group abundance contrasts across zones from the Bayesian multivariate model. The table reports the posterior median, 95% credible interval (CI), and the proportion of posterior samples supporting positive or negative differences. Estimates are derived from posterior draws of the zone-level model, with shipwrecks set as the reference level.

Functional Group	Comparison	Median	95% CI	P($\beta > 0$)	P($\beta < 0$)
Grazer	Nearshore – Wreck	0.7930 674	[0.182, 1.380] [0.122, 1.280]	0.9940 988	0.0060 012
	Pelagic – Wreck	0.6150 747	[0.033, 1.490] [-0.095, 1.320]	0.9800 958	0.0200 042

Fish on Shipwrecks vs. Pinnacles

	Pelagic – Nearshore	– 0.066 3	[-0.636, 0.584] [-0.662, 0.505]	0.453 404	0.547 596
	Nearshore – Wreck	0.804 670	[0.030, 1.540] [0.083, 1.440]	0.978 964	0.022 036
	Pelagic – Wreck	0.322 419	[-0.618, 1.210] [-0.770, 1.050]	0.767 606	0.233 394
Invertivore	Pelagic – Nearshore	-0.479 0.553	[-1.240, 0.277] [-1.290, 0.190]	0.099 063	0.901 937
	Nearshore – Wreck	0.897 843	[-0.588, 2.380] [-0.679, 2.290]	0.891 878	0.109 122
	Pelagic – Wreck	0.790 675	[-0.971, 2.590] [-1.170, 2.490]	0.824 782	0.176 218
Mesopredator	Pelagic – Nearshore	-0.119 0.163	[-1.580, 1.390] [-1.650, 1.340]	0.436 408	0.564 592
	Nearshore – Wreck	-0.525 0.499	[-1.770, 0.720] [-1.740, 0.773]	0.186 200	0.814 800
	Pelagic – Wreck	0.082 069	[-1.390, 1.560] [-1.450, 1.600]	0.541 539	0.459 461
HTLP	Pelagic – Wreck	0.082 069	[-1.390, 1.560] [-1.450, 1.600]	0.541 539	0.459 461
	Pelagic – Nearshore	0.605 569	[-0.595, 1.810] [-0.681, 1.840]	0.852 819	0.148 181

7. FIGURES

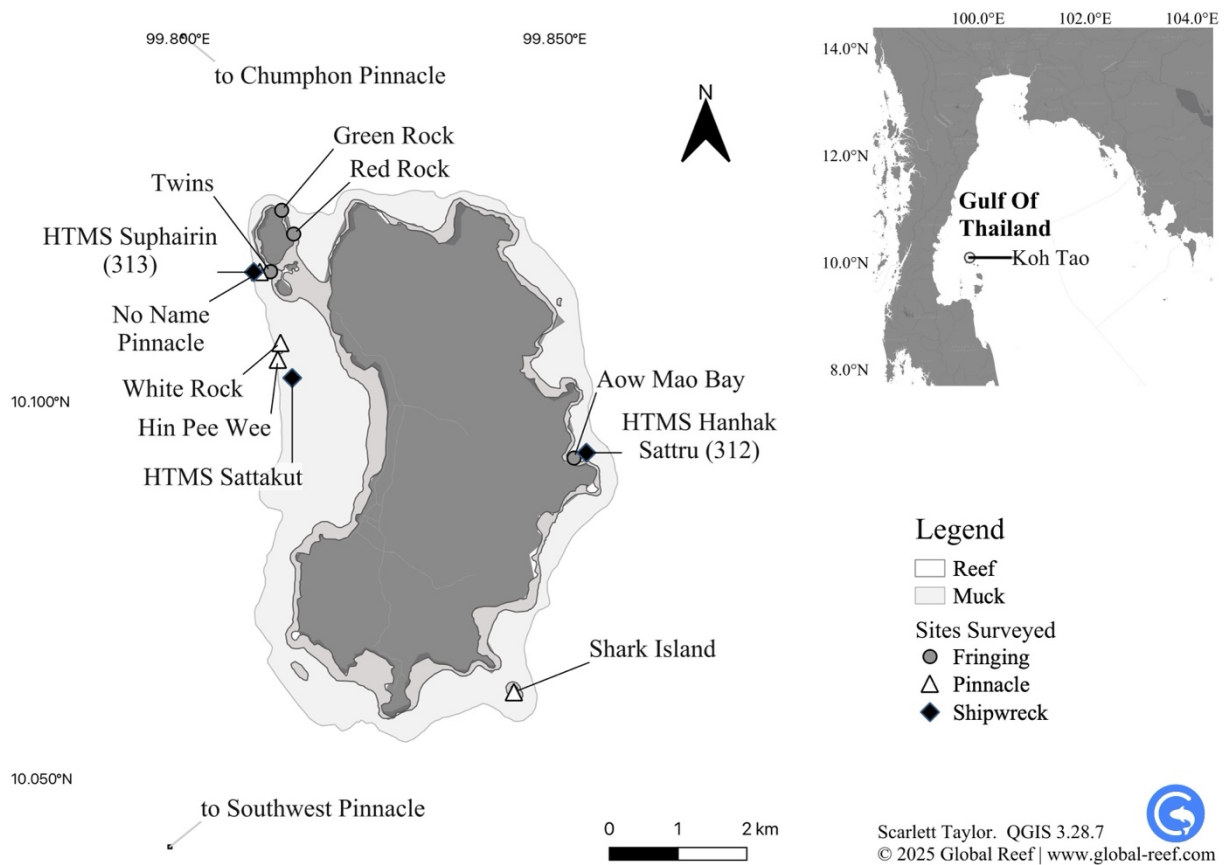
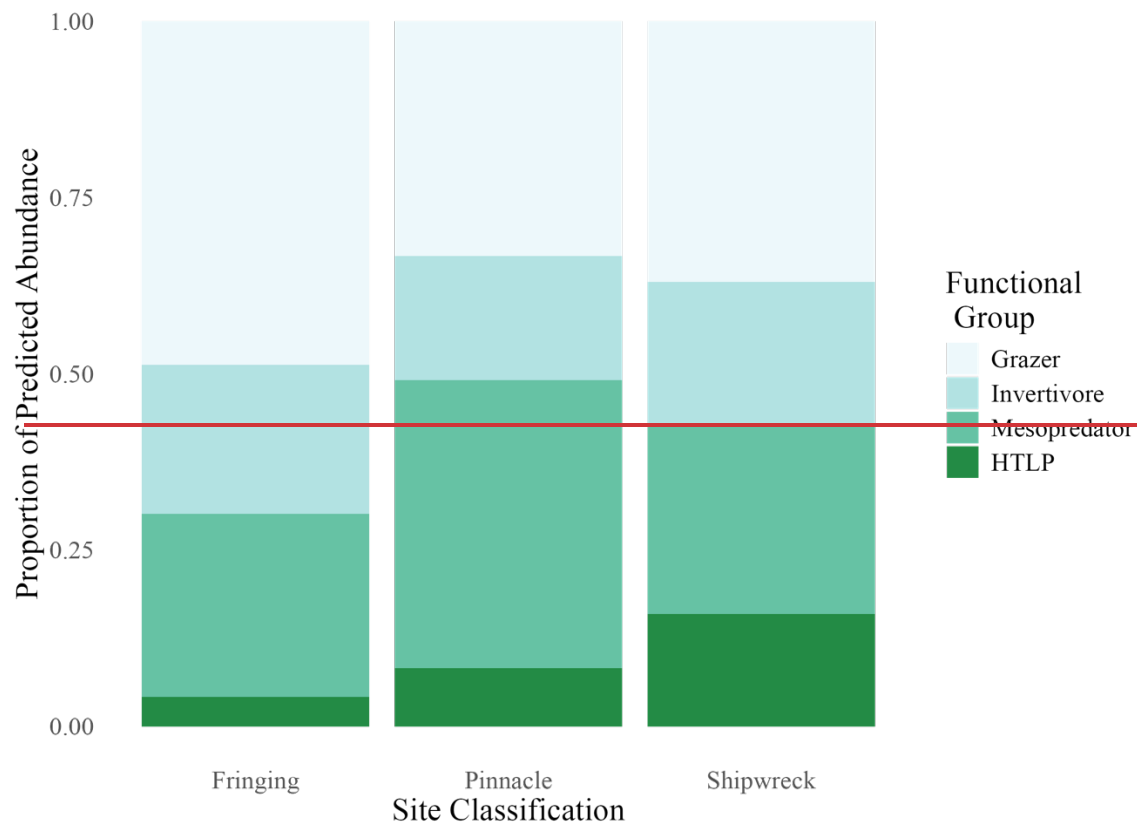
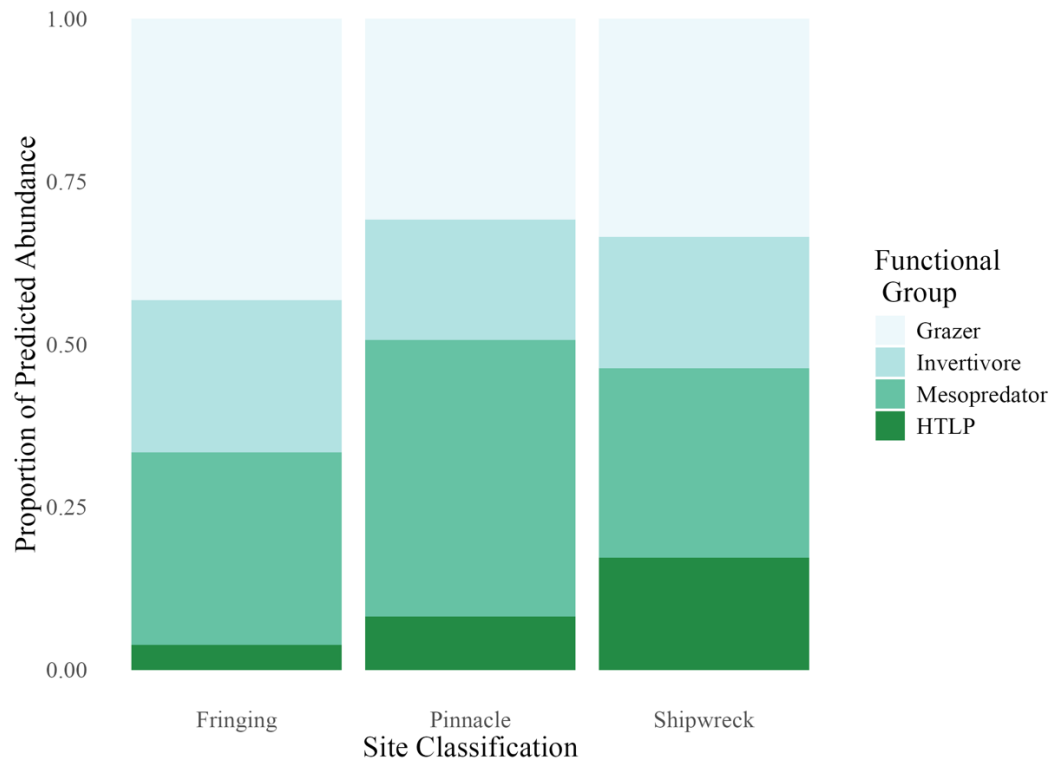


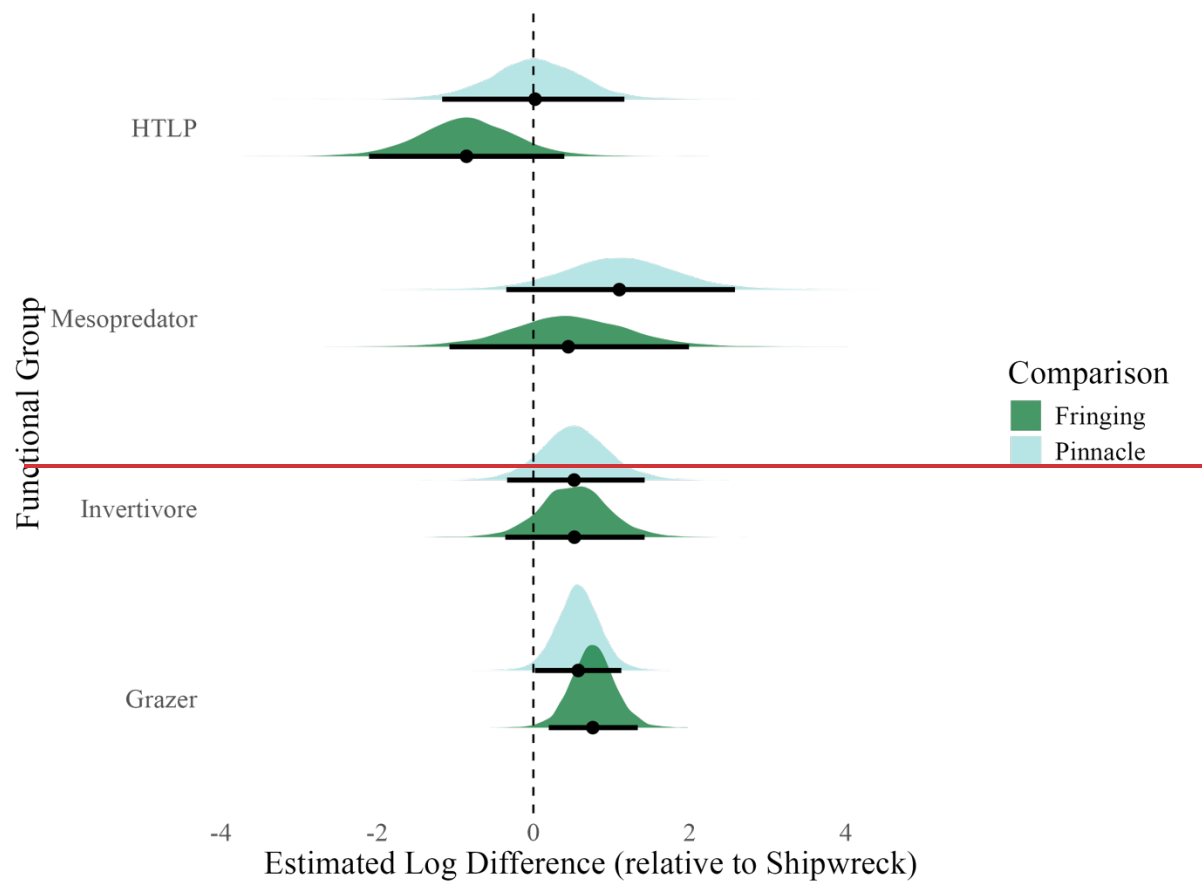
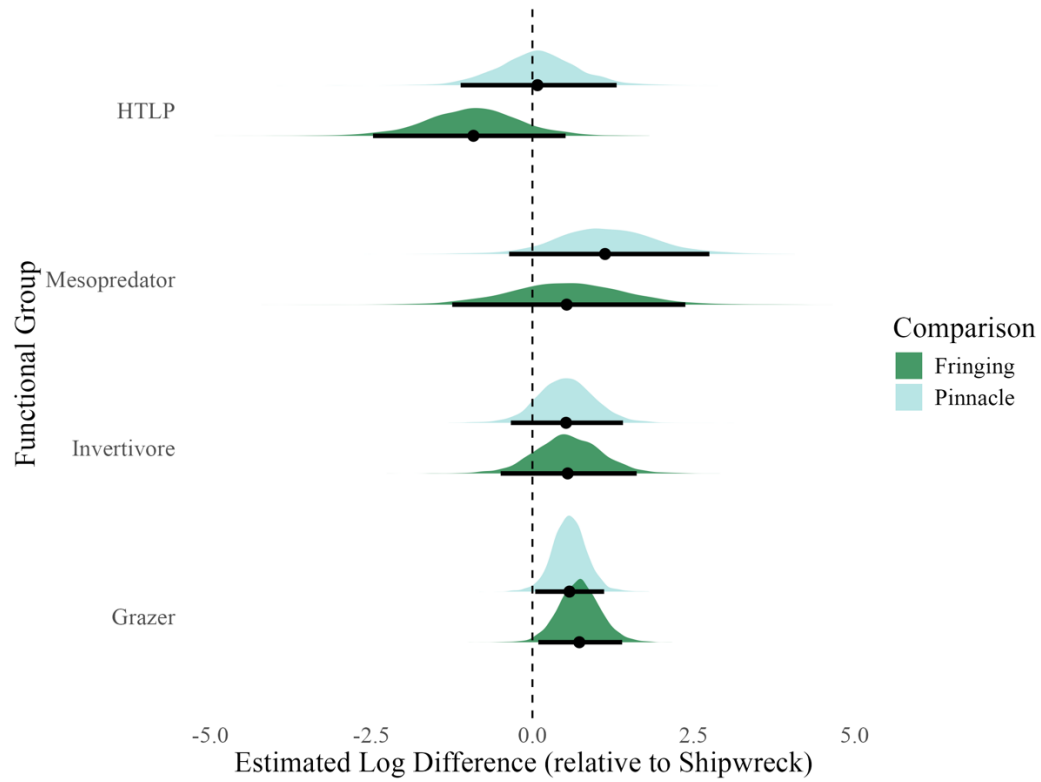
Figure 1 Dive sites surveyed around Koh Tao in the Gulf of Thailand (10.10980°N, 99.81180°E).

Fish on Shipwrecks vs. Pinnacles



761 Figure 2 Predicted mean abundances of four functional fish groups—grazers, invertivores,
762 mesopredators, and higher trophic level predators (HTLPs)—across three reef classifications:
763 fringing reefs, pelagic pinnacles, and shipwrecks. Each panel displays the posterior mean
764 abundance for one functional group based on multivariate negative binomial models.

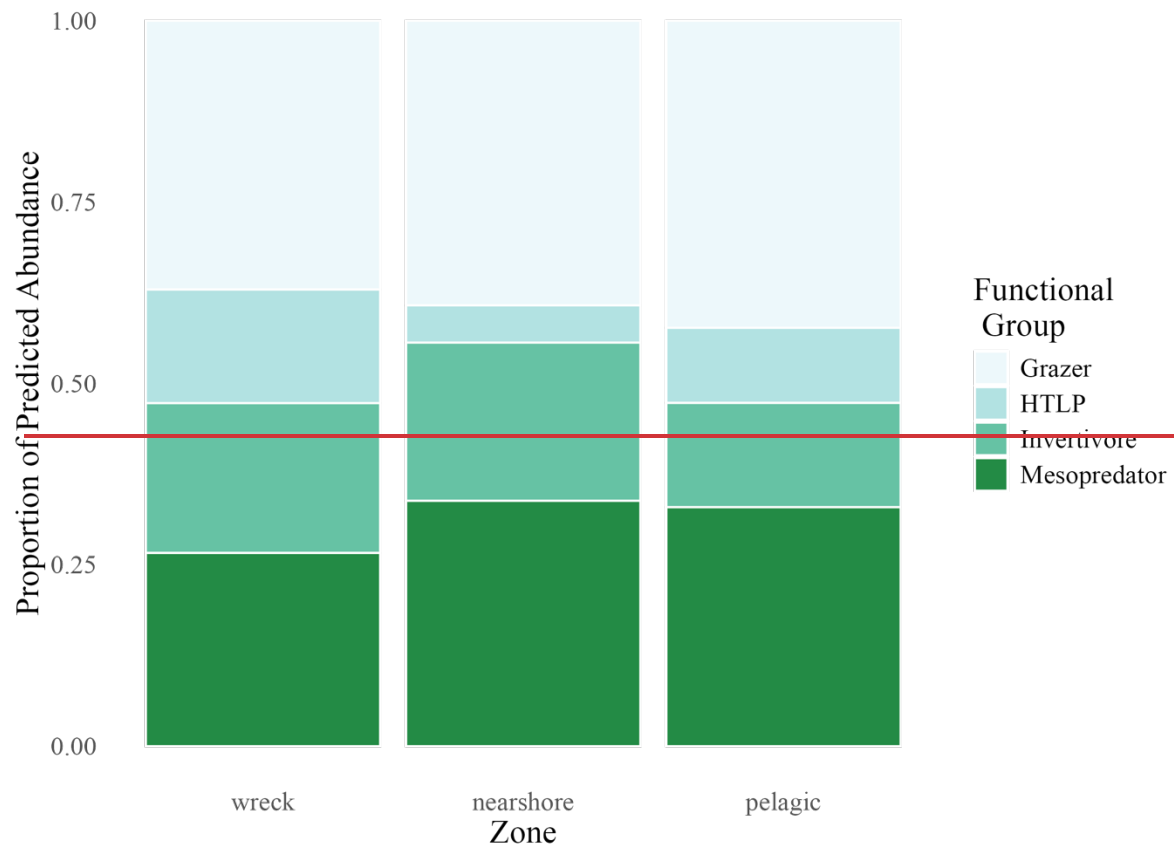
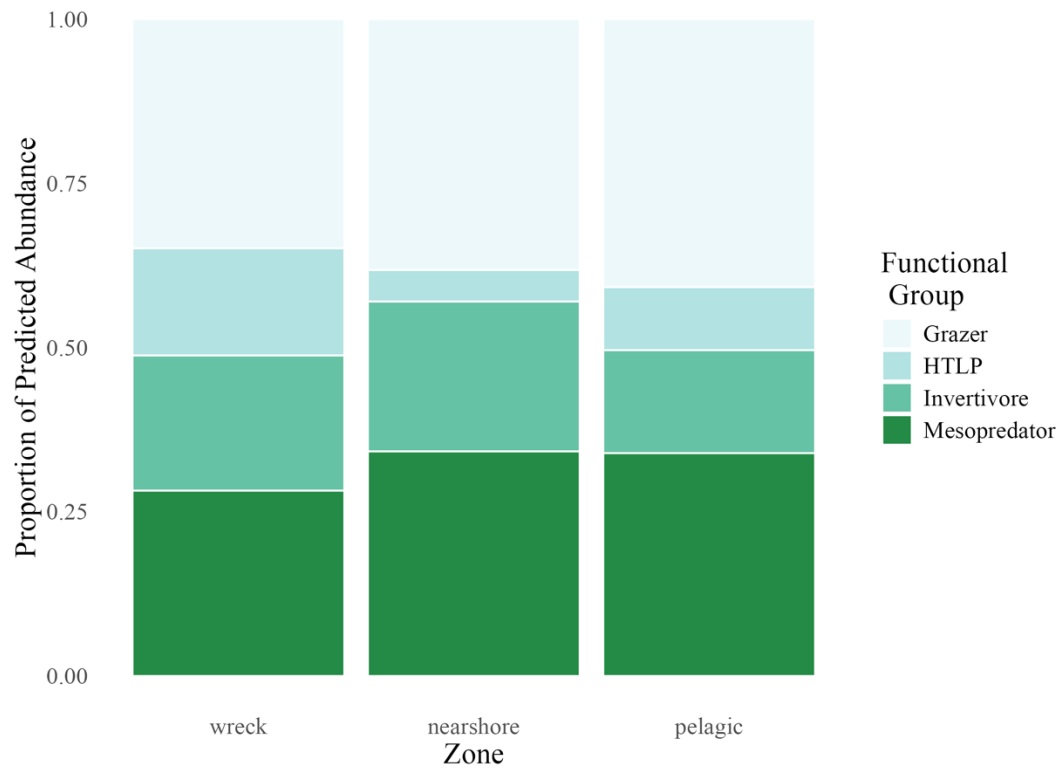
Fish on Shipwrecks vs. Pinnacles



Fish on Shipwrecks vs. Pinnacles

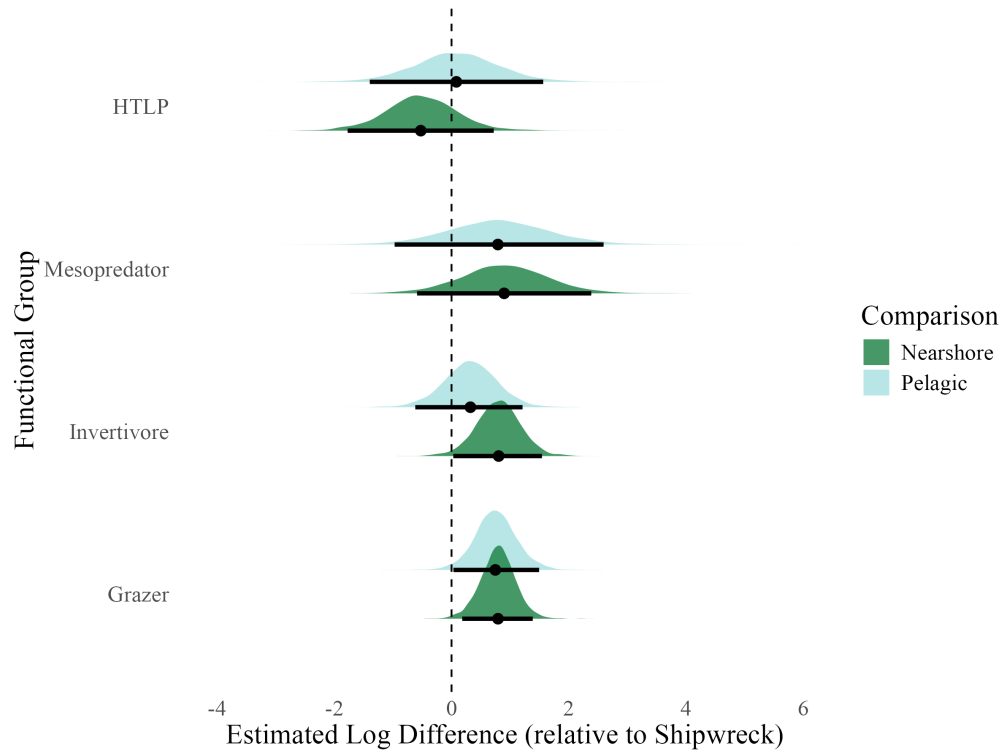
767 Figure 3 Posterior distributions show estimated log-scale differences in abundance for each
768 functional group on fringing reefs (dark green) and pinnacles (light blue), relative to shipwrecks.
769 Densities represent posterior uncertainty, with medians (dots) and 95% credible intervals (thick
770 bars) overlaid. Zero indicates no difference from shipwrecks; positive values denote higher
771 predicted abundance.

Fish on Shipwrecks vs. Pinnacles



774 Figure 4 Predicted proportional composition of four functional fish groups across three reef
775 zones (defined by proximity to shore): wreck, nearshore, and pelagic. Each bar represents the
776 relative contribution of grazers, invertivores, mesopredators, and higher trophic level predators
777 (HTLPs) to the total fish count within each zone. Proportions are normalised to 1.0 for
778 comparison across zones.

Fish on Shipwrecks vs. Pinnacles



779

Fish on Shipwrecks vs. Pinnacles

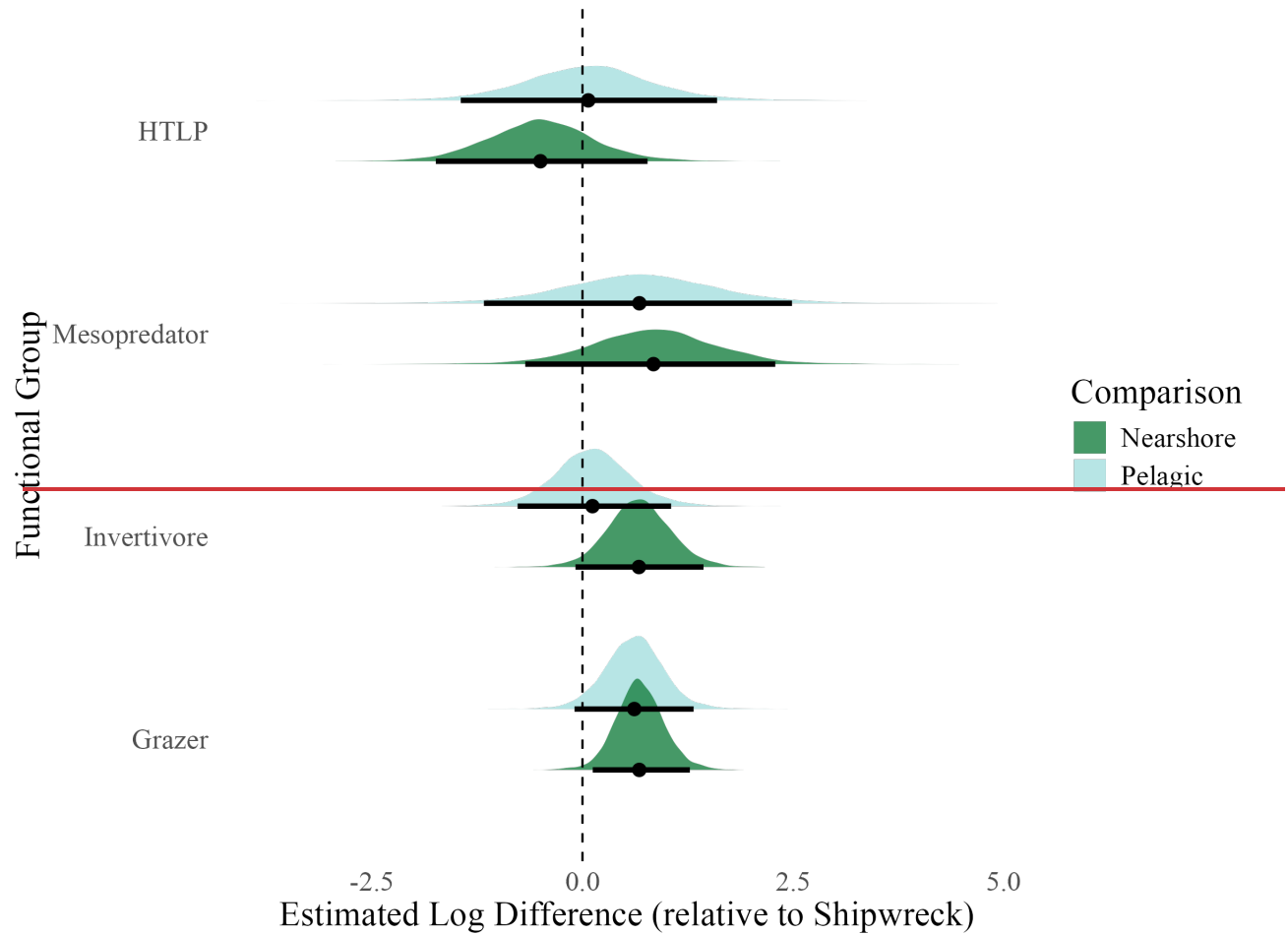


Figure 5 Posterior distributions show estimated log-scale differences in abundance for each functional group in nearshore (dark green) and pelagic (light blue) zones, relative to shipwrecks. Densities represent posterior uncertainty, with medians (dots) and 95% credible intervals (thick bars) overlaid. Zero indicates no difference from shipwrecks; positive values denote higher predicted abundance.

Fish on Shipwrecks vs. Pinnacles

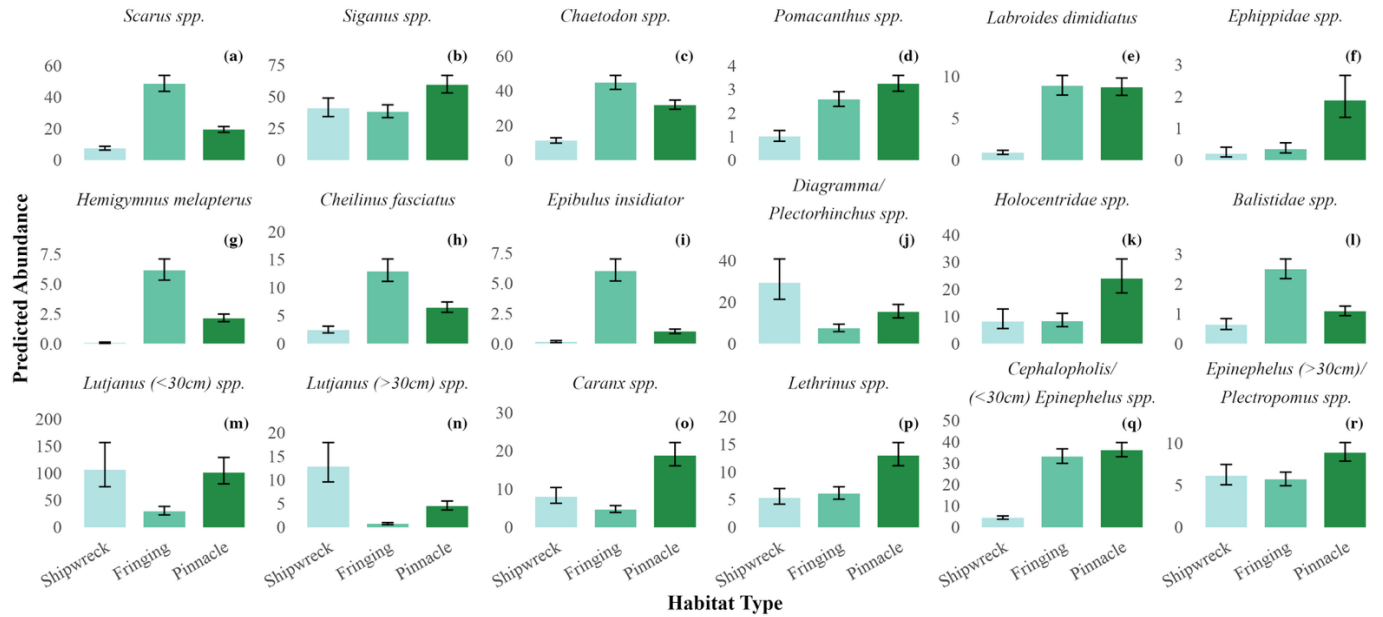


Figure 6 Predicted mean abundances of 18 reef fish taxa (a through r) across three reef classifications: fringing reef, pelagic pinnacle, and shipwreck. Each panel represents one species, with bars showing the posterior mean abundance from multivariate negative binomial models. Error bars represent Bayesian 95% credible intervals. Bars are coloured by reef classification to facilitate comparison of relative abundance patterns across reef types.

Fish on Shipwrecks vs. Pinnacles

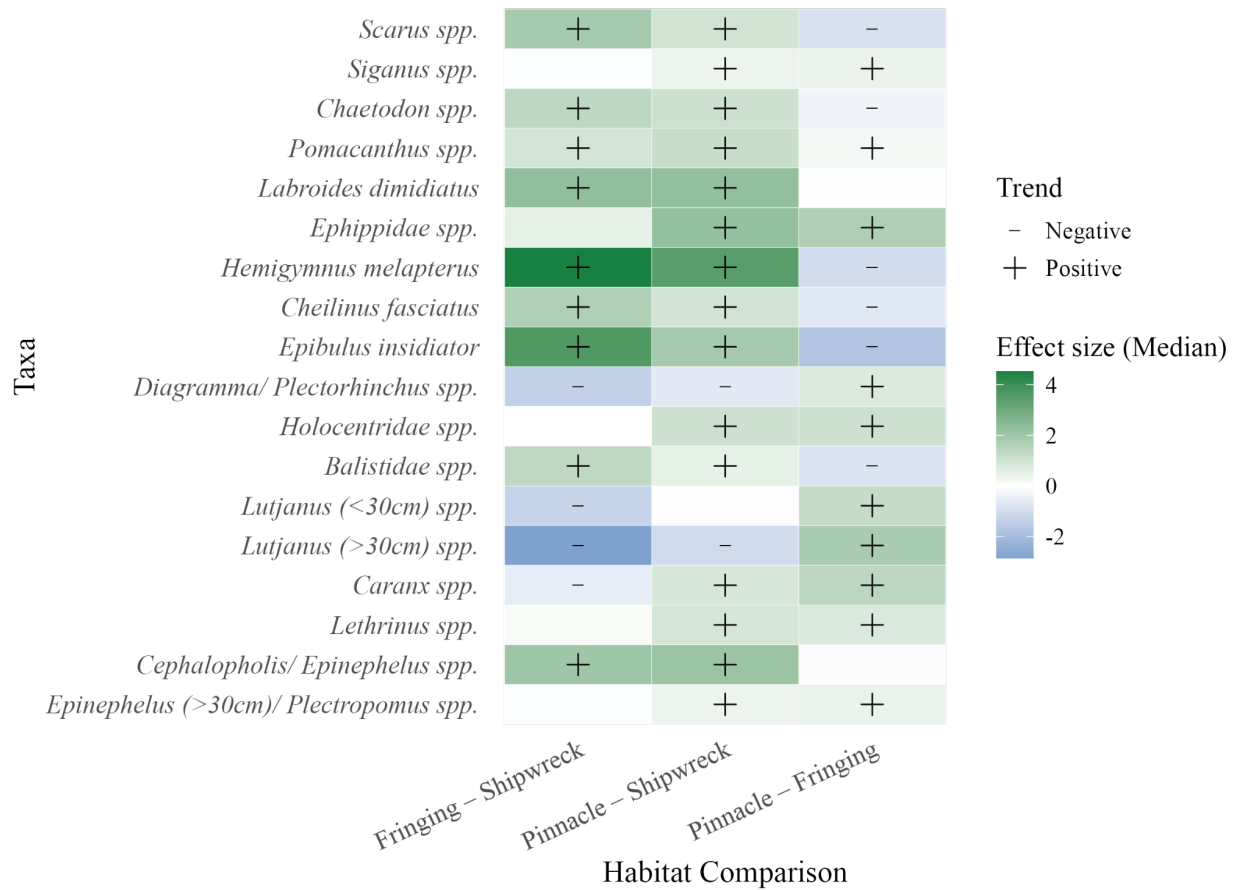


Figure 7 Species-level estimates of fish abundance differences across habitat types, relative to shipwrecks, from a Bayesian multivariate model. Each tile represents the median posterior effect size for a given species–habitat comparison (Fringing or Pinnacle reef vs. Shipwreck baseline). Positive values indicate higher predicted abundance on natural reefs relative to shipwrecks, and negative values indicate the opposite. Black symbols denote species with strong support for directional effects (posterior probability > 0.95 or < 0.05); plus signs indicate positive effects, and minus signs indicate negative effects.

8. ACKNOWLEDGEMENTS

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10. CONFLICT OF INTERESTS

Authors of the current study declare no conflict of interest.

11. ETHICAL APPROVAL

This study did not require formal ethical approval, and data collection followed standard scientific ethical guidelines. All necessary permits for observational research were secured.

12. DATA AVAILABILITY

All data and code used in this study are openly available through Zenodo from <https://doi.org/10.5281/zenodo.16784382> ~~<https://doi.org/10.5281/zenodo.16784464>~~, archived at the time of submission.

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Table S1 Sites surveyed around Koh Tao, Thailand (10.10980° N, 99.81180° E).

<u>Site</u>	<u>Type</u>	<u>Classification</u>	<u>Zone</u>	<u>First Date</u>	<u>Last Date</u>	<u>N Surveys</u>	<u>Days span</u>	<u>Median gap (days)</u>	<u>Mean Depth (m)</u>	<u>Depth Range (m)</u>
HTMS Hanhak Sattru	Artificial	Shipwreck	Wreck	2023-09-19	2025-06-09	19	629	14.0	20.6	17–26.5
HTMS Supharin	Artificial	Shipwreck	Wreck	2023-09-01	2025-06-17	28	655	14.0	20.7	14.8–24
HTMS Sattakut	Artificial	Shipwreck	Wreck	2022-11-01	2025-05-19	25	930	30.5	22.3	16–25
Aow Mao	Natural	Fringing	Nearshore	2023-07-01	2025-06-18	26	718	14.0	10.5	7.5–15
Green Rock Wall	Natural	Fringing	Nearshore	2020-06-01	2025-06-05	31	1830	31.0	10.6	8.5–13
Red Rock Wall	Natural	Fringing	Nearshore	2019-03-01	2025-06-05	38	2288	31.0	10.6	9.5–13
Twins Pinnacles	Natural	Fringing	Nearshore	2019-04-01	2025-06-16	42	2268	31.0	10.5	7–14
Hin Pee Wee	Natural	Pinnacle	Nearshore	2022-11-01	2025-05-19	29	930	17.5	16.6	11.8–23
Shark Island	Natural	Pinnacle	Nearshore	2025-02-05	2025-06-04	6	119	23.0	11.8	8–15
No Name Pinnacle	Natural	Pinnacle	Nearshore	2022-11-01	2025-06-17	32	959	30.0	16.1	12–24
White Rock	Natural	Pinnacle	Pelagic	2019-03-01	2025-05-08	43	2260	31.0	11.9	8–18
Chumphon Pinnacle	Natural	Pinnacle	Pelagic	2019-09-01	2025-06-17	11	2116	16.5	19.2	15.7–30
Southwest Pinnacle	Natural	Pinnacle	Pelagic	2019-09-01	2025-06-12	9	2111	28.0	18.8	13.5–25

Table S2 List of reef-associated taxa recorded during surveys, including common names, genus and species-level identifications (where possible), and assigned functional group classifications. Species chosen are the most commonly spotted in the waters surrounding Koh Tao, span a range of trophic levels and ecological roles, and include both site-attached and mobile predators.

Common Name	Genus	Species	Functional Group
Parrotfish	<i>Scarus</i>	spp.	Grazer
Rabbitfish	<i>Siganus</i>	spp.	Grazer
Butterflyfish	<i>Chaetodon</i>	spp.	Grazer
Angelfish	<i>Pomacanthus</i>	spp.	Invertivore
Cleaner Wrasse	<i>Labroides</i>	<i>dimidiatus</i>	Invertivore
Batfish	Ephippidae	spp.	Invertivore
Thicklip Wrasse	<i>Hemigymnus</i>	<i>melapterus</i>	Invertivore
Red Breast Wrasse	<i>Cheilinus</i>	<i>fasciatus</i>	Invertivore
Slingjaw Wrasse	<i>Epibulus</i>	<i>insidiator</i>	Invertivore
Sweetlips	<i>Diagramma</i> / <i>Plectorhinchus</i>	spp.	Invertivore
Squirrelfish and Soldierfish	Holocentridae	spp.	Invertivore
Triggerfish	Balistidae	spp.	Invertivore
Porcupinefish and Pufferfish	<i>Diodon</i> / <i>Tetraodon</i>	spp.	Invertivore
Ray	<i>Taeniura</i> / <i>Neotrygon</i>	spp.	Mesopredator
Brown Stripe Snapper	<i>Lutjanus</i>	<i>vitta</i>	Mesopredator
Russell's Snapper	<i>Lutjanus</i>	<i>russellii</i>	Mesopredator
Large (>30cm) Snapper	<i>Lutjanus</i>	spp.	HTLP
Eel	<i>Gymnothorax</i>	spp.	Mesopredator
Trevally	<i>Caranx</i>	spp.	HTLP
Emperorfish	<i>Lethrinus</i>	spp.	Mesopredator
Small (<30cm) Grouper	<i>Cephalopholis</i> / <i>Epinephelus</i>	spp.	Mesopredator
Large (>30cm) Grouper	<i>Epinephelus</i>	spp.	HTLP
Barracuda	<i>Sphyraena</i>	spp.	HTLP

Table S3 Predicted mean abundances (with 95% credible intervals) of 18 reef fish taxa across three reef classifications: shipwreck, fringing reef, and pelagic pinnacle. Estimates are derived from a Bayesian multivariate negative binomial model with reef classification as a fixed effect and site as a random intercept. Taxa are grouped by functional role (grazer, invertivore, mesopredator, HTLP), and values reflect the expected total count per survey.

Taxa	Functional Group	Shipwreck	Fringing	Pinnacle
<i>Scarus</i> spp.	Grazer	7.47 (6.39–8.78)	48.42 (43.69–53.81)	19.35 (17.62–21.34)
<i>Siganus</i> spp.	Grazer	40.88 (34.36–49.05)	38.19 (33.59–43.67)	59.39 (53.02–66.87)
<i>Chaetodon</i> spp.	Grazer	11.21 (9.80–12.88)	44.57 (40.79–48.84)	31.84 (29.37–34.62)
<i>Pomacanthus</i> spp.	Invertivore	1.01 (0.80–1.26)	2.57 (2.28–2.90)	3.24 (2.92–3.59)
<i>Labroides dimidiatus</i>	Invertivore	0.90 (0.68–1.17)	8.86 (7.77–10.12)	8.70 (7.74–9.81)
Ephippidae spp.	Invertivore	0.20 (0.10–0.41)	0.34 (0.22–0.54)	1.87 (1.35–2.66)
<i>Hemigymnus melapterus</i>	Invertivore	0.07 (0.03–0.14)	6.13 (5.31–7.09)	2.13 (1.84–2.48)
<i>Cheilinus fasciatus</i>	Invertivore	2.44 (1.92–3.12)	12.92 (11.12–15.12)	6.43 (5.60–7.45)
<i>Epibulus insidiator</i>	Invertivore	0.16 (0.09–0.26)	6.01 (5.19–7.01)	1.01 (0.84–1.21)
<i>Diagramma</i> / <i>Plectorhinchus</i> spp.	Invertivore	29.36 (21.39–40.85)	7.35 (5.84–9.42)	15.23 (12.46–18.93)
Holocentridae spp.	Invertivore	8.21 (5.56–12.77)	8.29 (6.27–11.17)	23.95 (18.68–31.17)
Balistidae spp.	Invertivore	0.64 (0.48–0.84)	2.49 (2.18–2.84)	1.09 (0.94–1.26)
<i>Lutjanus</i> spp. (<30cm)	Mesopredator	105.95 (74.84–156.53)	29.33 (22.74–38.45)	100.88 (80.22–128.99)
<i>Cephalopolis</i> / <i>Epinephelus</i> spp. (<30cm)	Mesopredator	4.46 (3.77–5.30)	33.18 (29.97–36.79)	36.20 (33.10–39.74)
<i>Lethrinus</i> spp.	Mesopredator	5.35 (4.17–7.00)	6.06 (5.07–7.32)	13.00 (11.14–15.33)
<i>Lutjanus</i> spp. (>30cm)	HTLP	12.82 (9.58–17.92)	0.74 (0.57–0.99)	4.47 (3.66–5.55)
<i>Epinephelus</i> spp. (>30cm)	HTLP	6.11 (5.06–7.48)	5.69 (4.95–6.57)	8.89 (7.88–10.10)
<i>Caranx</i> spp.	HTLP	8.02 (6.28–10.42)	4.67 (3.89–5.67)	18.81 (16.10–22.19)

Table S4 Posterior probabilities of pairwise differences in predicted log-abundance for 18 reef taxa across reef classifications (shipwreck, fringing reef, pinnacle). Values are derived from a Bayesian multivariate negative binomial model. The table reports the posterior median, 95% credible interval (CI), and the proportion of posterior samples supporting positive or negative differences.

Taxa	Functional Group	Comparison	Median	CI _{lower}	CI _{upper}	P($\beta > 0$)	P($\beta < 0$)
<i>Scarus spp.</i>	Grazer	Fringing – Shipwreck	1.87	1.68	2.06	1	0
		Pinnacle – Fringing	-0.917	-1.06	-0.78	0	1
		Pinnacle – Shipwreck	0.953	0.77	1.14	1	0
<i>Siganus spp.</i>	Grazer	Fringing – Shipwreck	-0.068	-0.287	0.153	0.272	0.728
		Pinnacle – Fringing	0.441	0.27	0.612	1	0
		Pinnacle – Shipwreck	0.373	0.152	0.579	0.999	0.001
<i>Chaetodon spp.</i>	Grazer	Fringing – Shipwreck	1.38	1.22	1.54	1	0
		Pinnacle – Fringing	-0.335	-0.462	-0.213	0	1
		Pinnacle – Shipwreck	1.04	0.887	1.2	1	0
<i>Pomacanthus spp.</i>	Invertivore	Fringing – Shipwreck	0.936	0.677	1.2	1	0
		Pinnacle – Fringing	0.231	0.074	0.387	0.999	0.002
		Pinnacle – Shipwreck	1.16	0.92	1.42	1	0
<i>Labroides dimidiatus</i>	Invertivore	Fringing – Shipwreck	2.29	1.99	2.59	1	0
		Pinnacle – Fringing	-0.017	-0.194	0.162	0.426	0.574
		Pinnacle – Shipwreck	2.27	1.98	2.57	1	0
<i>Ephippidae spp.</i>	Invertivore	Fringing – Shipwreck	0.539	-0.281	1.35	0.908	0.092
		Pinnacle – Fringing	1.7	1.14	2.25	1	0
		Pinnacle – Shipwreck	2.24	1.46	2.99	1	0
<i>Hemigymnus melapterus</i>	Invertivore	Fringing – Shipwreck	4.52	3.81	5.4	1	0
		Pinnacle – Fringing	-1.05	-1.26	-0.846	0	1
		Pinnacle – Shipwreck	3.47	2.75	4.36	1	0
<i>Cheilinus fasciatus</i>	Invertivore	Fringing – Shipwreck	1.67	1.37	1.97	1	0
		Pinnacle – Fringing	-0.698	-0.908	-0.484	0	1
		Pinnacle – Shipwreck	0.973	0.672	1.26	1	0
<i>Epibulus insidiator</i>	Invertivore	Fringing – Shipwreck	3.65	3.12	4.25	1	0
		Pinnacle – Fringing	-1.79	-2.03	-1.55	0	1
		Pinnacle – Shipwreck	1.86	1.32	2.46	1	0
<i>Diagramma/</i>	Invertivore	Fringing – Shipwreck	-1.38	-1.79	-0.997	0	1

Taxa	Functional Group	Comparison	Median	CI _{lower}	CI _{upper}	P($\beta > 0$)	P($\beta < 0$)
<i>Plectorhinchus spp.</i>		Pinnacle – Fringing	0.731	0.423	1.03	1	0
		Pinnacle – Shipwreck	-0.651	-1.04	-0.284	0	1
<i>Holocentridae spp.</i>	Invertivore	Fringing – Shipwreck	0.009	-0.503	0.499	0.515	0.485
		Pinnacle – Fringing	1.06	0.666	1.45	1	0
		Pinnacle – Shipwreck	1.08	0.574	1.54	1	0
<i>Balistidae spp.</i>	Invertivore	Fringing – Shipwreck	1.36	1.06	1.67	1	0
		Pinnacle – Fringing	-0.825	-1.02	-0.628	0	1
		Pinnacle – Shipwreck	0.536	0.225	0.857	1	0
<i>Lutjanus (<30cm) spp.</i>	Mesopredator	Fringing – Shipwreck	-1.29	-1.76	-0.851	0	1
		Pinnacle – Fringing	1.24	0.888	1.58	1	0
		Pinnacle – Shipwreck	-0.051	-0.508	0.374	0.405	0.595
<i>Lethrinus spp.</i>	Mesopredator	Fringing – Shipwreck	0.121	-0.204	0.442	0.768	0.232
		Pinnacle – Fringing	0.767	0.52	1	1	0
		Pinnacle – Shipwreck	0.889	0.581	1.19	1	0
<i>Cephalopholis/ Epinephelus spp.</i>	Mesopredator	Fringing – Shipwreck	2.01	1.81	2.2	1	0
		Pinnacle – Fringing	0.088	-0.05	0.226	0.89	0.11
		Pinnacle – Shipwreck	2.09	1.9	2.28	1	0
<i>Lutjanus (>30cm) spp.</i>	HTLP	Fringing – Shipwreck	-2.85	-3.28	-2.44	0	1
		Pinnacle – Fringing	1.8	1.46	2.12	1	0
		Pinnacle – Shipwreck	-1.06	-1.44	-0.692	0	1
<i>Caranx spp.</i>	HTLP	Fringing – Shipwreck	-0.542	-0.867	-0.229	0.001	0.999
		Pinnacle – Fringing	1.39	1.15	1.64	1	0
		Pinnacle – Shipwreck	0.849	0.544	1.15	1	0
<i>Epinephelus (>30cm)/ Plectropomus spp.</i>	HTLP	Fringing – Shipwreck	-0.074	-0.318	0.161	0.27	0.73
		Pinnacle – Fringing	0.446	0.26	0.634	1	0
		Pinnacle – Shipwreck	0.371	0.138	0.599	0.999	0.001